

TECHNICAL DESCRIPTION (PART B)

PROJECT	
Project name:	Scale-up biotech AI innovation by leveraging responsible data use under the European Health Data Space
Project acronym:	SCAILED
Coordinator contact:	Wei Gu, Luxembourg National Data Service (PNED G.I.E.)

Table of contents

PROJECT SUMMARY	2
1. RELEVANCE	2
1.1 Background and general objectives	2
1.2 Needs analysis	4
1.3 Complementarity with other actions and innovation — European added value	6
2. QUALITY	7
2.1 Concept and methodology	7
2.2 Consortium set-up	14
2.3 Project teams, staff and experts	16
2.4 Consortium management and decision-making	20
2.5 Project management, quality assurance and monitoring and evaluation strategy	22
2.6 Cost effectiveness and financial management	23
2.7 Risk management	24
3. IMPACT	26
3.1 Impact and ambition	26
3.2 Communication, dissemination and visibility	28
3.3 Sustainability and continuation	30
4. WORK PLAN, WORK PACKAGES, ACTIVITIES, RESOURCES AND TIMING	32
Cross WP Interaction:	42
<i>Subcontracting (n/a for prefixed Lump Sum Grants)</i>	62
5. OTHER	65
5.1 Ethics	65
5.2 Security	65
6. DECLARATIONS	65

List of figures

Figure 1 - The five building blocks to implement the ambition of the general objectives	8
Figure 2 - Work packages and mapping to the building blocks.	9
Figure 3 - The “Data Journey” and “AI Innovation Journey” serving as the guiding threads for the project to systematically identify gaps and needs.	11
Figure 4 - The synchronised iteration to align the tasks across work packages and reach the expected maturity of the outcome	11
Figure 5 - Consortium management structure	21

List of tables

Table 1 – Specific Objectives	5
Table 2 - Types of organisations in the consortium, their roles and value	14
Table 3 - Related EU initiatives and representation from the consortium	15
Table 4 - List of project teams and staff (full list available in annex)	16
Table 5 - List of indicators to measure the implementation of specific objectives defined in section 1.2 and target values (baseline: 0)	23
Table 6 - List of risks and mitigation measures	25
Table 7 - Target groups and benefits	27
Table 8 - List of communication and dissemination activities and KPIs	28
Table 9 - Target groups and ways to reach team	29

#APP-FORM-EU4H@#

#@PRJ-SUM-PS@#

PROJECT SUMMARY

Project summary

SCAILED (**S**cale-up Biotech **A**I Innovation by **L**everaging Responsible Data Use under the **E**uropean Health **D**ata Space) aims to accelerate the development and deployment of Artificial Intelligence (AI) and digital solutions in the European biotech sector by harnessing high-quality health data within the secure, interoperable framework of the European Health Data Space (EHDS).

The project addresses critical challenges faced by biotech innovators, including complex and evolving regulatory requirements (AI Act, Medical Device Regulations, etc.), fragmented data governance in legal and technical term sand respective barriers to AI deployment due to insufficient stakeholder collaboration. SCAILED responds directly to the EU4Health call by proposing a coordinated, multi-stakeholder approach that brings together industry, research infrastructures, health authorities, patient organisations, and AI Factories to foster responsible, scalable, and sustainable innovation based on recently adopted and upcoming legislation.

Objectives:

- Establish a multistakeholder platform and stakeholder forum to facilitate ongoing dialogue, knowledge-sharing, and co-creation across the biotech ecosystem.
- Conduct comprehensive landscape analyses and case studies to identify barriers, best practices, and actionable recommendations for integrating AI and health data infrastructures under the EHDS.
- Pilot the federation and integration of health data infrastructures (genomics, cancer imaging, brain health) within the EHDS, demonstrating harmonised governance and technical interoperability.
- Select and implement real-world use cases that address patient care, diagnosis, and treatment efficiency, validating the feasibility and impact of AI-driven solutions.
- Translate project outputs into production-level services through AI Factories, ensuring that results move beyond pilots to real-world adoption.

Methodology:

SCAILED employs an integrated, iterative approach, combining stakeholder engagement, technical pilots, legal, regulatory and ethical guidance, and real-world validation. The project is structured around five building blocks:

- stakeholder forums
- a pathfinder and navigation system
- EHDS-enabled data access
- biotech AI applications, and
- scale-up via AI Factories.

Cross-work package alignment ensures that legal and regulatory compliance, technical progress, and real-world impact are embedded throughout the project lifecycle.

Expected Impact:

SCAILED will deliver operational, policy, and collaborative breakthroughs that position Europe as a global leader in AI-driven biotech innovation. In the short term, the project will enhance collaboration, regulatory clarity, and data interoperability. Medium-term impacts include broader adoption of project outputs and integration of additional infrastructures. Long-term, SCAILED will accelerate AI-driven innovation, improve health outcomes, and strengthen European leadership in digital health and AI. The project's outputs will serve as reference models for future initiatives, supporting the sustainable growth of a cohesive European health data and AI ecosystem.

#\$PRJ-SUM-PS\$# #@REL-EVA-RE@# #@PRJ-OBJ-PO@#

1. RELEVANCE

1.1 Background and general objectives

Background and general objectives

Describe the background and rationale of the project.

How is the project relevant to the scope of the call? How does the project address the general objectives of the call? What is the project's contribution to the priorities of the call?

Europe: Towards a global leading AI continent powered by EHDS, AI factories, and European research

infrastructures

The European Commission's newly published AI Continent Action Plan¹ aims to establish the European Union (EU) as a global leader in artificial intelligence particularly by advancing trustworthy and competitive AI systems and strengthening Europe's technological sovereignty and innovation capacity. The complementary Apply AI Strategy² is designed to enhance the competitiveness of strategic sectors. In the life sciences and health domain, the EC's Strategy for European Life Sciences³, a key part of the EC's Competitiveness Compass⁴, seeks to make Europe the world's top destination for life sciences by 2030. AI and digital solutions in the biotechnology (hereinafter biotech) sector is at the crossing point of both strategies. As AI innovation must be fuelled with high-quality data, the new European Data Union Strategy aims to boost the availability and accessibility of data for AI development via simplification of EU data rules, and to strengthen the EU's global position on international data flows⁵. The convergence of health data and AI is set to revolutionise the biotech sector, unlocking unprecedented opportunities to enhance human health, stimulate economic development, and drive technological innovation.

The regulation (EU) 2025/327 on the European Health Data Space (EHDS)⁶ offers a once-in-a-lifetime opportunity for establishing a secure and efficient framework for health data access and reuse across the Union. European data sharing and research infrastructures are foundational to enabling access to high-quality health data in different domains, particularly in the fields of genomic, cancer, and brain research. The integration and federation of these infrastructures with the EHDS, as well as collaboration with the EU's AI Factories, are vital for fostering innovation, accelerating scientific discoveries, and promoting knowledge exchange, ultimately closing the gaps to deliver real-life impact via digital and AI innovation.

However, Europe risks not being able to fulfil the ambitions above, without decisive action and dedicated support. As highlighted in the Draghi report on EU competitiveness⁷, Europe is still lagging behind in terms of innovation and its competitiveness suffers from a certain number of shortcomings, which need to be addressed for Europe to use its assets, skills and capacities and to fully reap the benefits of AI, including AI in health.

Challenges faced by biotech innovators

The biotech industry still faces significant challenges in developing and adopting AI and digital solutions, including navigating complex regulations (including the AI Act, GDPR, Medical Device Regulation (MDR), In Vitro Diagnostics Regulations (IVDR)) with high compliance demands. Industry still lacks clarity on where and how to use health data for AI training within the different regulatory framework.

Access to consistent, interoperable health data is still hampered by fragmented governance and data standards. On the one hand, the EHDS builds on the concept of national nodes. On the other hand, pan-European data infrastructures for specific areas like EUCAIM, GDI, UNCAN are establishing or already established cross-country data governance. The potential roles of such pan-European data infrastructures within the EHDS ecosystem needs to be analysed in the context of the new legal roles introduced by the EHDS Regulation, such as authorised participants (Art. 75 EHDS), health data holders, single trusted data holders, cross-border registries, networks of registries (Art. 76 EHDS), intermediation entities (Art. 50 (3) EHDS) or national HDABs.

Companies also lack in-house AI expertise and computational infrastructures, struggle to validate tools, encounter cultural resistance, and face uncertainty around reimbursement and scaling. These hurdles collectively slow innovation, heighten risks, and make it especially difficult for small and medium-sized enterprises to benefit from AI and EHDS opportunities. Integrating technical (data, compute) and legal aspects, along with offering comprehensive landscape overviews and clear pathways within the EHDS, is crucial for realising the full potential of the European strategies and action plans mentioned above.

General objectives

The EU4Health⁸ Work Programme 2025 defines the general objective and the scope of the call EU4H-2026-SANTE-

¹ The AI Continent Action Plan, <https://digital-strategy.ec.europa.eu/en/library/ai-continent-action-plan>

² <https://digital-strategy.ec.europa.eu/en/policies/apply-ai>

³ Strategy for European Life Sciences, https://research-and-innovation.ec.europa.eu/strategy/strategy-research-and-innovation/jobs-and-economy/strategy-european-life-sciences_en

⁴ An EU Compass to regain competitiveness and secure sustainable prosperity, https://ec.europa.eu/commission/presscorner/detail/en/ip_25_339

⁵ European Data Union Strategy, <https://digital-strategy.ec.europa.eu/en/policies/data-union>

⁶ Regulation (EU) 2025/327 of the European Parliament and of the Council of 11 February 2025 on the European Health Data Space and amending Directive 2011/24/EU and Regulation (EU) 2024/2847, <https://eur-lex.europa.eu/eli/reg/2025/327/oj>

⁷ The Draghi report on EU competitiveness, https://commission.europa.eu/topics/competitiveness/draghi-report_en

⁸ 2025 EU4Health Work Programme (ANNEX), https://health.ec.europa.eu/publications/2025-eu4health-work-programme_en

PJ-08 (DI-g-25-31) “health data for biotech innovation leveraging the European Health Data Space” as:

The general objective of this action is to accelerate the development and deployment of AI and digital solutions in the biotech sector driven by health data, ensuring its secure and responsible use through the EHDS and leveraging AI Factories to augment these efforts.

The call document emphasises that it is essential to support a coordinated approach on how to leverage health data and research infrastructures in the context of the EHDS framework and collaboration with the EU's AI Factories. To address the general objective, the SCAILED consortium aims to **bring together the required expertise and capacity from biotech sector and industry associations, developers of AI solutions, Member States' regulatory authorities in the area of secondary use of health data, research/data infrastructures, patient organisations, and AI Factories to scale up the AI and digital innovation in the biotech sector. By doing so, the SCAILED consortium enable the creation of next-generation biotech solutions**, via:

- enabling access to high-quality data including those from data infrastructures under secure data access frameworks defined in the EHDS regulation;
- providing regulatory clarity;
- leveraging AI expertise and computational infrastructure in the AI Factories to overcome deployment challenges and enable scale up.

Contribution to the priority of the call

SCAILED will contribute to the implementation of the EU4Health Programme's general objective of strengthening health systems (Article 3, point (d), of Regulation (EU) 2021/522) through the specific objectives defined in Article 4, points (a) and (f), of Regulation (EU) 2021/522. The project will foster innovation to improve health outcomes and will enhance the competitiveness of the European biotech sector beyond the project's duration. The project aims to deliver the following outcomes that directly contribute to the priority of the call:

- **Leveraging EHDS for secure, efficient health data access and reuse:** The call prioritises a coordinated approach to leverage health data and research infrastructures in the EHDS context. SCAILED proposes piloting federation/integration of the genomics, cancer imaging, brain health data infrastructures under the EHDS framework, that is, alongside and integrated with Health Data Access Body (HDAB) services, addressing data readiness, governance, and data and technical interoperability. This operationalises EHDS as a vehicle for innovation.
- **Fostering broad collaboration:** The proposed establishment of a multistakeholder platform and forum will ensure active engagement from biotech, research, healthcare, HDABs, regulatory, patient, and AI Factory communities, and aligning with the EHDS Stakeholder Forum. This provides added value by building on major EU initiatives, engaging over 17 Member States, and supporting cross-border impact and reuse.
- **Providing regulatory clarity:** with actionable legal and ethical guidance, recommendations, contractual clauses and a navigation system to help stakeholders comply with the AI Act, and medical device and other applicable regulations through leveraging the EHDS.
- **Scaling up innovation:** by translating project outputs into production-level services in AI Factories, ensuring that results move beyond pilots to real-world adoption.
- **Delivering real-world evidence** through use cases and case studies that demonstrate impact on patient care, diagnosis, and treatment efficiency.
- **Augmenting EHDS services** through pilot service development, such as synthetic data samples within metadata catalogues, data augmentation to support training and validation.

SCAILED is highly coherent with the call's objectives by leveraging opportunities provided by EHDS, institutionalising collaboration, removing regulatory barriers, demonstrating real-world impact, and achieving scale-up through AI Factories. This positions the project as a strong contributor to accelerate health data-driven biotech innovation under the EHDS framework.

1.2 Needs analysis

Needs analysis and specific objectives

Describe how the objectives of the project are based on a sound needs analysis in line with the specific objectives of the call. What issue/challenge/gap does the project aim to address?

The objectives should be clear, measurable, realistic and achievable within the duration of the project. For each objective, define appropriate indicators for measuring achievement (including a unit of measurement, baseline value and target value).

Despite the benefits and impacts AI and digital innovation would bring to the biotech sector, several challenges and needs must be addressed/met to realise the potential of EHDS to enable such innovations at scale:

- **Unmet biotech industry needs**
 - **Complex Regulatory and Policy Landscape:** The introduction of the European Health Data Space (EHDS) and AI regulations, especially related to the biotech sector, present both opportunities and challenges. Stakeholders require clarity and practical guidance to navigate compliance with different legislation and regulations related to cross-border data sharing and AI development. The implications of EHDS on drug and medical device approval regulatory procedures call for more guidance and clarification.
 - **Gaps between Theoretical Development and Real-World Application:** Coordinated development of the EHDS for digital and AI solutions requires rigorous validation through real-world use cases and case studies. These should address unknown challenges, bring requirements to light, and show measurable benefits in patient care, diagnosis, and treatment efficiency.
 - **Lack of clear AI pathways through the regulatory approvals process.**
- **Unmet development and deployment needs**
 - **Technical Barriers to AI Deployment:** Advanced AI methods such as federated learning, de-identification, and data synthesis requests, access to standardised protocols, scalable tools, and high-performance computational infrastructure for secure sharing and processing of sensitive data.
 - **Lack of Guidelines and Best Practise to design, develop, deploy AI applications for different stakeholders in the eco-system:** Adaptable and targeted guidelines, concrete recommendations, and best practises, reusable templates are essential for the efficiency of future innovation.
- **Unmet infrastructure needs**
 - **The role under the EHDS Regulation for existing and upcoming infrastructures (health data infrastructure, AI Factories) founded to facilitate data sharing and AI model training on a large scale in specific areas is still not demonstrated with real examples that can be replicated.** There are clear opportunities, but the infrastructures need guidance on what role to choose and to what extent the choice entails necessary adjustments in terms of service offerings and operational or governance constraints.
 - **Aligned clauses for infrastructure statutes and contracts with data providers are still needed to facilitate the integration into the EHDS.**
- **Unmet collaboration needs:**
 - **Lack of Coordinated Stakeholder Engagement and Collaboration:** Addressing the above challenges and achieving effective innovation requires the active involvement of diverse stakeholders. Platforms and fora are needed to facilitate collaboration, knowledge-sharing, and co-creation.

Specific Objectives and Indicators

Based on the needs analysis and the general objective defined in the call, the project will seek to address the following specific objectives (SOs).

Table 1 – Specific Objectives

SO1	Establish a Multistakeholder Platform and Stakeholder Forum (Main Responsibility: WP2) Create and support a collaborative platform that brings together key actors from the biotech industry, research institutions, healthcare providers, patient organisations, regulatory bodies, HDABs, and AI Factories. This platform will facilitate ongoing dialogue, knowledge-sharing, and co-creation, ensuring that diverse perspectives and sectoral needs are represented throughout the project lifecycle.
SO2	Analyse Challenges and Barriers via Comprehensive Landscape Analysis, Case Studies and Develop Guidelines and a Strategic Roadmap (Main Responsibility: WP3, WP4 and WP8) Including data quality and availability, regulatory frameworks, ethical and privacy concerns, intellectual property ownership, technical barriers, and scientific validation, and identify potential solutions or strategies to address these challenges, guiding the effective utilization of the EHDS. Conduct case studies of successful and explainable AI applications in the biotech sector and promote identified best practices, pathways and knowledge. Develop and publish recommendations and guidelines on data access and readiness, legal agreements, governance mechanisms, compliance with European regulations and technical solutions. Develop a strategic roadmap for the integration of AI solutions with the EHDS and other European infrastructures. The roadmap will provide actionable recommendations for policymakers, industry leaders, and researchers and with methodology and tools (path-finder) to dynamically monitor and evaluate progress of the implementation.
SO3	Pilot Health Data Infrastructure Integration (Main Responsibility: WP5) Test and demonstrate the federation and integration of health data infrastructures (genomic, cancer

	imaging, brain and other infrastructures) under the EHDS framework. Harmonise data governance, operations, enhance data readiness and data and technical interoperability, providing the guidelines for the integration of data infrastructures into EHDS.
SO4	Prioritise AI Applications via Real-World Use Cases (Main Responsibility: WP6) Select and implement use cases that address real-world challenges, demonstrate feasibility and impact, and align with EU strategies and sectoral needs. Use cases will cover diverse disease areas and aspects of patient care, diagnosis, and treatment efficiency, and will leverage consortium resources including AI Factory expertise.
SO5	Scale Up AI Innovation via AI Factories (Main Responsibility: WP7) Translate the project's guidelines, best practices, templates, and tools into production-level services offered by various AI Factories. This objective aims to ensure that the outputs of the project are not only theoretical or pilot-level but are actively adopted and operationalised by AI Factories across Europe. By doing so, the project will accelerate and scale up AI innovation in the biotech sector, enabling rapid deployment, broader accessibility, and sustainable impact.

The indicators for measuring achievement of SOs

The indicators, including baseline and target value, are summarised in Table 5 of Section 2.5 “Project management, quality assurance and monitoring and evaluation strategy”.

#@COM-PLE-CP@#

1.3 Complementarity with other actions and innovation — European added value

Complementarity with other actions and innovation

Explain how the project builds on the results of past activities carried out in the field and describe its innovative aspects. Explain how the activities are complementary to other activities carried out by other organisations.

Illustrate the European dimension of the activities: trans-national dimension of the project; impact/interest for a number of EU countries; possibility to use the results in other countries, potential to develop mutual trust/cross-border cooperation among EU countries, etc.

Which countries will benefit from the project (directly and indirectly)? Where will the activities take place?

Building on Past Activities and Innovation

This effort is backed by previous work of consortium partners such as: [HealthyCloud Strategic Agenda](#) (coordinated (COO) by partners IACS), participation in the TEHDAS and [TEHDAS2](#) projects, and coordinating the development and operation of Health Data Infrastructures in the field of Genomic ([GDI](#), COO by partner ELIXIR), Cancer ([EUCAIM](#), COO by partner IIS La Fe), and Brain research ([EBRAINS](#), represented by partner EBRAINS AISBL) as well as AI Factories ([BSC-AIF Spain](#), [1HealthAI Spain](#), [L-AIF Luxembourg](#)). In addition, national competent authorities acting as HDABs under the EHDS or leading the HDAB direct grants (GÖG Austria, HDA Belgium, CIPH Croatia, CDR Denmark, FINDATA Finland, HDH France, LNDS Luxembourg, Ministry of Health Spain) as well as SMEs and professional organisations representing the Biotech industry and patients will bring their view to the table as partners in the Consortium.

This project also builds on the results of other past and ongoing European initiatives in health data, AI, and biotechnology, leveraging lessons learned and best practices from those projects coordinated or participated by SCAILED partners: common data access application form, metadata standards and secure processing environment from [HealthData@EU Pilot](#) (COO: HDH), and TEHDAS2 (participated by multiple partners), data quality label from [QUANTUM](#) (COO: IACS), case studies from [SHAIPED](#) (COO: HDH), domain learnings from [UNCAN-Connect](#) (COO: UTARTU), [CANDLE](#) (COO: Health-RI) and [EU-CiP](#) (COO: CHARITE).

By integrating these experiences, the project introduces innovative approaches to health data federation, AI deployment, and stakeholder engagement, moving beyond the status quo to address persistent gaps in data access, quality, interoperability, and regulatory clarity.

The project's activities are designed to be complementary to those carried out by other organisations, including national health authorities, research infrastructures, and industry platforms. Rather than duplicating existing efforts, the project aims to harmonise and federate health data infrastructures, develop interoperable tools and protocols, and foster cross-sectoral collaboration. This approach ensures that outputs are reusable, scalable, and aligned with broader European strategies for digital health and AI innovation.

European Dimension and Trans-national Impact, added Value for Europe

With a consortium spanning 17 Member States (including 8 member states authorities or HDAB direct grantees) and active engagement with multiple pan-European health and research data infrastructures (GDI, EUCAIM, EBRAINS, ELIXIR, FEGA) multiple cross-member-state AI Factories, biotech industry and stakeholder networks (industry associations, patient organisations) research institutes, AI developers, and healthcare institutions, the project will

build a strong European dimension. It addresses the trans-national challenge of health data fragmentation and limited interoperability, which have been highlighted in recent evaluations of EU health policy. By developing common mechanisms for data access, legal agreements, and technical standards, the project facilitates cross-border and cross-infrastructure data sharing, mutual trust, and cooperation among EU countries.

The project's results will be applicable and beneficial across the Union, enabling Member States to overcome national, regional, and local barriers to health data reuse. Activities will take place in several EU countries, with direct and indirect benefits for healthcare systems, research communities, and industry partners. The project's outputs—such as the multistakeholder platform, strategic roadmap, and validated use cases—will be disseminated widely, supporting the adoption of best practices and fostering a cohesive European health data ecosystem. The results will be used in multiple participating AI Factories across several member states to develop new services to support AI innovation. This network can be further expanded to cover all designated AI Factories.

In addition, the project will implement the federation of three major health data infrastructures (genomic, cancer imaging, and brain) under the EHDS framework as examples for other data infrastructures to follow. It proposes high-value proof of concepts that could be game changers in EHDS, including use of HDAB-leveraged data through an AI factory, enabling federated learning across HDAB-provided Secure Processing Environments (SPEs), benchmarking synthetic data applications in the EHDS context, testing the collaboration across major cancer mission platforms (UNCAN, CANDLE, EU-CiP) on enabling AI-driven biotech innovation in oncology, as well as piloting AI implementation in brain diseases through the EBRAINS infrastructure.

SCAILED is expected to achieve synergies with other projects under the same call for proposals such as the cancer screening programmes (EU4H-2026-SANTE-PJ-01, EU4H-2026-SANTE-PJ-02, EU4H-2026-SANTE-PJ-03) and the flagship initiative leveraging AI and health data for cardiovascular health and related non-communicable diseases (EU4H-2026-SANTE-PJ-04). The outcomes of SCAILED may provide significant added value to these initiatives.

#§COM-PLE-CP§# #§PRJ-OBJ-PO§# #§REL-EVA-RE§# #@QUA-LIT-QL@# #@CON-MET-CM@#

2. QUALITY

2.1 Concept and methodology

Concept and methodology

Outline the approach and methodology behind the project. Explain why they are the most suitable for achieving the project's objectives.

To fully unlock the potential of AI and digital solutions in the biotech sector, this project proposes a coordinated approach to leverage health data from research infrastructures within the EHDS framework, as well as integration with AI Factories. The concept of the project is designed to ensure that the consortium will have the:

- **Right aim** – the work and building blocks planned are relevant and fully aligned with the general objectives of the call.
- **Right people** – the consortium is equipped with competent partners bringing in all required expertise and offering strong representation across stakeholder groups.
- **Right plan** – there is an ambitious and realistic work plan to implement the objectives of the call.

Right aim: as described in the section 1.1 and 1.2, the general objectives and the planned contribution to the priorities of the call will be cascaded into five specific objectives. To realize these specific objectives, the deliverables of the project will be organised in the following five building blocks (shown in Figure 1 below):

- **Stakeholder forum:** Building a sustainable community and platform to coordinate and synergize field expertise, best practices feeding into the development and translation of the AI-enabled solutions in the biotech sector and generate impacts in the health care system;
- **A pathfinder and navigation system:** Conducting landscape analysis and determining a strategic roadmap to leverage the EHDS regulation for innovation, developing recommendations and promoting best practices from case studies of successful AI applications for the different stakeholders. Establishing a support system (pathfinder) will assist various stakeholders in navigating these outcomes, identifying their optimal courses of action, and accessing targeted recommendations and best practices. This approach will facilitate the identification of concrete technical, operational, and regulatory solutions tailored to different stakeholders. It will also support stakeholders with methods for monitoring and evaluation of the implementation of the strategic roadmap by different stakeholders during and beyond the project;
- **Data access through EHDS mechanisms:** Demonstrating how EHDS creates data access opportunities for integrated health data infrastructures via Healthdata@EU and/or national HDABs, equipped with strengthened services developed in this project. Facilitating innovation and compliance with a complex legal framework along the lifecycle of biotech industry products, by optimizing EHDS-offered opportunities and

integrating existing health data infrastructures and solutions under the EHDS framework;

- **Biotech AI applications and use cases:** Validating the development, demonstrating impact, gathering learnings via real-life development workflow targeting needs from biotech sectors in explainable AI application, using genomics, cancer, brain, metabolomics studies, proteomics databases, and clinical trial networks to reach better patient care;
- **Scale Up AI Innovation via AI Factories:** powered by AI factories to leverage high-performance and secure compute as well as interoperable services (data readiness, compliance support), to gather and optimize best available AI applications to meet the needs of biotech innovation.



Figure 1 - The five building blocks to implement the ambition of the general objectives.

Right people: from the expertise point of view, this consortium has involved partners representing a wide stakeholder community (industry and industry associations, Member States/local authorities, pan-European health and research data infrastructures, AI Factories, patient organisations, research institutes, AI developers, healthcare providers) with expertise and/or track-records in leading related past and ongoing initiatives needed to effectively implement the tasks towards reaching the objectives of the call (see details in Section 1.3, part “European Dimension and Transnational Impact” and Section 2.2 “Consortium set-up”). Besides the expertise represented in the consortium, the project will engage external experts through stakeholder forums (WP2) and advisory boards (see section 2.4 “Consortium management and decision-making”) to incorporate diverse perspectives. This approach ensures comprehensive community input and maximizes the project's impact on enabling AI and digital innovation in the biotech sector through responsible data use under EHDS.

Right plan: the tasks are then organized according to the five building blocks in nine Work Packages (WPs) to implement the actions required to meet the goal of the building blocks (see Figure 2). The WPs 2, 3, 4, 5, 7 (building block WPs), are implementing tasks related to a specific building block, while WP 8 will be a transversal WP linking to all the building block WPs for regulatory, legal, and ethical topics. Work package 1 and 9 are the two WPs supporting the building block WPs and transversal WP.

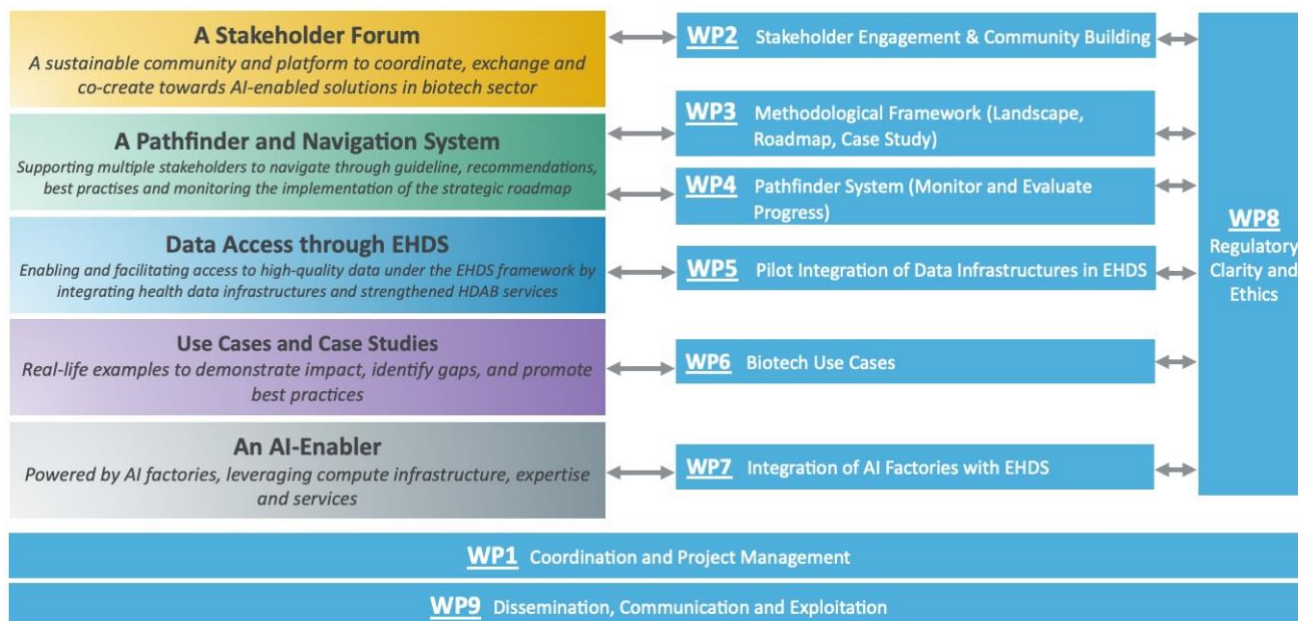


Figure 2 - Work packages and mapping to the building blocks.

Work Packages

WP1 – Coordination and Project Management: will ensure the achievement of the project goals and will carry out daily management of the project. This WP monitors progress against the work plan, ensuring the timely submission of reports, deliverables, milestones, and efficient use of the allocated budget. It coordinates the project's governance processes and the strategic development of the project and serves as the main interface with the EU Commission services. WP1 also guarantees that all ethical and legal issues related to the project management and their impact are being properly addressed and oversees risk identification and mitigation throughout the project.

WP2 – Stakeholder Engagement & Community Building: will establish an operational multistakeholder environment supporting broad stakeholder buy-in, knowledge transfer, fostering networking, and facilitating collaboration across biotech, healthcare, and regulatory sectors. It includes management of events, digital platforms, and communication strategies (together with WP9). It will also support other project work packages in stakeholder engagement for specific topics raised by those work packages. The forum serves as a community-supported structure to enable cross-sector communication and collaboration, within and beyond the lifespan of the project. This forum is ready to be linked with the key upcoming EU fora, including the EHDS Stakeholder Forum introduced by the EHDS Regulation (Art.93).

WP3 – Methodological Framework (Landscape, Roadmap, Case Study): aims to establish a robust methodological framework by analysing the current landscape (data and infrastructures, challenges and opportunities, potential solutions, case studies) to understand health data use for AI in biotech and to define a strategic roadmap describing targeted pathways for different types of stakeholders to scale up the development and deployment of trustworthy AI solutions across the EU biotech ecosystem. It also integrates technical, legal, ethical, and governance considerations from other WPs, drawing on continuous stakeholder feedback and collaboration with other work packages on specific topics for both the landscape analysis and roadmap development.

WP4 – Pathfinder System (Monitor and Evaluate Progress): will develop and pilot methodologies and tools to monitor and evaluate progress towards strategic roadmap objectives, aiming to provide insights for continuous improvement. It includes identifying stakeholder perspectives on the strategic roadmap with tailored solutions to stakeholder-specific innovation journeys (see Figure 3). WP4 collaborates closely with other WPs, especially WP3 on monitoring guidelines, ensuring comprehensive evaluation and alignment with the strategic roadmap, as well as jointly deliver the pathfinder and navigation system.

WP5 – Pilot Integration of Data Infrastructures in EHDS: Through the involvement of several HDABs/HDAB grantees or equivalent competent authorities (AT, BE, FR, LU) and diverse data infrastructures (GDI, FEGA, EUCAIM, EBRAINS, UNCAN(national cancer data nodes and central platform)) with different set-ups on Governance, Operations, Data provisioning and Processing and AI, and the piloting of key functionalities in these areas (e.g. publication of Data Infrastructures Data Sets on EHDS central services), we aim to identify the limitations and inefficiencies that could hamper the access to data and computational capacity available in these infrastructures and develop a guidance that could be used to ensure alignment and interoperability across all EHDS players involved. This guidance can then be used to streamline interactions across these data infrastructures and with the EHDS, as

well as to guide the integration of new data infrastructures into EHDS as they start to take shape.

WP6 – Biotech Use Cases: will deliver concrete AI applications in biotech sector and test crucial AI-supported or AI-supporting tools to augment EHDS and AI services. Four use cases will develop, test and deploy AI tools and services for biotech purposes and other four use cases will develop and pilot supporting tools and solutions that will accelerate and augment innovation in the biotech sector. Additional use cases will be onboarded via an open call. This WP supports the work of other work packages by providing real-world validation data and testing the practical application of project outputs, ensuring alignment with EHDS standards. WP6 manages project coherence through continuous cross-WP coordination, systematically providing 'lessons learnt' and operational feedback to WPs 5, 7, and 8. Ultimately, the WP will support the pathway for uptake and scale-up of solutions within the biotech sector (in collaboration with WP2, WP8 and WP9). Given the overall timeline of the project, WP6 use cases will start immediately to overcome some pre-identified issues in the ecosystem and to leverage some solutions rarely deployed so far in the EHDS ecosystem (e.g. federated learning, synthetic data, etc.).

WP7 – Integration of AI Factories with EHDS: WP7 focuses on establishing the technical and operational groundwork essential for AI Factories to provide high-performance, secure and trustworthy computational infrastructures and production level services to support AI development while meeting the EHDS requirements. It begins with a comprehensive assessment of Europe's current Secure Data Processing capabilities, leading to the creation of a practical toolkit that outlines standards for AI model usage, onboarding, and interoperability. WP7 further classifies secure computational infrastructures, defines governance and resource requirements, and sets procedures for deploying AI in compliance with regulatory frameworks. The frameworks developed are validated through real-world use-cases, with outcomes directly informing the roadmap for a federated network of AI factories, ultimately providing an example of production level implementation towards interoperable and secure services across AI Factories in Europe for scalable AI solutions development in health and biotechnology.

WP8 – Regulatory Clarity and Ethics: will outline the legal ways in which the EHDS contributes to promoting innovation and regulatory compliance by simplifying data access. In a multi-pronged approach, we will first lay the legal groundwork for integrating existing data infrastructures into the EHDS to contribute to the realisation of the one-stop concept of the EHDS. We will then analyse the regulatory challenges facing the industry in an increasingly complex legal framework. Subsequently, we will demonstrate how the EHDS can simplify regulatory processes in the various phases of development and commercialisation. The legal modalities of how AI factories can be included as SPE service providers in the process will be explored and supported by contractual clauses. We also address challenges posed by the EHDS and the use of AI, such as IP-related concerns from the ethical perspective. WP8 will build on the results of many projects, especially TEHDAS2, SHAIPEd, FORTIFY and main input from the industry partners. Finally, we will provide ethical guidance for the use cases and the coordinator.

WP9 – Dissemination, Communication and Exploitation: ensure the **visibility, uptake and long-term sustainability** of the project's results. It establishes a coherent approach to communication, dissemination and exploitation that supports engagement across the biotechnology, health-data and responsible artificial-intelligence communities. The WP covers the full lifecycle of outreach and impact management: strategic planning; continuous communication of progress and findings; alignment with European initiatives; management of innovation, knowledge and intellectual-property aspects; and the preparation of actionable policy recommendations for European and national authorities. Through an integrated set of activities, WP9 strengthens awareness and collaboration among stakeholders, supports transparency around the project's methodologies and outcomes, and creates the conditions for **long-term adoption, replicability and policy relevance** of the project's outputs within the evolving framework of the European Health Data Space.

Enabling cross-WP alignment and collaboration

A crucial success factor for the project is to ensure cross-WP alignment and collaborations and to avoid the WPs working in silos. **From the content perspective**, we will enable the cross-WP alignment and collaborations by following the same journeys as shared guiding threads to systematically identify all issues and gaps along the “end-to-end” lifecycle of data and AI innovation, the Work Packages will follow these journeys, when applicable, in designing and delivering targeted endpoints:

- **An AI innovation journey:** walking with innovators from development, translation, to implementation of innovation, highlighting key milestones, obstacles, stakeholders and solutions;
- **A Data journey:** a lifecycle of data from the data being collected/generated, through its processing, use/reuse and at the end archival.

Though these are two different journeys, they are nevertheless in large part overlapping with each other (see Figure 3). Those “crossing points” for both journeys are the steps where data requirements and innovation needs should be considered together. The AI innovation journey is the main journey for the projects to follow. However, some issues linked to data would need to be solved up-stream, therefore the data journey is still relevant to some of the work

packages.

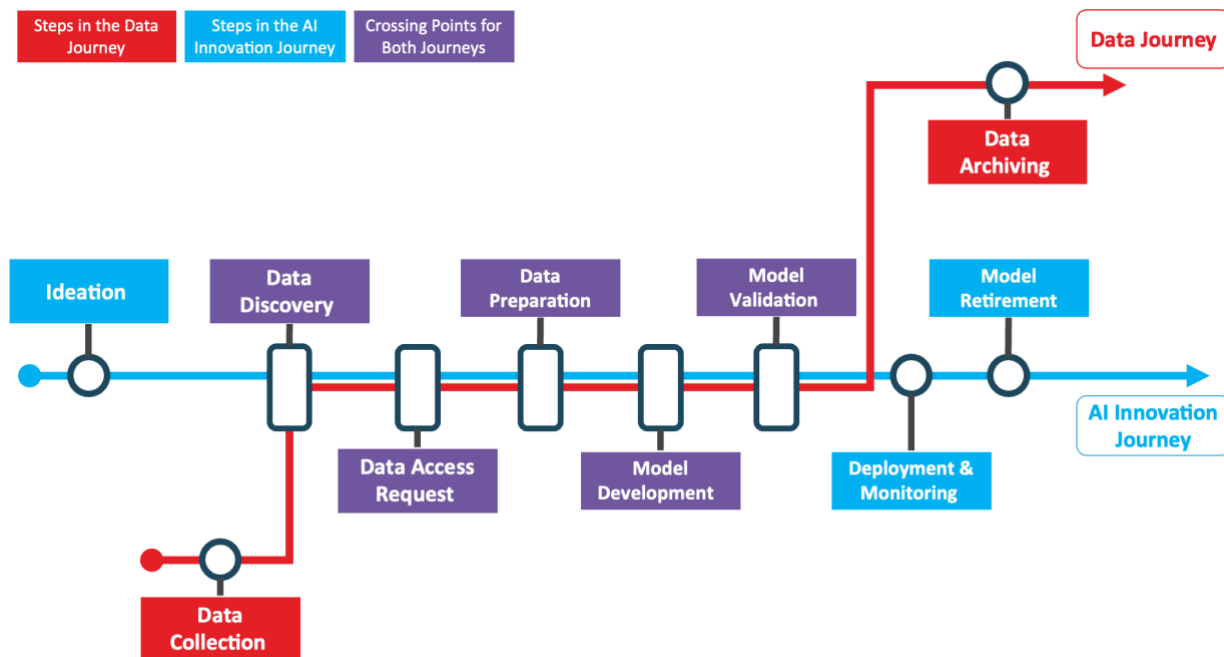


Figure 3 - The “Data Journey” and “AI Innovation Journey” serving as the guiding threads for the project to systematically identify gaps and needs.

From the timeline perspective, we will enable the cross-WP alignment by following a synchronised iteration of: **Analyse – Implement – Test – Scale** (see Figure 4), with a feedback loop among the first three steps (Analyse, Implement and Test) before moving to the Scale stage. WP1 will support those WPs by organising cross WP workshops.

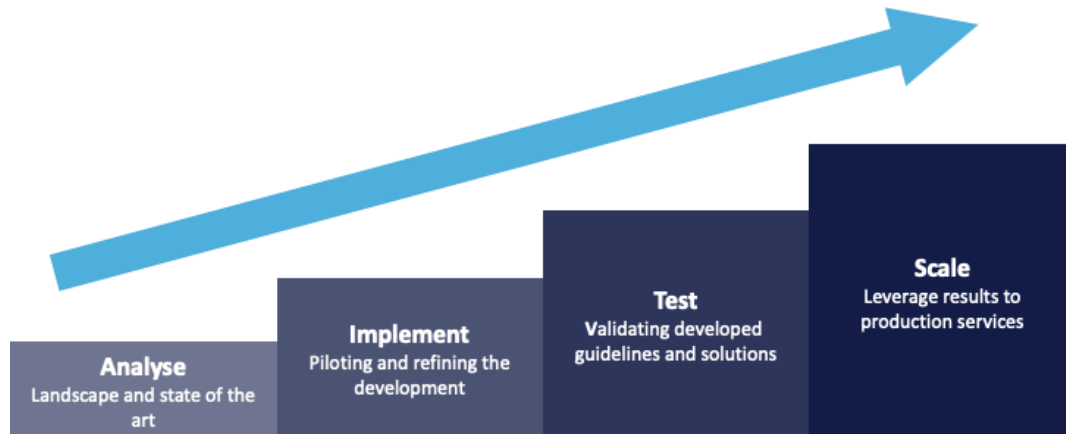


Figure 4 - The synchronised iteration to align the tasks across work packages and reach the expected maturity of the outcome.

- **Analyse:** Systematically assess the current landscape across all activity areas needed by different WPs, laying the foundation for effective implementation, testing, and scaling.
- **Implement:** Develop the expected outcomes from each WPs such as the strategic roadmap and its monitoring methods, piloting the roles for health data infrastructures under the EHDS, secure processing environment in AI Factory in line with EHDS requirements, recommendations for regulatory clarity, best practises from case studies, etc.
- **Test:** Test and validate developed guidelines and solutions in real-world settings (use cases and stakeholder review and feedback).
- **Scale:** Translate validated outcomes/results (guidelines, best practices, templates, and tools) to sustainable services in data infrastructures, HDABs and AI Factories, providing easy access to those materials via “pathfinder and navigation system” (an interactive knowledge management platform), disseminate the outcomes to the wider communities.

Suitability of the method

This integrated approach ensures that:

- Critical challenges identified are systematically addressed at each stage of the project from analysis to implementation, testing and scaling.
- Stakeholder engagement, regulatory compliance, technical progress, and real-world impact are embedded throughout the project lifecycle.
- All development is centred around supporting AI and digital innovation and gaps along the innovation and data journeys are visible and not missed.
- Strategic vision is translated into tangible outcomes and demonstrable impact, with theoretical insights informing practical solutions and real-world validation.
- Outputs are not only theoretical or piloted but also validated, scaled up and sustained through adoption by AI Factories.

By structuring the work using this approach, the method ensures that objectives are achieved efficiently and sustainably.

Use cases identified and to be identified

As described in the summary of WP6 above, the SCAILED project will respond to the call requirement to “**Identify and prioritise the most promising digital / AI applications for the biotechnology sector**” by selecting and implementing concrete AI applications in biotech sector and test crucial AI-supported or AI-supporting tools to augment EHDS and AI services. For identified use cases during the proposal preparation phase, several use case candidates have been proposed by biotech industry partners and data infrastructure partners. After evaluating the candidates, 8 use cases have been selected for pilot implementation (see below the list of selected use cases).

Recognizing the rapid advancements in AI applications, the consortium acknowledges the need to stay current with emerging technologies and use cases. To ensure our project remains aligned with these fast-paced developments, we plan to initiate an **open call for new use cases** during project implementation. This approach is supported by a dedicated budget and task allocation in WP6 to manage the open call process. Upon selection, the new use case teams will receive the necessary resources.

The selection criteria, to be formulated in WP6 with contributions from other WPs, will focus on factors such as potential health outcomes, feasibility, technological maturity, and alignment with sectoral needs, as highlighted in the call text as well as their alignment with the initial work of the project's technical WPs. The organisations proposing new use cases via the open call should also demonstrate sufficient financial capacity to execute the use case, with the support from the open call. With the consortium's three AI Factories and ongoing development in WP7, these new use cases will benefit from enhanced resources and services. This support will facilitate rapid prototyping, testing of AI-driven biotech solutions, and expedite the transition of innovations from concept to healthcare implementation.

Below is the list of identified use cases to start.

UC1 – Generating synthetic sample for early access through EHDS metadata catalogue

This UC establishes a certified DataGen methodology within the HDH Secure Processing Environment (SPE), leveraging AI models from ALIA SANTE, to generate privacy-preserving synthetic samples from diverse data sources (HDH, Sciensano, THL). The generated samples and their fidelity metrics are then embedded directly into the HealthDCAT-AP catalogue structure, validating an EHDS-aligned metadata pathway. The primary action involves executing feasibility studies using these samples to empirically prove the reduction in time-to-data-readiness for biotech developers. This validates an efficient, compliant pre-access pathway for secondary use of health data.

This UC's key added value is demonstrating a practical, compliant pre-access pathway that eliminates initial project delays, directly validating the utility of WP5 metadata standards and providing relevant WP8 regulatory clarification by achieving DPO validation of the synthetic data generation methodology.

UC2 - Enabling the secondary use of data in HPC environment according to EHDS specifications

This UC documents the methodology and requirements for securely integrating High-Performance Computing (HPC) resources (LUMI AI Factory supercomputer) with EHDS-compliant Secure Processing Environments (SPE). The use case will demonstrate the requirements with national SPE by testing the secure transfer and processing of large, sensitive data, specifically synthetic genomic data sourced from the GDI project (THL), within this integrated system. The execution validates that the system maintains data protection and EHDS compliance during complex, high-capacity computation jobs. This provides a proven, transferable operational model and documentation for other SPEs aiming to scale up their capacity.

This UC's key added value is demonstrating the feasibility of using HPC for large genomic and AI datasets within SPEs, thereby validating the technical capacity of the WP7 AI Factory and clarifying relevant WP8 regulatory

interpretations for data protection during high-capacity computing.

UC3 – Exploring cross-border Federated Learning for secondary data use

This UC establishes the necessary security and governance alignment between two distinct national Secure Processing Environments (HDH and Finnish SPEs) to enable algorithm transfer. The core action involves conducting secure, iterative Federated Learning (FL) training of an AI model on e.g. imaging datasets across these cross-border environments, utilizing a biotech-provided algorithm. The process includes reviewing the algorithm updates to ensure continuous compliance with security and GDPR standards. The goal is to compare FL to distributed learning and produce a cross-border FL protocol.

This UC's key added value is demonstrating the feasibility of scaling AI training by eliminating data silos, directly validating WP5 interoperability protocols and WP7 AI Factory capacity while providing relevant WP8 regulatory clarification for FL.

UC4 – GenAI-Powered Virtual Patients for Optimising Early-Phase Clinical Trials in Glioblastoma

This UC develops an advanced computational framework leveraging GenAI and machine-learning algorithms (led by VHIO) trained on complex MRI imaging biomarkers to generate diverse Virtual Patient trajectories. These synthetic trajectories, mirroring an upcoming Phase I trial of Batea Oncology's GlioHook device, will be generated using real-world data accessed via the EUCAIM federated network. The primary action is comparing the predicted outcomes (survival, relapse) against the real clinical data generated during the 15-patient Phase I study. This process aims to refine trial endpoints, improve planning for future clinical phases, and validate a methodology for potentially reducing control arm requirements.

This UC's key added value is demonstrating a non-invasive methodology for optimizing early clinical trials and reducing control arm size, directly validating the utility of EUCAIM's federated infrastructure and accelerating the clinical translation of AI solutions for the biotech sector.

UC5 – AI – driven prevention, diagnostic and management for delirium in ICU patients

This UC focuses on developing and validating a multimodal AI algorithm, integrating clinical, lab, and other relevant data for early delirium detection in critical care. A Federated Infrastructure for data standardization (FHIR IPS & SNOMED CT) and secure cross-border data exchange, provided by Clinomic, will be established and validated across at least four European ICU centres to ensure secure, EHDS-compliant data access. The goal is to deploy a prototype Clinical Decision Support System (CDSS) that provides a definitive diagnosis and treatment plan in under 12 hours. This process includes developing technical recommendations for the harmonized querying of unstructured data into the EHDS data catalogue.

This UC's key added value is demonstrating the real-time application of federated AI for critical care diagnostics, providing essential WP5 data standardization input and validating the capacity of WP7 AI tools to significantly reduce patient morbidity and ICU costs.

UC6 - Polygenic Risk Scores

This UC federates and harmonizes cross-border multimodal datasets (genomic, clinical, population-level) from participating Member States (BE, ES) via the EHDS and Genomic Data Infrastructure (GDI) frameworks. The core activity involves developing and validating Polygenic Risk Score (PRS) models using explainable AI, assessing their transferability across diverse populations. Utilizing the LUMI AI-Factory infrastructure, the project will prototype an AI-powered personalized prevention/screening matching platform (developed by Curewiki). This platform will match individuals to personalized programs based on their integrated genomic and socio-demographic profiles.

This UC's key added value is demonstrating the pathway from diverse multimodal data to personalized prevention tools, validating the GDI/EHDS data access framework (WP5) and leveraging the high-capacity computing services of the WP7 AI Factory to enhance AI model interpretability and transferability.

UC7 - Generating high-fidelity synthetic medical images for AI development and evaluation

This UC focuses on developing and validating a methodology (Inria, INSA LYON) for generating high-fidelity synthetic medical images (cancerology, cardiology) using advanced biophysical model simulation. The methodology ensures physiological realism, particularly for images concerning rare conditions and fragile populations. A multimodal extension is included to generate associated clinical data (ECGs, reports), creating comprehensive synthetic datasets. These open-source datasets will be used by industrial partners (Philips, Siemens, BrightHeart, echOpen) to train and evaluate AI products for improved robustness and generalization across diverse imaging quality and pathologies.

This UC's key added value is solving the 'Data Scarcity for Equitable AI' gap by delivering an EHDS-compliant

synthetic image generation methodology, providing technical input for WP5 standardization of imaging formats and validating the utility of WP7 AI tools for industrial benchmark development.

UC8 - Generating synthetic control arms to accelerate rare disease trials

This UC develops and applies multimodal AI methods to generate Synthetic Control Arms (SCAs) for evaluating treatments in rare diseases like Myotonic Dystrophy Type 1. The execution leverages EHDS-aligned infrastructure, specifically the linked SNDS (French National Health Data System) and BNDMR (National Rare Diseases Database), containing extensive longitudinal data. Partners (SANOFI, INRIA, HDH) will develop rigorous methodological guidance and predictive models to construct SCAs, aligning their aggregate characteristics with those of the treated arm. The goal is to provide evidence that can inform regulatory authorities and accelerate clinical research where conventional control groups are infeasible.

This UC's key added value is providing a robust, AI-based methodology for addressing rare disease trial constraints, demonstrating the high impact of EHDS secondary use data to accelerate clinical translation and regulatory evidence generation.

#§CON-MET-CM§# #@CON-SOR-CS@#

2.2 Consortium set-up

Consortium cooperation and division of roles (if applicable)

Describe the participants (Beneficiaries, Affiliated Entities and Associated Partners, if any) and explain how they will work together to implement the project. How will they bring together the necessary expertise? How will they complement each other?

In what way does each of the participants contribute to the project? Show that each has a valid role and adequate resources to fulfil that role.

Note: When building your consortium you should think of organisations that can help you reach objectives and solve problems.

The SCAILED consortium brings together 49 organizations from 17+ Member States, including 8 Member State authorities or HDAB direct grantees, multiple pan-European health data infrastructures (Genomics, Cancer, Brain), 3 AI Factories, 7 biotech industry and industry associations, 2 patient organisations, multiple research institutes, AI developers, and healthcare organisations. Besides the coverage of entities required by the call text as describe above, SCAILED also involves partners who play pivotal roles (coordinators and WP leaders) in other synergistic initiatives such as 1+MG (GDI, B1MGplus), EUCAIM, EBRAINS, SHAIPEd, THEDAS2, QUANTUM, UNCAN-Connect, CANDLE, EOSC, and HealthyCloud. Several key partners are involved in multiple of those initiatives providing valuable cross-initiative perspectives to the project (see Table 3).

The SCAILED consortium benefits from this diverse network of partners, providing the necessary expertise to fulfil the core objectives outlined in the call. This wide representation across stakeholder groups ensures that the project incorporates perspectives from all relevant parties.

SCAILED is coordinated by the Luxembourg National Data Service (LNDS), which coordinates the Luxembourg HDAB direct grant, leads the development of the future 1+MG EDIC, and leads data service innovation within the Luxembourg AI Factory. LNDS is also a member of ELIXIR (Luxembourg Node).

The project consists of 9 WPs, each led and co-led by designated organisations chosen for their expertise, experience in previous initiatives, roles within the EHDS ecosystem, and commitment to representing stakeholder diversity. Within each Work Package, contributing partners define the tasks under the guidance of WP leads and co-leads. Task leads and co-leads are selected collaboratively by the WP leads/co-leads, in consultation with the project coordinator and other WP contributors.

This structure ensures a balanced distribution of expertise, responsibilities, and budget, promoting effective representation and accountability throughout the consortium.

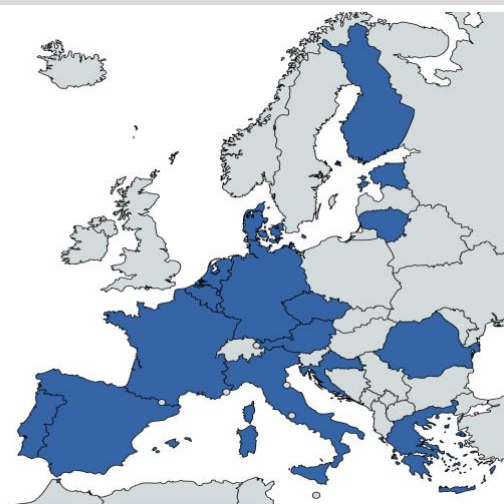


Table 2 - Types of organisations in the consortium, their roles and value

Type of organisations and key perspectives	Partners	Role and added value
Biotech industry	CLINOMIC (DE), BATEA (ES), BAHIA (ES), ALIA(FR), SANOFI (FR)	Bring hands-on expertise in AI-enabled biotech solutions and real-world development pipelines; co-lead and validate use cases; provide technical and regulatory insights from industry perspective.

Biotech industry associations	ASEBIO (ES), DE (BE)	Ensure industry-wide alignment and outreach; mobilise biotech ecosystems across Member States; support dissemination and adoption of project outputs; ASEBIO acts as Board Member of EUROPABIO.
Health Data Access Bodies, Competent authorities (HDAB direct grantees), National and regional authorities	GÖG (AT), HDA (BE), SCIENSANO (BE), CDR (DK), MoH-ES (ES), THL-FINDATA (FI), HDH (FR), CIPH (HR), LNDS (LU)	Core implementers of EHDS governance; contribute expertise on secure data access, compliance, SPE operations, and national implementation pathways. Provide policy alignment, regulatory insight, and ensure institutional embedment of recommendations; key actors for sustainability and uptake across Member States
Health data infrastructures coordinators	ELIXIR (IO), IISLAFE (ES), HRI (NL), CRG (ES), BIODATA (PT), EBRAINS (BE)	Provide mature research infrastructures, data assets, technical capacity and operational models for integrating infrastructures into the EHDS, including EUCAIM, GDI, EBRAINS, FEGA and the ELIXIR node.
Healthcare professionals	MCN (RO)	Provide frontline clinical expertise; validate use cases in real hospital settings; ensure clinical relevance, feasibility and safe integration into routine care workflows.
AI Factories or AI Antenna	LNDS (LU) – Luxembourg AI Factory (L-AIF) BSC (ES) – BSC AI Factory CESGA (ES), USC (ES) – 1HealthAi Factory SCIENSANO (BE) - BE-AIFA (AI Antenna)	AI Factories bring compute, secure processing, model development workflows and AI governance capabilities; key to scaling outputs into sustainable AI services aligned with EHDS. AI Antenna's build an AI ecosystem to support SMEs, startups, public bodies and researchers with easier access to AI-ready compute power and quality data.
Patient Organisations	POLA (LT), ELLOK (EL)	Ensure patient-centric perspectives, ethical acceptability, and co-creation with end-users; contribute to evaluation of acceptability and societal impact.
Science and Academia	IACS (ES), BSC (ES), CHARITE (DE), TMF (DE), CERTH (EL), USC (ES), CESGA (ES), SCIENSANO (BE), CNR (IT), MU(CZ), VHIO (ES), UNILU (LU), UTARTU (EE), INRIA (FR) UNIC (FR) CHU-B (FR), INSA LYON (FR)	Provide scientific excellence, methodological rigour, evaluation frameworks, regulatory analysis, and contribute to case studies, pilots and roadmap design.

Table 3 - Related EU initiatives and representation from the consortium

Related initiatives	Partners involved and their role (COO: Coordinator)
HDABs community of practice	GÖG, HDA, SCIENSANO, HRI, CDR, MoH-ES, THL-FINDATA, HDH, CIPH, LNDS
HealthyCloud	IACS (COO), and multiple partners
TEHDAS2	Multiple partners
Genomics Data Infrastructure (GDI)	ELIXIR (COO), and multiple partners
B1MGplus	ELIXIR (COO), and multiple partners
EUCAIM	IISLaFe (COO), and multiple partners
EBRAINS	EBRAINS (COO), and multiple partners
AI Factories	Multiple partners
HealthData@EU Pilot1	HDH (COO), and multiple partners
QUANTUM	IACS (COO), and multiple partners

SHAIPED	HDH (COO), and multiple partners
UNCAN-Connect	UTARTU (COO), and multiple partners
CANDLE	HRI (COO), and multiple partners
EU-CiP	CHARITE (COO), and multiple partners
EOSC-ENTRUST	ELIXIR (COO), and multiple partners

2.3 Project teams, staff and experts

Project teams and staff

Describe the project teams and how they will work together to implement the project.

List the staff included in the project budget (budget category A) by function/profile (e.g. project manager, senior expert/, junior expert, trainers/teachers, technical personnel, administrative personnel etc. — use the same profiles as in the detailed budget table, if any (n/a for prefixed Lump Sum Grants)) and describe briefly their tasks. Provide CVs of all key actors (if required).

Table 4 - List of project teams and staff (full list available in annex).

Name and function	Organisation	Role/tasks/professional profile and expertise
Wei Gu, Head of Service	LNDS	Project Coordinator (PC). Dr Wei Gu brings extensive experience in EU health data initiatives (GDI), AI Factory (L-AIF), research infrastructures (ELIXIR LU), EHDS (HDAB-LU, TEHDAS2) and managing multidisciplinary teams as well as service innovation. Expertise: health data governance, interoperability, AI-related infrastructures, and complex multi-stakeholder coordination.
Ruth Muleiro Alonso, Project Manager	LNDS	Quality and Risk Manager (QRM), >10 years of experience in EU project management: EU4H, HE, DE, IHI and PCP. Expertise: quality control and compliance with EC requirements. Support in consortium coordination, reporting, budget and effort monitoring, and ensuring methodological consistency.
Davit Chokoshvili, Senior ELSI Specialist	LNDS	Ethics & Data Protection Manager. Dr Davit Chokoshvili will help identify potential ethical and GDPR-related risks and provide guidance on safeguards, compliance measures and mitigation strategies. Expertise in ELSI, GDPR, responsible data-sharing EU health-data initiatives, including GDI, B1MGplus and EPND.
Valentine Facque	HDH	Lead WP6 , providing strategic oversight for all UCs execution. Dr. Facque is European and International Affairs Project Manager. PhD in Cognitive Neuroscience, expertise in scientific rigor and innovation design in digital health.
Maari Parkkinen	FINDATA	Contributor to WP6, Use Cases 2 and 3, expertise in developing infrastructures that enable the secure and effective secondary use of health data Development Manager at the Finnish Social and Health Data Permit Authority (Findata), leads the national FinHITS project aimed at strengthening Finland's health data ICT infrastructure for secondary use. .
Persephone Doupi	THL	Contributing to WP6, WP8 and WP9. Senior Medical Officer, Data Governance Team Leader. MD (Public health), PhD (Medical informatics), leading THL's international activities related to the EHDS, incl. TEHDAS I/II JAs, Xt-EHR JA, HealthData@EU-pilot, QUANTUM, SHAIPED (WP4 Lead) and EOSC-ENTRUST projects. Vice President of the European Federation for Medical Informatics (EFMI).
Juan González-García	IACS	Lead WP2. PhD in Computer Architecture. Head of Health Data Strategy and Biocomputing Unit Manager. Juan will coordinate the overall activities of WP2, liaison with the rest of WPs and the narratives construction, and engagement with external stakeholders, with special interest in the policy alignment.
Alberto Montolio-Marco	IACS	PhD in Mechanical Engineering. Biocomputing Unit lead of BIGAN operations, the Aragón platform for secondary use of health data. Alberto will lead the WP2 activities devoted to exchange with technological partners involved in the HealthData@EU infrastructural development.
Christian Fynbo Christiansen	CDR	Contributors use case 1 in WP6. Professor, Daily manager of Centre for Clinical and Genomic Data (CONNECT) and Consultant in Clinical Epidemiology, Department of Clinical Epidemiology, Aarhus University Hospital. Building on experience from e.g. EHDS Pilot, TEHDAS2, SHAIPED.
Nienke Schutte	SCIENSANO	Lead WP3 and Task 5.2. Innovation in Health Information Systems unit lead at Sciensano, aiming to advance innovative, privacy-preserving health information

		systems and secure data infrastructures. Coordinating Sciensano's active contribution to Joint Action TEHDAS2, SHAIPEd, PHIRI, QUANTUM, xSHARE
Beatriz Barros	SCIENSANO	Lead WP3 and Task 5.2. Scientist at Innovation in Health Information Systems unit at Sciensano. Expert in health policy analysis and advancing the implementation of the EHDS through TEHDAS2 JA (WP lead) and SHAIPEd
Lifang Liu	HRI	Contributing to WP5, co-leading Task 5.3 on synergizing infrastructure solutions along the EHDS blueprint digital capabilities, crystalize cancer-specific infrastructure through the CANDLE project and UNCAN.eu platform. Medical doctor by training, PhD in epidemiology and public health in oncology. Coordinator CANDLE project. Head of the International Team of Health-RI and Dutch National Cancer Data Node coordinator.
Salvador Capella-Gutierrez	BSC	Lead WP7. Director of Technologies for Biomedical Research Laboratory at the BSC, and the Coordination team lead of the Spanish National Bioinformatics Institute (INB/ELIXIR-ES). Expertise in developing long-term infrastructures, innovative techniques to use research data across different centralized and federated infrastructures at local, regional, national and European levels.
Juan Arenas Marquez	ELIXIR	Lead WP5. Leading the deployment of European Genomic Data Infrastructure, ensuring alignment with HealthData@EU. His role in shaping and implementing the 1+MG initiative, including contributions to European GDI and its alignment with the EHDS, shows deep understanding of interoperability, governance, regulatory compliance (GDPR), technical integration at European scale.
Luis Martí Bonmatí	IISLAFE	Director of the Clinical Area of Medical Imaging at La Fe Hospital and Director of the Biomedical Imaging Research Group, he will contribute as senior clinical and imaging expert. With over 40 years of experience in radiology, he serves as Scientific Director of EUCAIM and has led major European initiatives on cancer imaging, data standardisation, and AI validation.
Eriona Dashja	EMPIRICA	Senior Research Consultant, contributing to WP3 and WP5 with her ongoing work with two grant relevant EC studies: analysis of the regulatory framework for biotechnology and biomanufacturing in the EU (DG GROW and SANTE), and landscape analysis of the biotechnology sector in the Union with a view to foster its competitiveness and innovation (DG CONNECT).
Ion Arocena	ASEBIO	CEO of AseBio. Holds a Biology degree and Executive MBA, experience in technology transfer, IP valorisation, and life-science venture capital, strong investment and market expertise in the Spanish and global biotech sector.
Ivan Pristaš	CIPH	Public health physician and researcher. CIO involved in numerous European projects. Led work packages on PARENTJA and InfAct. Leads projects unCoVer and PHIRI. Works on TEHDAS and EHDS for secondary uses.
Jordi Rambla	CRG	Leads the European Genome Phenome Archive (EGA) at CRG, a renowned resource for human genomics data sharing. Key contributor to collaborative efforts to develop the GA4GH Beacon discovery technology, main focus of his PhD; leads work to create Beacon networks for data discoverability. Leads a multidisciplinary team of 24 persons.
Harald Wagener	CHARITE	Lead WP4. Group Lead Cloud and HPC at the Centre of Digital Health at Berlin Institute of Health. WP6 Lead (Technical Implementation) in EU-CIP and Member of the Executive Committee of the ELIXIR Compute Platform. WP Co-Lead in MyHealth@MyHands and Task Lead in CANDLE.
Irene Schlünder, legal and researcher	TMF	Lead WP8. Lawyer and senior expert in EU data protection law and data governance. She has been involved in various EU projects and was leading the legal WP7 in the EHDS2pilot and now leads WP3 of UNCAN-Connect on behalf of BBMRI-ERIC. On behalf of TMF she is leading and co-leading several tasks in the TEHDAS2 joint action for the German Ministry of Health.
Fotis Psomopoulos	CERTH	Contribute primarily in T3.2 (connecting to the ELIXIR AI activities), T4.1 (connecting to the AHEAD indicators), T8.2 (connecting to CANDLE and the National Cancer Data Nodes), and T7.2 (connecting to AI standards). Senior Researcher at INAB CERTH, co-lead of the ELIXIR AI Focus Group, Coordinator of the EOSC EVERSE project.
Jorge Oliveira	BIODATA	PhD in Biotechnology/Bioinformatics. Technical officer and Trainer at the Portuguese Infrastructure for Biological data. Focused on sensitive data infrastructures and projects, as FEGA and the European Genomic Data Infrastructure.

Luciana Peixoto	BIODATA	PhD in Chemical and Biological Engineering (2012) from University of Minho (UMinho). Training Officer in BioData.pt, Node coordinator and Training coordinator of ELIXIR Portugal.
Juan Carlos Vidal Aguiar	USC	Associate Professor at CiTIUS-USC. His research focuses on knowledge discovery, semantic annotation, semantic modelling of workflows and services, and the use of artificial intelligence for business intelligence.
David García Selfa	CESGA	Ph.D. in Physics and M.Sc. in Physics of Complex Systems. Coordinates AI research with particular focus on application and technology transfer, coordinating projects in this area for the agroforestry and industrial sectors. Experience in biophysics and neuroscience. Co-coordinates 1Health AI Factory.
Šarūnas Narbutas, Chief Advisor	POLA	M.Sc., LL.M Life Sciences lawyer, patient advocate, cancer patient with 19 years of experience, WP and task lead in a dozen EU-funded projects, latest are EU-CAYAS-NET, YARN, EU-CIP, MELIORA, ONCOSCREEN, ONCODIR, READI
Lorenz Dolanski-Aghamanoukjan	GÖG	Data scientist, programmer, statistician with longstanding experience in managing both national and international projects in public health, especially concerning process development, standardisation, and data quality in the area of health data use.
Martin Weibrecht	CLIN	M.Sc. Electrical Engineering, expertise in Med Tech Research and Business Creation. Head of data business, driving latest AI technologies and business focussing on secondary use of data for Med Tech and Pharma industry.
Jorge Barbazán	BATEA	CSO of Batea Oncology. PhD in molecular medicine with more than 15 years of experience in tumour microenvironment, biomaterials and mechanobiology.
Sergio Figueiras Gómez	BAHIA	Extensive experience in R&D project management and technology transfer, specializing in digital health and biotechnology. He will contribute to an in-depth review of technical and operational standards for developing AI systems in the biotech sector and the use of AI factories and EHDS (WP8).
Ségolène Combettes	ALIA	Leads the company's R&D strategy in health data and AI. Oversees the scientific direction of the use case on synthetic data generation from partners' clinical databases. Ensures alignment with overall EU project objectives, ethical/GDPR compliance, and exploitation of results (publications, product roadmap, innovation transfer). Expertise: AI for healthcare, data governance, innovation management, multi-partner scientific coordination.
Nathan da Silva Carvalho	DE	Senior EU projects and innovation manager specialising in digital transformation, digital health and data governance. He leads multi-country initiatives under Horizon Europe, Digital Europe Programme and EU tenders and maintains strategic policy engagement with European institutions. He holds a PMP® certification, an MSc in Public Sector Innovation & eGovernance, and executive training from Harvard Medical School, and has authored several publications on interoperability and digital transformation.
Dr. Francesca De Leo	CNR	Francesca De Leo is Technologist Director at the National Research Council of Italy (CNR) and Deputy Head of Node of ELIXIR-Italy, specialising in life science research infrastructures and innovation.
Pr E. Louis	ULIEGE	Pr E.L. is a distinguished physician-scientist leading a major Belgian university hospital department in gastroenterology. His internationally recognized research program investigates the genetics, mechanisms, and novel treatment strategies for inflammatory bowel disease (IBD) to improve clinical outcomes.
Ciprian Ciulpan	EBRAINS	Background in Computational Mathematics and Modern Information Technology, and 22 years of experience and success stories in ICT projects, contributes with the ability to identify gaps and blockers. His business analysis and technical content management knowledge will enable finding the best approach and reaching the most optimal outcome for this project.
Katerina Nikitara	ELLOK	Contribution to WP3, WP3 and WP8. Public health researcher specialising in evidence-based science, with a strong focus on patient advocacy, participatory research, and behavioural interventions, applying psychological and social science insights.
Vojtech Bystry	MU	ELIXIR CZ Strategic Lead establishing Czech FEGA/OmiCZ nodes and managing SPE integration. Coordinates federated analysis deployment, orchestrates AI factory connections, ensures EHDS compliance.

Raquel Perez Lopez	VHIO	Team Leader of the Radiomics Group at VHIO, leads research in AI-driven imaging biomarkers and precision oncology. Leadership roles in European research—as PI, co-PI, or WP lead in flagship Horizon Europe/H2020 projects (DART, TANGERINE, ODELIA). Holding prominent positions in organizations such as the ESR Research Working Group, RECIST WG, and ESAC.
Inge Franki	HDA	Lead of the EU Office at the Belgian Health Data Agency, coordinating Belgium's strategic engagement with European partners and steers key activities related to the implementation of the EHDS secondary-use framework. Expertise in structuring public-health data processes, experienced researcher with leadership of multidisciplinary projects. Solid scientific background in paediatric neurological rehabilitation, with many years of academic and clinical research.
Venkata Satagopam	UNILU	Senior Research Scientist and Head of Clinical and Translational Informatics Group working on biomedical informatics (data integration and knowledge management, data curation, harmonisation and FAIRification, dynamic visual analytics, text mining, and deep learning, advanced ML and AI) to bridge bench-to-bedside gaps. Technical coordinator of ELIXIR (LU) Node, since 2016, leading data hub development, ensuring GDPR-compliant clinical data sustainability.
Alexander Zlotnik	MoH ES	Technical lead in Spain for the implementation of the EU Patient Summary (PS) and ePrescription/eDispensation, the Digital COVID Certificate, and is Spain's representative in the eHMSEG, the eHN and is Spain's senior technical expert in the EU Council in matters related to digital health. He is Deputy Vice-Director of international projects of Spain's Ministry of Health and a career civil servant of Spain's State Administration. Dr. Zlotnik has a Degree on Telecommunications Engineering, master's in electrical engineering; PhD in artificial Intelligence; Master's in Public Administration and master's in public leadership.
Anne-Mari A. Willmore	UTARTU	Coordination team of UNCAN-Connect. Contributing to T5.1. PhD in nanomedicine, BS, MS in chemical engineering. 7 years' experience in biotech including process development, quality, control, data science as well as cancer research.
Constantin Bodolea	MCN	Prof. Dr. Bodolea MD, PhD, DEAA is Consultant in Anaesthesia and Intensive Care, Head of 2nd ICU Discipline at the University of Medicine and Pharmacy from Cluj-Napoca, Romania. Specialized in managing multi-morbid patients with highly complex health status. Representative in the board of the Romanian Society of Anaesthesia and Intensive Care
Ana Miguel	MAT	Involved in WP2, T2.2 in the work to define the governance model for the stakeholders' forum. Currently leading the EUCAIM EDIC Working Group, with + 15 Member States, on drafting the statutes of the future European infrastructure. With in depth understanding of the needs from health stakeholders.
Lu Zhao	EPIDATA	Co-Lead WP4, CEO of Epidata, with long-standing quality control, quality assurance, and monitoring expert from inspection and certification business sector. Contribution to WP4, in developing recommendations, guidelines and ensure a workable SOP, and technical capabilities intended Pathfinder system.
Olivier Bernard	INSA LYON	Olivier Bernard is a specialist in machine learning applied to cardiac imaging. His work focuses on image simulation, segmentation, motion estimation, uncertainty modelling, and multimodal data fusion.
Jean-Sébastien Gosuin	CUREWIKI	Technology entrepreneur, Founder of Curewiki, background in digital health innovation. Expertise in building and scaling large-scale digital health platforms, including QVAX, Belgium's national vaccine appointment matching system, reaching millions of users under strict regulatory, security, and privacy requirements. Expert in intelligent matching technologies and the application of AI to improve patient access, inclusion, and healthcare system efficiency.
Lise Bosquet	UNIC	Senior Scientific Project Manager and Medical & Scientific Advisor with over 20 years of experience in clinical research, real-world data (RWD), medical writing, quality management, and scientific program coordination. Proven expertise in managing large-scale research projects, coordinating multidisciplinary stakeholders, and delivering high-impact scientific outputs in academic, public, and pharmaceutical environments.

Julien Guérin	Institut Curie	Chief Data Officer with extensive experience (since 2017) in defining institutional data strategy, leading multidisciplinary teams (up to 15 people), and building health data platforms, demonstrating expertise in data valorization, technical implementation of infrastructure for massive data processing (including Federated Learning and NLP/ML for oncology cohorts), and significant involvement in national and international consortia like UNICANCER and DigiCore.
Carole Dufouil	CHU Bordeaux	Inserm Research Director and Deputy Director at the Bordeaux School of Public Health (ISPED), specializing in Cognitive Aging, Alzheimer's Disease, and Dementia (ADRD) through Lifecourse Epidemiology and Causal Inference. With expertise in biostatistics and handling longitudinal missing data, this leader co-directs the MELODEM initiative (focusing on methodological challenges in dementia research), leverages large national cohorts (e.g., MEMENTO, 3C-Dijon), and investigates the influence of social and policy-amenable determinants (like education) on neurological outcomes, resulting in a high research output (H-Index 85, 245 publications).
Rana Raich	SANOFI	Since 2020, Rana has led modelling and simulation projects across multiple Sanofi clinical programs. She has a strong background in applied mathematics and statistics, with extensive expertise in machine learning, meta-analysis, Bayesian modelling, and causal inference methods, leveraging diverse data sources including historical, clinical, and real-world data.

A full list of project teams and their expertise is available in the annex.

Outside resources (subcontracting, seconded staff, etc)

If you do not have all skills/resources in-house, describe how you intend to get them (contributions of members, partner organisations, subcontracting, etc.).

If there is subcontracting, please also complete the table in section 4.

Some project activities require highly specialised technical and regulatory expertise that is not fully available within the beneficiary organisations. In particular, this concerns the development and operation of secure HPC- and cloud-based environments enabling the secondary use of health data in line with EHDS specifications, the development of dedicated digital tools, specific sequencing services, and targeted support on data protection and regulatory compliance. These limited and well-defined activities will therefore be supported through subcontracting to external specialised providers, while overall responsibility for the action remains with the beneficiaries. Subcontracted tasks are strictly technical in nature, complement in-house capacities, and do not cover coordination, governance or decision-making functions.

All subcontracting will be implemented in accordance with the beneficiaries' internal procurement procedures and the EU4Health MGA, ensuring transparency, best value for money and the absence of conflicts of interest. Further details on the scope and justification of the subcontracted tasks are provided in Section 4.

2.4 Consortium management and decision-making

Consortium management and decision-making (if applicable)

Explain the management structures and decision-making mechanisms within the consortium. Describe how decisions will be taken and how regular and effective communication will be ensured. Describe methods to ensure planning and control.

Note: The concept (including organisational structure and decision-making mechanisms) must be adapted to the complexity and scale of the project.

Considering the number of partners in SCAILED consortium, the selected project management approach is well balanced in terms of providing formal structures and processes while avoiding unnecessary complexity that may affect the smooth running of the project. Consortium bodies and key actors in the management structure are depicted below (also shown in Figure 5):

Roles and responsibilities

Duties, responsibilities, and rights will be codified and agreed in the Consortium Agreement (CA).

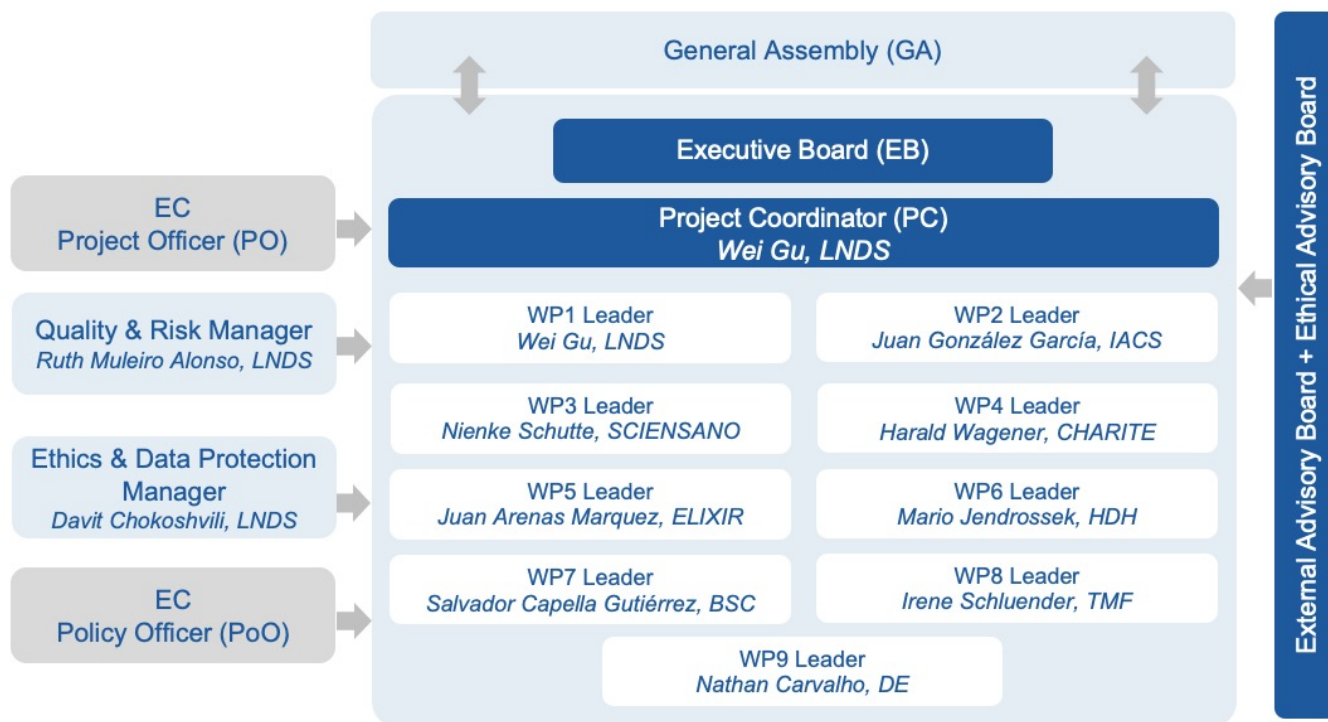


Figure 5 - Consortium management structure.

The General Assembly (GA), chaired by the Project Coordinator, is the highest decision-making body in the project, comprising one management-level representative per partner, each having one vote. The GA ensures that the partners give continued support and adequate resources to the project; sponsors and promotes the results and acts as the final level of escalation, should serious conflicts arise during the project. In particular, the GA is responsible for: i) any major changes to the consortium, plan or budget, in particular those which are subject to approval of the EC; ii) review, quality check and approval of project deliverables and, more in general, of all public output issued by the project (via the work of the Quality and Risk Manager), iii) guaranteeing confidentiality of internal results and project output designated as restricted to consortium; iv) establishing a controlling and reporting system for resource consumption and progress against schedule.

The Executive Board (EB) is the project's driving force for project activities. It is chaired by the PC, reports to the GA and is composed of leaders of each WP. Where necessary, other team members might be invited to individual EB meetings, e.g., more WP team members or the Quality and Risk Manager (QRM). The EB is in charge of project progress and in particular: i) implementation of all action plans; ii) coordination of work package dependencies; iii) maintaining communication and ensuring reporting; iv) providing guidance on Intellectual Property issues; v) supporting implementation of the quality assurance system; vi) ensuring guidance on ethical and (legal) data protection issues is followed; vii) creation of efficient team structures; viii) establishing flexible effective communication and an appropriate meeting schedule.

The Project Coordinator (PC), as defined in the Article 7 of the EU4H MGA, is responsible for the overall management, administrative, financial, communication, and coordination aspects of the entire project, and chairs and ensures proper operation of the key consortium bodies, GA and EB. Particular PC responsibilities include: i) execution of all project obligations vis-à-vis the EC and the funding programme, incl. submission of deliverables; ii) acting as the intermediary between the partners and the EC / Project Officer (PO) and Policy Officer (PoO); iii) calling GA and EB meetings, informing & chairing GA and EB, minutes taken & approved; iv) resolution of financial and contractual issues; v) ensuring implementation of administrative processes including data collection from project partners.

The Quality and Risk Manager (QRM) reports directly to PC and the Executive Board, providing regular updates on quality assurance procedures, risk monitoring, and any issues requiring remedial action. The QRM is responsible for establishing and enforcing the project's quality assurance processes, including the internal review of deliverables. When necessary, the QRM also informs the GA of critical quality or risk-related matters that require strategic attention.

The Ethics & Data Protection Manager (EDPM) ensures that the ethical and legal environment is taken into account in project activities, especially those related to data access and reuse. The EDPM may raise potential ethical and data protection issues related to the use of personal data and suggest appropriate measures for addressing such issues.

The Advisory Board: The project will benefit from the support of two high-level advisory bodies: an External Advisory

Board (EAB) and an Ethical Advisory Board (EthAB). These boards bring independent expertise and strategic oversight to the project. Both boards will participate in one advisory meeting per year and will provide constructive feedback on the consortium's presentations, documentation and planned activities.

- The EAB includes experts with recognised experience in digital health innovation, AI, the biotech industry and sector-level collaboration. The following candidates have been identified:
 - **Mr Pieter Jeekel**, Chairman of the Dutch AI Coalition for Well-being and Healthcare (NL AIC).
 - **Mr Steffen Hess**, Head of Health Data Lab, BfArM (Federal Institute for Drugs and Medical Devices)
 - **Mr Ioannis Pandis**, Site Lead, Pfizer
- The EthAB is composed of specialists in ethics, research governance and data protection, ensuring that the project adheres to the highest ethical and legal standards. The following candidates have been identified:
 - **Dr Michaela Th. Mayrhofer**, Senior Social Science Researcher (Health & Medicine), University of Innsbruck
 - **Dr Janos Meszaros**, Postdoctoral Researcher, KUL Clinical Pharmacology and Pharmacotherapy.
 - **Mr Adrian Thorogood**, Data Governance Manager, Terry Fox Research Institute

Together, these advisory bodies reinforce the project's transparency, quality and accountability, while providing independent perspectives that support robust decision-making and long-term impact.

#§CON-SOR-CS§# #@PRJ-MGT-PM@#

2.5 Project management, quality assurance and monitoring and evaluation strategy

Project management, quality assurance and monitoring and evaluation strategy

Describe the measures planned to ensure that the project implementation is of high quality and completed in time.

Describe the methods to ensure good quality, monitoring, planning and control.

Describe the evaluation methods and indicators (quantitative and qualitative) to monitor and verify the outreach and coverage of the activities and results (including unit of measurement, baseline and target values). The indicators proposed to measure progress should be relevant, realistic and measurable.

Project management structure: LNDS, as Project Coordinator (PC), ensures the effective day-to-day management of the project, facilitating communication, coordination and consistency across all partners. The project governance is structured around the General Assembly (GA) as the highest decision-making body and the Executive Board (EB) as the operational steering body composed of all WP leaders and chaired by the PC. The GA meets at least once per year to validate strategic decisions and review overall progress, while the EB meets quarterly (or as needed) to coordinate inter-WP dependencies, address operational issues, and align upcoming activities. This governance structure ensures clear responsibilities, timely decision-making and smooth coordination throughout the project.

Planning and risk management: Project planning is coordinated by the PC, who maintains the detailed implementation plan covering milestones, deliverables and interdependencies up to Month 36. The plan, supported by a Gantt chart and inputs from WP leaders, is continuously monitored, with progress reviewed at each EB meeting to identify deviations early and apply corrective measures when needed. Risk management is overseen by the Quality & Risk Manager (QRM), who maintains a project-wide risk register detailing the likelihood, impact and mitigation measures for each identified risk, together with their responsible owners. Risks are reviewed quarterly by the EB and escalated to the GA when strategic decisions or resource reallocations are required. This integrated planning and risk management approach ensures that the project remains on track, resilient and able to adapt proactively to emerging challenges.

Quality assurance and ethical oversight: Quality assurance is coordinated by the Quality & Risk Manager (QRM), who implements internal quality procedures for all project outputs. This includes conducting peer reviews of deliverables, verifying compliance with quality standards, ensuring consistency across WPs and monitoring the implementation of corrective actions. The QRM reports regularly to the EB and GA on the status of deliverables, identified quality issues and mitigation progress. Ethical and data protection compliance is overseen by the Ethics & Data Protection Manager (EDPM), who monitors adherence to ethical guidelines, GDPR requirements and applicable EU/national regulations. This includes reviewing data flows, proposed data processing activities, and safeguards for sensitive or personal data handled in other WPs. The EDPM will also support partners leading the Use Cases in WP6 with assessing the compliance requirements for specific use cases, such as ascertaining the need for a Data Protection Impact Assessment (DPIA). Independent oversight is further ensured by the Ethical Advisory Board (EthAB), which conducts annual reviews and provides expert recommendations on ethical and legal matters.

Monitoring and evaluation: A clear monitoring and evaluation framework is in place to track the project's performance through defined Key Performance Indicators (KPIs) linked to the project's Specific Objectives (SO). Each KPI has a baseline value and a target value, with a timeframe for achievement. KPI progress is reviewed quarterly by the EB and formally assessed at mid-term and final review points by both the EB and the GA. Monitoring draws on periodic technical and administrative reports, WP contributions and progress evidence collected by the PC.

Independent scientific and strategic guidance is provided by the External Advisory Board (EAB), which reviews progress annually and provides feedback on the project's direction, coherence and alignment with European and international developments. Together, the internal governance bodies and external advisory boards ensure that implementation remains robust, high-quality and continuously aligned with project objectives.

To further support monitoring efforts, the project has structured its KPIs around the Specific Objectives (SO) outlined in the call text. This objective-based matrix links each indicator to a measurable and time-bound target. The table below reflects the initial indicators, which will be refined and expanded in collaboration with partners and stakeholders during the early implementation phase. The KPIs to measure communication and dissemination activities are provided in Section 3.2.1.

Table 5 - List of indicators to measure the implementation of specific objectives defined in section 1.2 and target values (baseline: 0)

ID	Indicator (KPI)	Measurement / Target
Indicators related to SO1: Establish a Multistakeholder Platform and Stakeholder Fora		
1-1	Number and diversity of stakeholders engaged	≥30 organisations from ≥ 5 organisation types and all country diversity (central, eastern, mediterranean, Nordic)
1-2	Number of fora meetings or thematic events organised annually	≥ 2 events per year
1-3	Frequency of platform interactions	≥ 50 interactions per quarter
1-4	Average participant satisfaction score	≥ 80% positive values of respondents
Indicators related to SO2: Analyse Challenges and Barriers via Comprehensive Landscape Analysis, Case Studies and Develop Guidelines and a Strategic Roadmap		
2-1	Core datasets and European initiatives and infrastructures mapped	≥50 by Month 12
2-2	Number of case studies identified	≥20 by Month 12
2-3	Stakeholder-specific strategic recommendations formulated	≥10 by Month 36
2-4	Methodology and tools to monitor and evaluate progress developed	1 methodology and tool developed
Indicators related to SO3: Pilot Health Data Infrastructure Integration		
3-1	Piloting data infrastructure data sets in EHDS central catalogue (test)	>3 by Month 12
3-2	Piloting infrastructures integrating EHDS DAMMs	>3 by Month 18
3-3	Piloting data infrastructure providing data for use cases	>2 by Month 18
3-4	Piloting data processing infrastructure for analytics and learning on data infrastructures	>2 by Month 30
3-5	Participants' engagement in WP5 attendance to WP meetings and workshops	>85% attendance to WP5 meetings and workshop
3-6	Individual participation from individual partners	>85% attendance to WP5 meetings and workshop
Indicators related to SO4: Prioritise AI Applications via Real-World Use Cases		
4-1	Use cases successfully completed	≥ 7 (out of 11) by Month 36, including new use cases from the open call
4-2	New use cases selected and onboarded via open call	≥ 3 by Month 12
Indicators related to SO5: Scale Up AI Innovation via AI Factories		
5-1	AI Factory services adopting project outputs: guidelines, templates, and tools	≥5 new services available across ≥3 AI Factories

#\$PRJ-MGT-PM\$# #@\$FIN-MGT-FM@\$#

2.6 Cost effectiveness and financial management

Cost effectiveness and financial management (n/a for prefixed Lump Sum Grants)

Describe the measures adopted to ensure that the proposed results and objectives will be achieved in the most cost-effective way.

Indicate the arrangements adopted for the financial management of the project and, in particular, how the financial resources will be allocated and managed within the consortium.

 *Do NOT compare and justify the costs of each work package, but summarize briefly why your budget is cost effective.*

The distribution of resources in SCAILED has been designed to maximise impact, ensure efficient use of funds, and reflect the logical structure of the project's WPs. Resource allocation follows six strategic categories:

- **Coordination and Project Management (~9%):** Resources allocated to **WP1 (Coordination and Project Management)** ensure effective consortium management led by LNDS, streamlined governance processes, and smooth coordination across all technical and pilot activities. This budget line remains intentionally lean while guaranteeing strong oversight, quality control, risk management, and administrative efficiency.
- **Strategic Intelligence, Methodological Frameworks & Roadmap (~15%):** This category covers **WP3 (Landscape Analysis)** and **WP4 (Monitor and Evaluate)**, which together provide the evidence base, methodological foundations and strategic direction of the project. **WP3** maps the current state of AI in biotech across Europe, while **WP4** develops the Pathfinder system and tools that translate the roadmap into actionable pathways for stakeholders. These WPs guide all technical, infrastructural and regulatory work, making this investment central to SCAILED's long-term relevance and sustainability.
- **Infrastructure, Integration Pathways & AI Factory Enablement (~27%):** This category includes **WP5 (Integration of Health Infrastructure in EHDS)** and **WP7 (Integration of AI Factories with EHDS)**. These WPs develop recommendations, alignment pathways and technical conditions needed for infrastructures, AI Factories, and secure processing environments to operate in line with the EHDS framework. They constitute the operational backbone enabling compliance, interoperability and secure processing — critical for a future-ready biotech and AI ecosystem.
- **Use Case Validation & Real-World Learning (~20%):** Covering **WP6 (Biotech Use Cases and Case Studies)**, this category represents the project's empirical validation engine. **WP6** conducts case studies, implements all use cases and translates lessons learnt into evidence that informs **WPs 3, 4, 5 and 7**. This investment ensures that the project delivers applied, practice-based results rather than purely conceptual outputs, generating knowledge with direct value for regulators, industry and healthcare providers.
- **Stakeholder Engagement, Community Building & Regulatory Alignment (~23%):** This category corresponds to **WP2 (Stakeholder Engagement)** and **WP8 (Regulatory, Legal and Ethics)**. **WP2** creates the multistakeholder platform and mobilises biotech, healthcare, patient organisations, regulators and AI Factories, ensuring alignment, buy-in and coordinated knowledge exchange. **WP8** ensures regulatory clarity, addresses ethical and legal barriers and provides cross-WP guidance. Together, they build the ecosystem needed for adoption and long-term impact.
- **Communication & Dissemination (~6%):** Corresponding to **WP9 (Communication and Dissemination)**, this budget supports stakeholder-oriented communication, visibility actions, and EU-level dissemination to ensure uptake, but remains proportionate and cost-efficient.

Leveraging Existing Tools and Infrastructure:

SCAILED maximises cost-efficiency by leveraging existing tools, infrastructures and prior work already developed and validated by consortium members. This avoids redundant development, reduces costs, and accelerates delivery. By building on proven capabilities instead of creating new systems from scratch, the project ensures optimal use of resources and faster deployment of results.

Efficient Procurement and Cost Control: All partners apply transparent, competitive procurement processes, following internal and EU best-practice guidelines to guarantee best value for money. Whenever feasible, joint procurement and bulk purchasing will be used to reduce costs. The consortium implements strict cost-control measures with continuous monitoring of expenditures against the budget, and regular financial reporting to detect and correct any deviations early.

Robust Financial Oversight: LNDS, as Project Coordinator, leads financial management across the consortium, ensuring compliance with EU4Health rules and internal accountability. Each partner manages its budget share and reports costs using standardised procedures. Budget disbursements are linked to progress on milestones and deliverables, ensuring alignment between spending and actual achievements. The General Assembly validates any budget adjustments to maintain overall cost effectiveness.

#FIN-MGT-FM\$# #@RSK-MGT-RM@#

2.7 Risk management

Critical risks and risk management strategy

Describe critical risks, uncertainties or difficulties related to the implementation of your project, and your measures/strategy for addressing them.

Indicate for each risk (in the description) the impact and the likelihood that the risk will materialise (high, medium, low), even after taking into account the mitigating measures.

Note: Uncertainties and unexpected events occur in all organisations, even if very well-run. The risk analysis will help you to predict issues that could delay or hinder project activities. A good risk management strategy is essential for good project management.

For risk management, SCAILED applies the classic 4T model - Tolerate, Treat, Terminate, Transfer. The process consists of a full set of steps to identify, analyse and prioritise risks, plan mitigation actions, track progress and control outcomes, as illustrated. The Quality and Risk Manager will contact WP leads and, as the need arises, single partners to estimate the probability and the consequences of potential threats and to collect their assessment of additional risks perceived. Risks are reported in management meetings.

Table 6 - List of risks and mitigation measures

Risk No	Description	WP	Proposed risk-mitigation measures
1	Regulatory and policy uncertainty: EHDS implementing Acts and national legislation are ongoing, which may introduce new regulatory uncertainties during the project and may complicate data sharing and AI deployment. Impact: High; Likelihood: Medium	1, 3, 8	Starting with areas and Member States that those requirements are more matured; constant monitoring of the regulatory developments; active participation in policy fora via the consortium partners involvement (e.g., TEHDAS2, EHDS2-HDAB Community of Practise); regular legal reviews; flexible project planning to adapt to new requirements.
2	Limited uptake or sustainability of project outputs: Project results (tools, guidelines, templates) may not be adopted or maintained after project completion. Impact: High; Likelihood: Low	2,4, 6,9	1) Early engagement with AI Factories and end-users; co-design of outputs; training and support; integration with existing European infrastructures; sustainability planning. 2) Strengthen early outreach through targeted communication campaigns, direct engagement via the multistakeholder platform, and coordinated participation in external events. 3) Adjust communication channels and messages based on monitoring indicators. Mobilise partners' networks to amplify dissemination and ensure representation of stakeholders across all WP9 activities.
3	Disputes between the partners within the project over terms in the contract and/or deliverables. Impact: Medium; Likelihood: Medium	1	1) Create and use clear and detailed contracts with defined deliverables, responsibilities and penalties; 2) Maintain open communication with and across all involved partners
4	Stakeholder conflict, such as disagreements or conflicting priorities and objectives among stakeholders in the field. Impact: Medium; Likelihood: Medium	1, 2, 3	1) Facilitate regular stakeholder meetings and feedback sessions (within the project and with the EC, where applicable); 2) Create and use a stakeholder engagement plan that employs conflict resolution techniques and neutral facilitators
5	The integration of infrastructures will depend on statutes of infrastructures as well as implementing acts and/or national law that is still to be adopted Impact: High; Likelihood: High	3, 4,8	Work together across WPs where infrastructures as well as HDABs are represented to constantly inform about necessary adaptation measures. Develop a contingency plan in the first year of the project if needed.
6	Availability of SPE Services at affordable cost Impact: High; Likelihood: High	6, 8	Early outreach to existing SPE providers (e.g. through FINDATA) as well as to AI factories, planning budget for using SPEs by use cases. Alternatively, the partners providing services in AI-Factories can also leverage their resources related to service provision in the AI-Factory
7	Stakeholder event execution risks: inability to attract attendants and speakers limiting the impact of the gathering. Impact: High; Likelihood: Medium	2	Advanced planning and scheduling with WPs' leads involved. Clear agreements on partners responsibilities for contents, logistics and administrative support. For in-person events, backup plan for virtual alternative.
8	Risk for duplication in mapping and landscaping exercise Impact: Medium; Likelihood: Low	3	Consortium partners are involved in key projects in which these exercises have been established (TEHDAS and TEHDAS2 joint actions, HealthyCloud, UNCAN, BY-COVID, EHDS2-HDAB Community of Practice). Projects outputs from those projects will be leveraged.
9	Loss of key personnel at one or more	all	Consortium partner back named personnel by qualified

	consortium partners Impact: Medium; Likelihood: Low		teams that can take over tasks. In time-critical cases, coordinators will work on re-assigning work to other partners (amendment required)
10	Lack of buying in from data infrastructures Impact: High; Likelihood: Low	5	The consortium has ensured diversity of infrastructures taking part (GDI, EUCAIM, EBRAINS, UNCAN, FEAGA) and will promote active participation of additional data infrastructures to ensure guidelines produced area fit for purposed and that the pilots can be used as exemplars
11	Inflexibility of Data Infrastructures set up and/or EHDS regulation to deliver true interoperability across all levels Impact: High; Likelihood: High	5	The diverse type of HDABs, DIs would allow to explore the potential roadblocks and inefficiencies and face them in a constructive environment where alternative solutions can be analysed and documented to inform policy makers and to allow for DIs to evolve their governance and operations, if required

#\$RSK-MGT-RM\$# #SQUA-LIT-QL\$# #IMP-ACT-IA@#

3. IMPACT

3.1 Impact and ambition

Impact and ambition — Progress beyond the state-of-the-art

Define the short, medium and long-term effects of the project.

Who are the target groups? How will the target groups benefit concretely from the project and what would change for them?

Does the project aim to trigger change/innovation? If so, describe them and the degree of ambition (progress beyond the status quo/state-of-the-art).

The project aims to generate impacts during and beyond the period of the project towards accelerating AI and digital innovation, enhancing European competitiveness in biotechnology sector and improving health outcomes.

Short-term impacts (during the project):

- Increase effective collaboration, dialogue, exchange and knowledge among diverse stakeholders in the biotech and healthcare sectors by the establishment of the stakeholder platform and fora, directly supporting the call's aim to create an active network of stakeholders and encourage collaboration and knowledge-sharing among the biotech and healthcare sectors.
- Demonstrate the potential of accelerating innovation and supporting the development and deployment of AI and digital solutions in the biotech industry by integrating project outcomes into operational services within AI Factories.
- Showcase the value and feasibility of enhancing health outcomes via implementation of real-world use cases and case studies as required by the call's emphasis on enhancing patient care, diagnosis accuracy and treatment efficiency across the Union.
- Offer visibility and clarity to regulatory and technology challenges, requirements, opportunities, and action plans via the delivery of a comprehensive landscape analysis and strategic roadmap for biotech stakeholders and health data and research infrastructures, in line with the call's requirement for mapping the current state and providing actionable recommendations and pathways toward integration into the EHDS framework.
- Demonstrate the benefit of accelerating data access and enhancing interoperability needed for AI innovation via piloting the integration of selected health data infrastructures into EHDS.

Medium-term impacts (1–3 years post-project):

- Support the EHDS Stakeholder Forum and EHDS Board to better reach the biotech sector and the wider community identified in the project.
- Widen the adoption of project outputs (guidelines, tools, templates, services) by stakeholders, leading to the operationalisation of production-level AI services in the biotech sector, thus supporting the call's ambition to accelerate innovation and competitiveness in the Union's biotech industry.
- Expand integrated health data infrastructures and AI Factories under EHDS framework beyond the ones integrated during the project. Make Available diverse (multiple types of data, populations across Member States) and high-quality data facilitating cross-border research and innovation.
- Ensure continuous growth of participants in the stakeholder fora as a sustained platform collaboration and knowledge-sharing among diverse stakeholders, fostering a cohesive European health data ecosystem and supporting the call's goal of increased collaboration and knowledge-sharing.
- Increase the regulatory clarity and harmonisation, supporting responsible and compliant use of health data and AI.

Long-term impacts (beyond 3 years post-project):

- Accelerate AI-driven innovation in the European biotech sector.
- Enhance health outcomes via facilitating a broader adoption of digital / AI solutions in the biotechnology sector enhancing patient care, diagnosis accuracy and treatment efficiency across the Union.
- Ensure a sustainable integration of project outputs into EHDS framework, European infrastructures towards ongoing impact and scalability.
- Strengthen European leadership in digital health, AI, and biotechnology, contributing to economic growth and global competitiveness, in line with the call's ambition for the Union's biotech sector.

Target groups and benefits

The target groups and how will they benefit concretely from the project (changes relevant to them) are summarised in the following table.

Table 7 - Target groups and benefits

Target group	Benefits
Biotech industry and AI developers	Easier access to harmonised, high-quality health data, production-ready AI services and infrastructure, and clearer compliance guidelines, all contributing to faster R&D and reduced time-to-market for innovative solutions
Healthcare providers and patients	Enhanced diagnostic and treatment options, better patient outcomes, and increased involvement in innovation processes.
Health Data Access Bodies and HealthData@EU	Strengthen the services provided by HDABs and boost EHDS deployment and compliance.
Health Data Infrastructures	Accelerate integration under EHDS, generate higher impact via increased data reuse under the EHDS framework
Regulatory bodies and policymakers	Clearer frameworks, guidelines, and best practices for responsible data use and AI deployment, supporting evidence-based policy and compliance.
AI Factories	Integration of new tools, templates, and services into their portfolios, enabling rapid scaling and broader adoption of AI in biotech.
Research institutions	Improved collaboration opportunities, enabling cutting-edge research and cross-border projects.

Ambition and Progress Beyond the State-of-the-Art

The project seeks to achieve ambitious goals and advance beyond current standards in the following areas:

- **Pan-European data federation under the EHDS framework:** The project will pilot the integration and federation of health data infrastructures (starting with genomic and cancer imaging) under the EHDS framework, enabling harmonised, cross-border data sharing and access of high-quality datasets. This goes beyond the health data held by national data holders and currently siloed data infrastructures in specific domains, creating a unified European resource for AI-driven biotech innovation.
- **Regulatory and policy harmonisation:** The project will deliver actionable recommendations and frameworks that directly inform EU and national policy, supporting regulatory clarity and harmonisation for AI and health data use. This will reduce barriers for innovators and accelerate compliance across Member States.
- **Operational support for AI development and deployment:** By leveraging AI Factory resources, the project will move from proof-of-concept and pilot studies to production-level deployment of AI tools and services. This means that validated guidelines, templates, and best practices will be actively adopted by multiple AI Factories, resulting in multiple new operational services and production-level deployments across different AI factories by project end.
- **Stakeholder engagement at scale:** The project will establish and sustain a multistakeholder platform, engaging at least 30 organisations from industry, research, healthcare, patient groups, and regulators. This will foster ongoing collaboration, knowledge-sharing, and co-creation, ensuring that solutions are relevant and widely adopted.
- **Real-world impact and case studies:** At least 8 use cases will be implemented and validated, with additional 3 added over open call demonstrating measurable improvements in patient care, diagnosis accuracy, and treatment efficiency. More case studies (≥ 20) will be carried out and will be disseminated to promote best practices and accelerate adoption across the sector.

In summary, the project will not only advance the technical and organisational state-of-the-art, but will deliver operational, policy, and collaborative breakthroughs that position Europe as a global leader in AI-driven biotech innovation.

3.2 Communication, dissemination and visibility

Communication, dissemination and visibility of funding

Describe the communication and dissemination activities which are planned in order to promote the activities/results and maximise the impact (to whom, which format, how many, etc.). Clarify how you will reach the target groups, relevant stakeholders, policymakers and the general public and explain the choice of the dissemination channels.

Describe how the visibility of EU funding will be ensured.

3.2.1 Communication and dissemination activities to maximise impact

The project's communication and dissemination activities, led by DE and AseBio in WP9, aim to ensure that the project's objectives, progress and outcomes reach all relevant stakeholder groups across Europe as widely and effectively as possible. This process will be achieved through the selection of appropriate communication channels and tools, supported by engaging and timely promotional actions. The dissemination and communication strategy will be designed to raise awareness of the project's role in advancing responsible data-driven and artificial-intelligence-enabled biotechnology, increase visibility for its outputs, and strengthen the multistakeholder ecosystem required to support the implementation of the EHDS. As the project evolves, communication activities will be continually informed by emerging evidence, lessons learned and validated results, helping clarify opportunities, address misconceptions and promote informed engagement from biotech industry actors, data holders, public authorities, developers and patient organisations. The dissemination, communication and stakeholder engagement plan (D9.1) will outline the project's communication objectives, key messages, audiences, tools and outreach strategy. It will set a clear course of action to maximise visibility, engagement and the impact of project outcomes, ensuring a strong and coherent project voice throughout the three years. The plan will also incorporate communication relating to the implementation and evolution of the EHDS. Key performance indicators will be used to measure communication effectiveness and to support periodic adjustments of the strategy. The plan will be implemented through targeted promotional actions and coordinated use of the communication channels of both the project and its partners. The table below outlines the key channels that will be activated throughout the project, describing their intended audience, expected impact and operational focus. It also specifies measurable indicators (KPIs) that will be monitored throughout implementation to ensure effectiveness, track performance and enable ongoing optimisation of the project's communication approach.

Table 8 - List of communication and dissemination activities and KPIs.

Channel	Description, impact and intended SCAILED target audience	Main KPIs (baseline value)
Brand identity	The project will establish a coherent and recognisable visual identity that reflects its mission of advancing responsible, data-driven and artificial-intelligence-enabled biotechnology in Europe. A strong and consistent brand will ensure clarity and trust across all communication materials and channels, supporting visibility among biotechnology developers, data holders, research infrastructures, regulators, and patient organisations. This identity will reinforce the project's position within the wider European Health Data Space ecosystem and enhance long-term stakeholder engagement.	Branding manual: 1 Project presentation templates: 2
SCAILED Website	A dedicated project website will create and serve as the central hub for all updates, outputs and resources related to SCAILED. DE will create the mentioned project website and ASEBIO will manage and update the content. It will provide accessible information on objectives, activities, events, deliverables and opportunities for engagement. Designed with user experience and accessibility in mind, the website will support broad visibility and enable stakeholders to easily access project news, results and materials.	No. of total visitors by M36: 6000
Social media channels	SCAILED will maintain an active online presence through LinkedIn and X (formerly Twitter) to broaden visibility and stimulate engagement around responsible AI in biotechnology and the EHDS. Social media will be used to share updates, insights, expert messages, and highlights from project activities and events. LinkedIn will support networking and professional interaction; X will deliver timely updates and visual summaries; Partners will amplify communication through their own channels to maximise reach.	Average no. of posts on LinkedIn: 4 per month Average no. of posts on X: 4 per month Average impressions per month across all channels: 800 No. of followers across all channels: 1000
Email	Targeted email updates will be used to inform stakeholders about	Total no. of email campaigns

marketing	project milestones, opportunities for engagement, consultations, events and key results. Campaigns will direct readers to the project website and social media channels, supporting broader visibility and continued involvement. A subscription form on the website will allow interested stakeholders across biotechnology, digital health and data governance communities to stay informed. Consortium partners will also use their own mailing lists to extend outreach across Europe.	(newsletter): 6 Number of subscribers: 150
Partners, policy and media outreach	The consortium will issue press releases and news updates to announce major project milestones, findings and recommendations. These materials will target policymakers, regulators, research infrastructures, journalists and sector-specific platforms, helping position SCAILED as a reference initiative for trustworthy AI in biotechnology and health data use. Media outreach will strengthen the project's visibility across Europe and support engagement from diverse stakeholder communities.	No. of press releases: 3 No. of policy briefs: 1 No. of mentions in other websites (blogposts, media coverage, related posts): 6
Visual assets	A mix of digital and printable visual materials will support the project's communication activities, including infographics, banners, leaflets, presentation templates and branded materials for events. These assets will make complex concepts understandable and attractive, facilitating outreach to biotechnology stakeholders, policymakers, data holders and the wider public.	No. of communication and dissemination materials produced: 6
Third-party events	Project partners will participate in external events organised by European initiatives, research infrastructures, industry associations and public authorities to present the project, raise awareness and engage relevant communities. Participation in these events will strengthen visibility, build new connections and support alignment with ongoing European efforts such as HealthData@EU, TEHDAS2 and AI-related initiatives.	No. of third-party events attended: 20

3.2.2 How the consortium will reach the target groups

Effective engagement with the project's stakeholder community is essential to ensure the visibility, uptake and long-term relevance of the project's results. Each target group plays a distinct role in the European biotechnology and health data landscape and therefore requires tailored communication approaches that reflect their needs, expectations and potential contributions. The consortium will draw on the strengths of its diverse membership, including industry associations, research infrastructures, national authorities, AI Factories, patient organisations and academic institutions, to ensure targeted, credible and high-impact outreach. DE will play a central role in enabling wide geographical reach by mobilising its network of National Trade Associations, ensuring consistent engagement across Member States and associated countries. The table below outlines the primary target groups and the specific mechanisms through which the consortium will reach and involve them throughout the project. ASEBIO will assume a pivotal role in ensuring national and European representation of biotech sector by activating its extensive network of biotechnology companies, research organisations, and regional innovation clusters. Through targeted outreach, coordinated communication channels, and established collaborations with public and private stakeholders, ASEBIO will facilitate the active participation of key actors in the bio-based ecosystem.

Table 9 - Target groups and ways to reach team.

Target groups	How the consortium will reach them
Biotech industry and AI developers	ASEBIO and DE will approach Biotech companies and developers. DIGITAL EUROPE's 120 corporate members, can mobilise national ecosystems via its network of 41 National Trade Associations. As the voice of the Spanish biotechnology sector, ASEBIO brings together more than 340 entities, including innovative biotech SMEs and large companies, with 67% of its technology-using members active in the health sector. ASEBIO hosts an Artificial Intelligence and New Technologies Working Group comprising 61 biotech companies and 45 technology-driven firms, developing or deploying AI-based solutions, ensuring structured and sustained engagement with AI-driven industrial stakeholders. SMEs in the consortium (CLINOMIC, BATEA, BAHIA, ALIA, MAT) will showcase concrete pipelines and demonstrators, opening doors to their own networks. Research and technology organisations: BSC, INRIA, CESGA and CSC will engage technical developer communities through hands-on events (hackathons, code sprints, technical webinars) and participation in AI and biotech conferences.
Healthcare	Hospitals and care providers will primarily be reached through clinical and health-system partners

providers and patients	(MCN, IISLAFE, VHIO, HRI, CDR), who can relay project outputs within their institutional networks and professional societies. Patient organisations POLA and ELLOK will co-design easy-to-understand materials and help organise feedback activities with patient communities. Ministries and agencies (MoH-ES, SCIENSANO, CIPH) will connect the project to national health structures, while DE's National Trade Associations will help link to broader e-health and provider networks at country level.
Regulatory bodies and policymakers	For regulators and policymakers, the project will rely on direct participation of competent authorities and Health Data Access Bodies (LNDS, HDH, FINDATA/THL, GÖG, HDA, CIPH) as well as the Spanish Ministry of Health. These institutions already sit inside national and European governance processes and will host or co-host briefings, closed roundtables and consultation sessions. Policy-oriented partners (CHARITE, TMF, CERTH, UNILU, CNR) will translate technical outcomes into concise, policy-ready messages. DE will complement this with Brussels-level advocacy channels and, through its National Trade Associations, structured outreach to ministries, regulators and parliaments across Member States.
AI Factories	AI Factories will be engaged directly through partners that operate or anchor them: LNDS (AI Factory), BSC, CESGA and USC (1HealthAI Factory), together with CSC in Finland. These actors will be involved in co-designing workflows, testing templates and integrating selected outputs into their service catalogues. Technical exchanges, joint architecture sessions and proof-of-concept integrations will be used to make the work tangible for other compute and model-development hubs, while DE and ASEBIO will promote these results across wider AI and digital-innovation communities.
Research institutions	Research institutions will be reached through the strong cluster of research infrastructures and universities in the consortium, including ELIXIR, EBRAINS, CRG, IISLAFE, BIODATA, HRI, IACS, MUNI, UNILU, UTARTU, USC, VHIO, Sciensano, CNR, and ASEBIO. These partners will circulate methods, data-governance models and results via their established working groups, scientific networks, training activities and conference tracks. Outputs will also feed into domain-specific communities in genomics, cancer, neuroscience and health data. DE's National Trade Associations will be used to connect this work to national research and innovation ecosystems beyond the immediate consortium.

3.2.3 Visibility of the EU funding

All visual assets produced for the project, whether digital or printed, including website banners, roll-up banners, backdrops, leaflets, social media visuals and other communication materials, will prominently display the European Union emblem. This ensures that EU support is clearly and consistently acknowledged across all communication and dissemination outputs. All written deliverables, such as reports, briefs and guidance documents, in both online and printed formats, will include the EU emblem together with the standard funding disclaimer. The project's public website will display the EU emblem and disclaimer prominently and will clearly indicate the type of funding received. Social media campaigns and promotional activities will direct users to the website, ensuring full visibility of the EU contribution. Likewise, all promotional and informational content will feature the EU emblem and an explicit acknowledgement of financial support and will be disseminated widely across digital platforms to maximise transparency and reach.

#§COM-DIS-VIS-CDV§# #@SUS-CON-SC@#

3.3 Sustainability and continuation

Sustainability, long-term impact and continuation

Describe the follow-up of the project after the EU funding ends. How will the project impact be ensured and sustained?

What will need to be done? Which parts of the project should be continued or maintained? How will this be achieved? Which resources will be necessary to continue the project? How will the results be used?

Are there any possible synergies/complementarities with other (EU funded) activities that can build on the project results?

The impact of SCAILED will extend well beyond the project's lifetime. The core outputs: the strategic roadmap, the monitoring and evaluation methods, the best practises, the guidelines, and the regulatory clarity analysis will be made available in the Pathfinder and Navigation System for wide adoption.

The technical development and governance recommendations for integrating health data infrastructures and AI Factories into the EHDS will be embedded in operational environments of the piloted data infrastructure as services and therefore sustained beyond the project's duration. These pilots can be used as successful examples to allow the technical development and governance recommendations to be adopted and further developed by national authorities, and other existing and future research infrastructures in EHDS ecosystem.

The developed AI applications in the use cases are exploitable by the biotech industry as potential commercial products or services and therefore sustained.

After EU funding ends, components of the work will continue through existing structures, particularly the AI Factories, health data access bodies, health data infrastructures and biotech communities engaged throughout the project, who will maintain and update the tools, guidelines and services that have proven effective during implementation.

The Stakeholder Forum Platform created in WP2 and the strong ties between actors involved in the EHDS construction (HDABs, RIs and EDICs, the AI Factory network), are expected to heavily contribute the EHDS Stakeholder Forum (EHDSR Art.93), offering a natural continuation pathways, ensuring that the methodologies, standards and recommendations become embedded in the regional, national and European practice. The validated use cases and case studies will remain available as reference models for regulators, developers and healthcare actors, supporting further uptake and scale-up. Many project partners are active in related EU initiatives such as TEHDAS2, QUANTUM, xSHARE, SHAIPEd, GDI, EUCAIM, 1+MG, UNCAN Connect, FORTIFY, which ensures strong complementarity and provides mechanisms to reuse, refine and expand SCAILED outputs in future calls and policy processes. In this way, the project lays the groundwork for a sustained European capacity to develop trustworthy biotech-oriented AI, leveraging the EHDS and maintained collaboratively across the organisations already committed to its long-term success.

Part of the work planned in WP9 will focus on enabling long-term exploitation pathways, strengthening policy alignment, supporting regulatory and organisational uptake, and fostering continued collaboration among stakeholders. The activities there include developing sustainability models, aligning results with EU initiatives (such as EHDS, HDAB networks, and European research infrastructures), promoting integration into national and European workflows, and ensuring that stakeholders have the capacity, incentives, and guidance needed to continue using and evolving the project's outcomes.

#§SUS-CON-SC§#

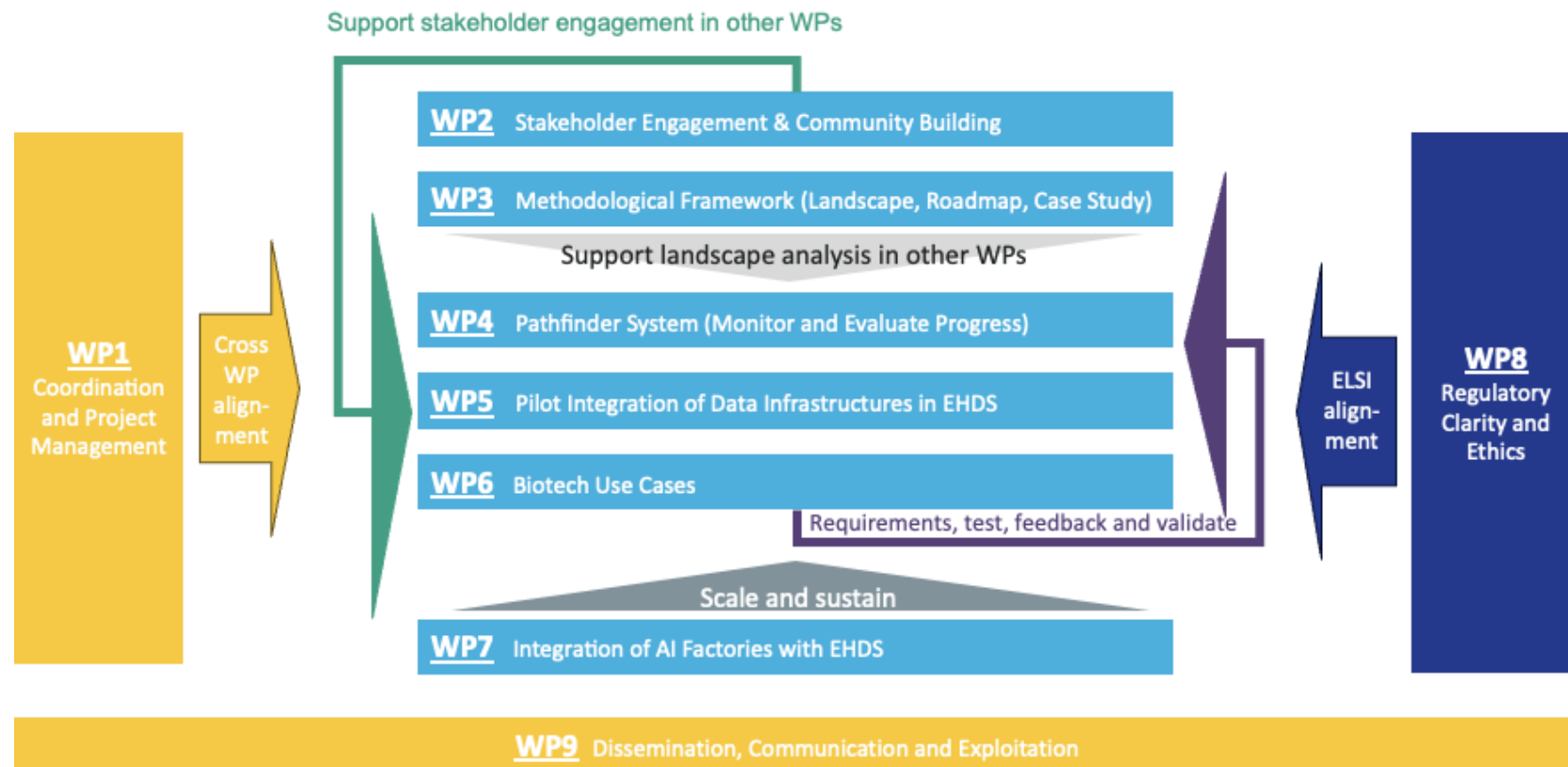
4. WORK PLAN, WORK PACKAGES, ACTIVITIES, RESOURCES AND TIMING

4.1 Work plan

Work plan

Provide a brief description of the overall structure of the work plan (list of work packages or graphical presentation (Pert chart or similar)).

The work of SCAILED is organized in 9 Work Packages (WPs) to implement the actions required to meet the goal of the project. The WP2, 3, 4, 5, 7 (building block WPs), are implementing tasks related to a specific building block, while WP8 is a transversal WP for regulatory and legal topics. WP1 will support all the WP in project management and cross-WP alignment. WP9 will support communication and dissemination of results from all work packages.



#\$WRK-PLA-WP\$#

4.2 Work packages, activities, resources and timing

WORK PACKAGES

Work packages

This section concerns a detailed description of the project activities.

Group your activities into work packages. A work package means a major sub-division of the project. For each work package, enter an objective (expected outcome) and list the activities, milestones and deliverables that belong to it. The grouping should be logical and guided by identifiable outputs.

Projects should normally have a minimum of 2 work packages. WP1 should cover the management and coordination activities (meetings, coordination, project monitoring and evaluation, financial management, progress reports, etc) and all the activities which are cross-cutting and therefore difficult to assign to another specific work package (do not try splitting these activities across different work packages). WP2 and further WPs should be used for the other project activities. You can create as many work packages as needed by copying WP1.

For very simple projects, it is possible to use a single work package for the entire project (WP1 with the project acronym as WP name).

Work packages covering financial support to third parties (⚠ only allowed if authorised in the Call document) must describe the conditions for implementing the support (for grants: max amounts per third party; criteria for calculating the exact amounts, types of activity that qualify (closed list), persons/categories of persons to be supported and criteria and procedures for giving support; for prizes: eligibility and award criteria, amount of the prize and payment arrangements).

⚠ Enter each activity/milestone/output/outcome/deliverable only once (under one work package).

⚠ Ensure consistence with the detailed budget table/calculator (if applicable) (n/a for prefixed Lump Sum Grants).

Objectives

List the specific objectives to which the work package is linked.

Activities and division of work (WP description)

Provide a concise overview of the work (planned tasks). Be specific and give a short name and number for each task.

Show who is participating in each task: Coordinator (COO), Beneficiaries (BEN), Affiliated Entities (AE), Associated Partners (AP), indicating in bold the task leader.

Add information on other participants' involvement in the project e.g. subcontractors, in-kind contributions.

Note: In-kind contributions: In-kind contributions for free are cost-neutral, i.e. cannot be declared as cost. Please indicate the in-kind contributions that are provided in the context of the work package.

The Coordinator remains fully responsible for the coordination tasks, even if they are delegated to someone else. Coordinator tasks cannot be subcontracted.

If there is subcontracting, please also complete the table below.

Milestones and deliverables (outputs/outcomes)

Milestones are control points in the project that help to chart progress (e.g. completion of a key deliverable allowing the next phase of the work to begin). Use them only for major outputs in complex projects, otherwise leave the section empty. Please limit the number of milestones by work package.

Means of verification are how you intend to prove that a milestone has been reached. If appropriate, you can also refer to indicators.

Deliverables are project outputs which are submitted to show project progress (any format). Refer only to major outputs. Do not include minor sub-items, internal working papers, meeting minutes, etc. Limit the number of deliverables to max 10-15 for the entire project. You may be asked to further reduce the number during grant preparation.

For deliverables such as meetings, events, seminars, trainings, workshops, webinars, conferences, etc., enter each deliverable separately and provide the following in the 'Description' field: invitation, agenda, signed presence list, target group, number of estimated participants, duration of the event, report of the event, training material package, presentations, evaluation report, feedback questionnaire.

For deliverables such as manuals, toolkits, guides, reports, leaflets, brochures, training materials etc., add in the 'Description' field: format (electronic or printed), language(s), approximate number of pages and estimated number of copies of publications (if any).

For each deliverable you will have to indicate a due month by when you commit to upload it in the Portal. The due month of the deliverable cannot be outside the duration of the work package and must be in line with the timeline provided below. Month 1 marks the start of the project and all deadlines should be related to this starting date.

The labels used mean:

Public — fully open (⚠ automatically posted online on the Project Results platforms)

Sensitive — limited under the conditions of the Grant Agreement

EU classified — RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision [2015/444](#). For items classified under other rules (e.g. national or international organisation), please select the equivalent EU classification level.

Work Package 1

Work Package 1: Coordination and Project Management

Duration:

M1 – M36

Lead Beneficiary:

1 – LNDS

Objectives

- **O1.1:** To ensure the achievement of the project goals and the management on a daily basis.
- **O1.2:** To guarantee the proper development of tasks and the submission of deliverables on time.
- **O1.3:** To effectively monitor the project, in administrative, technical and financial terms and to ensure its strategic management.
- **O1.4:** To offer the necessary interface to the EU services and external actors.
- **O1.5:** To effectively coordinate and support the performance of the use case activities.
- **O1.6:** To guarantee that ethical and legal issues related to project activities and impact are being properly considered and relevant conventions are being respected.
- **O1.7:** To identify project risks by performing effective risk management.

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contr. & Subcontr.
			Name	Role	
T1.1	Project Management	This task comprises the work of the project coordinator, in collaboration with all WP leaders, to ensure the quality of the project results. The necessary work includes: • Assure that a) the project requirements defined are well aligned with the project objectives and considers the user needs and expectations; b) the validation procedures are adequate and are accomplished as defined; and c) the requirements are reviewed in response to risks and internal and external opportunities identified. • Monitor the progress of all partners, ensuring that the deliverables and milestones are met and taking the necessary actions if they are not. • Collect, review to verify the consistency, and submit reports and other deliverables (including financial statements and related certifications) to the European Commission. • Organize the project meetings, elaborate the minutes, and distribute them to the consortium. This includes preparing and convening the General Assembly and Executive Board meetings, as well as organizing the annual meetings of the External Advisory Board and the Ethical Advisory Board, ensuring that agendas, supporting materials and follow-up actions are managed consistently. • Create and organize internal communication channels and tools for internal disseminating information about project results. • Initial identification of risks and the definition of a risk management process, aiming to continuously identify and categorize risks, as well as their occurrence of chance and impact on the project.	LNDS ALL	COO BEN	No
T1.2	Administrative and financial management	This task comprises the work of the Project Coordinator to simplify the communication between the partners and/or other participating entities, including the following subtasks: • Setting up the communication procedures, contact list, repository for documents, and code sharing. • Maintaining the Consortium Agreement and Grant Agreement. • Transmitting documents and information connected with the Project to any other Parties concerned. • Administering the financial contribution of the Union and fulfilling the financial tasks described in. • Distributing funding to partners according to their budget and expenditure in previous periods. • Agreeing and overseeing transfer budgets between partners if necessary.	LNDS ALL	COO BEN	No
T1.3	Data Management	The task involves developing and managing the project's Data Management Plan (DMP) (Deliverable D1.2) by M6. The DMP will outline how data is collected, processed, stored, and shared (in line with FAIR principles and	LNDS	COO BEN	No

GDPR). It will specify roles, responsibilities, storage solutions, data quality procedures, and plans for data sharing and long-term preservation. The DMP will be updated at least annually.

ALL3

Milestones and deliverables (outputs/outcomes)

Milestone No	Milestone Name	WP No	Lead Beneficiary	Description			Due Date	Means of verification
MS1.1	Kick-off meeting and project launch	1	LNDS	Kick-off meeting and project launch: Formal start of the project, establishment of governance bodies, and agreement on initial work plan and responsibilities.			M1	Signed attendance list and meeting minutes confirming the kick-off meeting, and approved work plan circulated to all partners.
MS2.1	Mid-term review and technical assessment	1	LNDS	Mid-term review and technical assessment: Evaluation of project progress, risk status, and technical alignment at project mid-point, including review of deliverables and risk register			M18	Submission of a mid-term review report, updated risk register, and confirmation of deliverables submitted, as well as minutes from the review meeting.
MS3.1	Project Completed	1	LNDS	Project completed: Final assessment of project implementation, all activities and tasks completed, all deliverables submitted and approved. Closure of all coordination, administrative, and reporting obligations.			M36	Project Completed. All project deliverables are completed. Final Project Progress Report (D1.4)
Deliverable No	Deliverable Name	WP No	Lead Beneficiary	Type	Dissem. Level	Due Date	Description	
D1.1	Project Management Plan	WP1	LNDS	R	SEN	M04	Project Management Plan. Internal procedures adopted for the correct development of the project, including the corporate identity, review process, communication protocols, risk management, and appointment of the members of the different committees, among others. It will also refer to the quality standards and best practice policies agreed by the consortium. This information will be subsequently updated in the Periodic Reports that will be submitted to the EC. [LNDS, ALL]	
D1.2	Data Management Plan	WP1	LNDS	R	SEN	M06	Draft and deliver the project's Data Management Plan by month 6. The DMP will define how project data (including sensitive health/biotech data) are handled: data collection methods, processing workflows, storage/back-up solutions, access controls, and sharing policies in line with FAIR principles and GDPR. It will assign responsibilities for data stewardship and describe how data will be archived or made available to the EHDS where appropriate. The DMP will be updated (versioned) as the project evolves as an annex to D1.3, D1.4, and D1.5. [LNDS, ALL]	
D1.3	Initial Project Progress Report (PPR)	WP1	LNDS	R	SEN	M12	A yearly report containing all technical and financial details about the project and risk measures, countermeasures required to avoid identified risks, and the measures needed to ameliorate the consequences of adverse events. Additionally, it will include the required updates on the innovation, knowledge and Data Management Plan updates, and any deviation of the project specifications due to identified opportunities. [LNDS, ALL]	
D1.4	Intermediate	WP1	LNDS	R	SEN	M24	Report updated in the 2nd year [LNDS, ALL]	

	Project Progress Report (PPR).						
D1.5	Final Project Progress Report (PPR)	WP1	LNDS	R	SEN	M36	Report updated in the 3rd year [LNDS, ALL]

Estimated budget — Resources

See detailed budget table/calculator (annex 1 to Part B).

Work Package 2

Work Package 2: Stakeholder Engagement and Community Building

Duration:	M1 – M36	Lead Beneficiary:	4 - IACS
------------------	----------	--------------------------	----------

Objectives

The primary objectives of WP2 are to establish a core operational environment for the project, with a clear commitment of service towards the rest of WPs, to ensure a broad stakeholder buy-in and effective knowledge transfer across the EU's health and biotech ecosystems, in line with the call's foundational requirements:

- **02.1:** Create and sustain a functioning Multistakeholder Platform to foster an active network between the biotech industry, research institutions, healthcare providers, patient organisations, regulatory bodies, and AI Factories. The platform will support the exchange activities of the project (events) and beyond, supporting the setup of EHDS related governance bodies (e.g., Stakeholder Forum).
- **02.2:** Facilitate systematic collaboration and knowledge exchange to identify and analyse critical challenges and barriers related to data access, quality, ethics, IP, and regulatory issues in the use of AI for biotech.
- **02.3:** Ensure effective communication and dissemination of project outputs (e.g., case studies, roadmap recommendations) to relevant stakeholders and policymakers, guaranteeing broad adoption and impact.

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contr. & Subcontr.
			Name	Role	
T2.1	Stakeholder Mapping and Mobilisation	Objective: align partner and prior-initiative mappings (HealthyCloud, TEHDAS 1 & 2, EUCAIM, EOSC Health Data Task Force) to identify and frame stakeholder profiles and initial Forums for further development in T2.3. 1) Stakeholder identification and categorisation: build a consolidated map across segments (industry, regulators, researchers, patient organisations, healthcare authorities), classifying by role (health data holders/users), influence, and interest. 2) Needs analysis: capture expectations and priorities per group to define conditions for meaningful, sustained involvement. 3) Mobilisation strategy: design targeted engagement plans with invitations, incentives, channels, and practical participation mechanisms to maintain commitment. 4) Initial conceptualisation of fora: outline objectives, themes, structure, membership, cadence, and governance for the Project, Industry, and Policy Forums, to be refined in T2.3.	IACS IISLAFE CIPH ELLOK CDR ASEBIO	BEN BEN(CoL) BEN BEN BEN BEN	No

T2.2	Platform Establishment (Governance & Digital Hub)	Objective: definition of stakeholder platform a definition of the governance and methodological approach for its use, ensuring tool sustainability to support the EHDS Stakeholder Forum long term. 1) Selection of bi-directional communication tools: evaluate and choose digital solutions (collaborative portals, video conferencing, online forums), leveraging prior community experience such as Confluence in the HDAB Community of Practice and Simply CRM in EUCAIM. 2) Governance model design: define roles, responsibilities, and decision-making processes for effective forum management. 3) Sustainability assessment: analyse technical, financial, and organisational feasibility to guarantee continuity beyond the project, and propose alignment mechanisms with the EHDS Stakeholder Forum in line with EHDSR, Art. 93. 4) Integration with other WPs: ensure coordination and compliance with technical and regulatory requirements across the project.	IISLAFE MATICAL IACS ELLOK CDR DE	BEN BEN(CoL) BEN BEN BEN BEN	No
T2.3	Community Building & Fora Organisation	Objective: define, organise, and evaluate stakeholder fora to build the EHDS biotech community, delivering at least three major events per year (Projects, Industry, Policy) and additional on-demand sessions with WP leads. 1) Fora definition: design the conceptual structures—Project, Industry, and Policy Forums—specifying thematic focus, objectives, and stakeholder groups. 2) Annual event planning: co-plan concrete activities (workshops, webinars, conferences) with WP leaders to operationalise the fora, enabling cross-forum participation (e.g., legal representatives from biotech, health providers, and regulators). 3) Operational management: coordinate logistics, communications, and follow-up to ensure smooth delivery and knowledge capture. 4) Impact evaluation: measure participation, satisfaction, and outcomes to refine strategy and improve subsequent programming.	IACS ASEBIO IISLAFE Sciensano HDH CIPH CDR CHA DE	BEN BEN(CoL) BEN BEN BEN BEN BEN BEN BEN	No
T2.4	Strategic Communication and Impact	Along with the WP3, WP4 and WP9, create specific narratives to address stakeholders' profiles through the different Forums, preparing adapted communication materials to maximise the engagement. The specific activities will be the following: 1) Narrative development: Create key messages for each stakeholder group (industry, regulators, patients, etc.); 2) Tailored materials: Prepare presentations, infographics, newsletters, and digital content to foster engagement; 3) Dissemination coordination: Align communication actions (WP9) with the overall project dissemination and exploitation strategy. Support the fine-tune generation of contents for the strategic roadmap (WP3) and pathfinding activities (WP4), and 4) Impact measurement coordination: Support the activities to track reach, engagement, and perception metrics to optimise communication campaigns (WP9) and guidelines adoption monitoring (WP4).	IACS DE IISLAFE ELLOK CHA ASEBIO DE	BEN BEN(CoL) BEN BEN BEN BEN BEN	No

Milestones and deliverables (outputs/outcomes)

Milestone No	Milestone Name	WP No	Lead Beneficiary	Description	Due Date	Means of verification
MS2.1	Selection Stakeholder Platform	2	IISLAFE	Selection of the software tool to be used along the project to exchange with Stakeholders	M06	Platform available for its use online
MS2.2	Y1 Event planning	2,3,4,5,6,7,8	IACS	Listing of the events, topics and dates to be carried out in Y1	M06	Report available under request
MS2.3	Y2 Event planning	2,3,4,5,6,7,8	ASEBIO	Listing of the events, topics and dates to be carried out in Y2	M12	Report available under request

MS4	Y3 Event planning	2,3,4,5,6,7,8	ASEBIO	Listing of the events, topics and dates to be carried out in Y3			M24	Report available under request
Deliverable No	Deliverable Name	WP No	Lead Beneficiary	Type	Dissem. Level	Due Date	Description	
D2.1	Stakeholder mapping & Fora Planning	2	IACS	R	PU	M06	Report providing a comprehensive stakeholder mapping of key actors (biotech industry, AI developers, regulatory bodies, patient groups, etc.) and detailing the strategic plan for the subsequent engagement events, or 'fora'.	
D2.2	Y1 Event Proceedings	2, 7	IACS	R	PU	M12	Yearly deliverables will document the execution and outcomes of the Stakeholder Fora, highlighting collaboration, knowledge-sharing, and the challenges of using AI and health data in biotech. They will distil best practices and stakeholder input for the Strategic Roadmap and policy recommendations. Identified events: Workshop on guidelines revision, monitoring and adjustments (M9, Lead: WP5)	
D2.3	Y2 Event Proceedings	2, 7	ASEBIO	R	PU	M24	Second periodic deliverables documenting the execution and key outcomes of the organized Stakeholder Fora (covering the second year of the project). Identified events: Stakeholder Event on Strategic Roadmap & Monitoring and Evaluation Guidelines (M17, Lead: WP4), Workshop on guidelines revision, monitoring and adjustments (M21, Lead: WP5)	
D2.4	Y3 Event Proceedings	2, 7	IACS	R	PU	M32	Third periodic deliverables documenting the execution and key outcomes of the organized Stakeholder Fora (covering third year of the project) Identified events: Workshop on guidelines revision, monitoring and adjustments (M33, Lead: WP5)	

Estimated budget — Resources

See detailed budget table/calculator (annex 1 to Part B).

WP3**Work Package 3: Methodological Framework (Landscape, Roadmap, Case Study)**

Duration:	M1 – M36	Lead Beneficiary:	6 – SCIENSANO
------------------	----------	--------------------------	---------------

Objectives

The primary objectives of WP3 are to establish a robust methodological framework for understanding the current use of health data for AI in biotechnology and to define strategic pathways for scaling trustworthy, effective, and sustainable AI solutions across the EU biotech ecosystem.

- **O3.1:** Conduct a landscape analysis to map the current state of health data use for AI in biotech, identify relevant datasets and infrastructures, and assess technical, considerations to inform the design of pilots, use cases, AI Factory integration, and regulatory guidance
- **O3.2:** Develop a strategic roadmap outlining priority AI applications, deployment pathways, governance models, and sustainability mechanisms, providing actionable guidance for policymakers, industry, and research stakeholders.

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contr. & Subcontr.
			Name	Role	
T3.1	Landscape Analysis of Health Data Use for AI in the Biotech Sector	Task 3.1 will provide an overview of the current and emerging ecosystem for AI development in biotechnology. The analysis will draw on continuous feedback from stakeholders engaged via WP2 and integrate insights from other WPs, including research infrastructures (WP5), use case needs (WP6), and AI Factory capabilities (WP7). This task will build on the work in previous EU projects, such as both TEHDAS joint actions (TEHDAS1 WP5, TEHDAS2 WP4) and HealthyCloud (WP3). The task comprises two sub-tasks: Task 3.1.1 Mapping the data, infrastructures and AI Factories ecosystem for AI in Biotechnology This sub-task will map the health data, infrastructures, tools, and AI Factories relevant for AI development in the biotech sector. It will characterise datasets by availability, suitability, quality, maturity, representativeness, completeness, and alignment with AI Act requirements, while exploring how the EHDS secondary-use framework may expand future access. The semantic dimension will be assessed, including interoperability and evaluation of biomedical ontologies, vocabularies, and metadata standards to identify gaps limiting cross-border reuse. Existing European initiatives and research infrastructures providing data, compute, or services (e.g. EUCAIM, TEF-Health, EOSC, regulatory sandboxes) will be reviewed, along with emerging ecosystem elements from initiatives such as the Data Union Strategy and Apply AI Strategy (e.g. Data Labs, Simpl cloud middleware, EOSC federation, Cloud and AI Development Act). Finally, the task will examine how AI Factories can support development, benchmarking, scaling, and deployment of AI solutions, aligning their capabilities with EHDS infrastructures. The analysis will provide a structured overview of current assets, emerging capabilities, and future resources to inform the strategic roadmap in Task 3.3. Task 3.1.2 Analysis of barriers, enablers, and opportunities for AI development in biotech sector This sub-task will build on the findings of Sub-task 3.1 to identify the main barriers, limitations, enablers, and emerging opportunities affecting AI development in biotechnology. It will focus on practical challenges related to data availability and quality, interoperability, technical readiness, access to infrastructures and tooling, organisational constraints, skills gaps, and pipeline maturity, while also considering positive drivers such as new infrastructures, methodological advances, emerging governance models, and EU-level initiatives. The outcome will be a consolidated, stakeholder-validated overview of factors that hinder or support AI development, providing a direct input to the first version of the strategic roadmap in Task 3.3.	SCIENSANO USC UNILU CHARITE IACS EMPIRICA ELLOK EBRAINS DIGEU CRG CESGA POLA	BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN	No
T3.2	Case study success Criteria	This task will benchmark successful AI applications in the Biotech sector, drawing on existing efforts, initiatives and projects and the capabilities of the AI Factories. Its objective is to identify concrete technical, regulatory, and ethical success indicators that can guide the development of case studies and support the project's evaluation framework. The work will begin with the selection and analysis of a defined set of successful AI-driven Biotech implementations. Each exemplary case will be examined for its methodological proposition, technical implementation, regulatory pathway, use of research data available through different sources	BSC CHARITE HDH CRG ELIXIR CESGA	BEN BEN BEN BEN BEN BEN	No

		including the integration of Research Infrastructures under the EHDS, the use of AI Factory services, and demonstrated real-world impact. The task would pay attention to potential biases present in the proposed datasets. Based on this analysis, the task will consolidate findings into a set of actionable success criteria that describe what constitutes feasibility in practice. The outputs of this task will serve as empirical backbone for the WP6 use cases, will provide input for task 3.3, will supply WP4 with the success criteria needed to design the project's evaluation and monitoring methodology and serve as reference for further developments of the methodological toolkit in task 7.2			
T3.3	Strategic Roadmap for Scaling AI in the Biotech Sector	This task will develop a Strategic Roadmap for Scaling AI in the Biotech Sector, enabling the European biotech ecosystem to fully leverage the EHDS for access to high-quality data, software, computing environments, and AI models. Building on insights from Task 3.1, the roadmap will define priority AI applications, essential data foundations, deployment pathways, semantic interoperability and the role of emerging infrastructures, including AI Factories and research platforms. The roadmap will define clear responsibilities and actions for all stakeholders (policymakers, research infrastructures, industry actors, SMEs, and innovation hubs) highlighting how their contributions, interactions, and synergies collectively address gaps, unlock resources, and achieve the overarching objectives. It will also emphasize the strategic role of AI Factories, illustrating how these hubs can provide integrated pipelines, benchmarking environments, tooling, and lifecycle management for AI models while fostering collaboration with data providers, infrastructures, and industry actors. The output will be actionable, stakeholder-specific recommendations, with short-, medium-, and long-term actions guiding coordinated development, deployment, and scaling of AI solutions. The roadmap will be developed iteratively, with structured feedback from all relevant WPs and WP4 on monitoring, ensuring continuous refinement based on pilot results, use cases, AI Factory integration.	SCIENSANO CERTH USC UNILU LNDS CHARITE ELLOK EBRAINS DE HDH IACS POLA CRG CESGA	BEN BEN BEN BEN COO BEN BEN BEN BEN BEN BEN BEN BEN BEN	No

Milestones and deliverables (outputs/outcomes)

Mile-stone No	Milestone Name	WP No	Lead Beneficiary	Description				Due Date	Means of verification
MS3.1	Initial Strategic Roadmap for Scaling AI in the Biotech Sector	3	SCIENSANO	First version of the Strategic Roadmap based on the landscape analysis and early stakeholder engagement, providing a preliminary overview of proposed actions, priority datasets, key AI applications, deployment pathways, and governance recommendations, serving as the foundation for iterative refinement through monitoring and evaluation activities (WP4)				M16	Report outlining the strategic roadmap, its main focus and key actions.
MS3.2	Case studies	3	BSC	Milestone reporting on the evaluated successful AI applications in the Biotech sector, drawing on existing projects and AI Factory available services. It will outline preliminary success criteria for technical feasibility, and real-world impact, providing early empirical input for task 3.3.				M12	Report listing successful case studies on AI applications in Biotech
Delive-rable No	Deliverable Name	WP No	Lead Beneficiary	Type	Dissem. Level	Due Date	Description		

D3.1	Landscape Analysis of Health Data Use for AI in the Biotech Sector	3	SCIENSANO	R	PU	M12	Comprehensive report mapping current and emerging datasets, infrastructures, and initiatives for AI in biotech. It will assess dataset suitability, identify barriers and enablers, and include a stakeholder-validated catalogue of high-priority datasets to inform pilot implementations, use cases, AI Factory integration, and the Strategic Roadmap.
D3.2	Strategic Roadmap for Scaling AI in the Biotech Sector	3	SCIENSANO	R	PU	M36	Final Strategic Roadmap providing an actionable plan for policymakers, industry, and research organisations. It will define priority AI applications, deployment pathways, governance structures, and integration with EHDS infrastructures and AI Factories, offering policy guidance and a sustainability plan to support long-term adoption and impact.

Estimated budget — Resources

See detailed budget table/calculator (annex 1 to Part B).

Work Package 4

Work Package 4: Pathfinder System (Monitor and Evaluate Progress)

Duration: M1 – M36 **Lead Beneficiary:** 16 - CHARITE, 43 - EPIDATA

Objectives

- Develop Self Evaluation Tools and Operating Principles and Procedures (“Pathfinder system”)

▪ Validate and Iterate Pathfinder system

With the input of the **Strategic Roadmap** that builds on the landscape analysis, this work package will develop **self-evaluation tools, operating principles & procedures**, and **navigation tools** to allow all stakeholders involved to **identify** their current state in regard to the roadmap and effective next steps to move forward to **realization of the strategic roadmap objectives, in full alignment of the EHDS blueprint digital capabilities**.

The tools developed here will be **interactive and informative**, allowing an **iterative** approach to adopting advanced operating principles and procedures. We foresee a **high degree of collaboration across work packages** as we intend to receive input from WP2 (Stakeholders), WP3 (Strategic Roadmap), and seek to **validate our approach** with the Piloting Integration of Data Infrastructure into EHDs (WP5), Use Cases (WP6), and AI Factories (WP7). To maximize the applicability of these tools, we will model them according to different stakeholder needs.

O4.1: Develop the self-evaluation tools and operating principles and procedures, with guidelines on how to implement them. This comprises the Pathfinder system complementing the Strategic Roadmap.

O4.2: Validate and Iterate the Pathfinder system by practical application in the project and by a constructive exchange with project-external representatives of public service and industry.

Cross WP Interaction:

Collaborate with WP2 for the industry / public services stakeholder forum on evaluation and monitoring guidelines

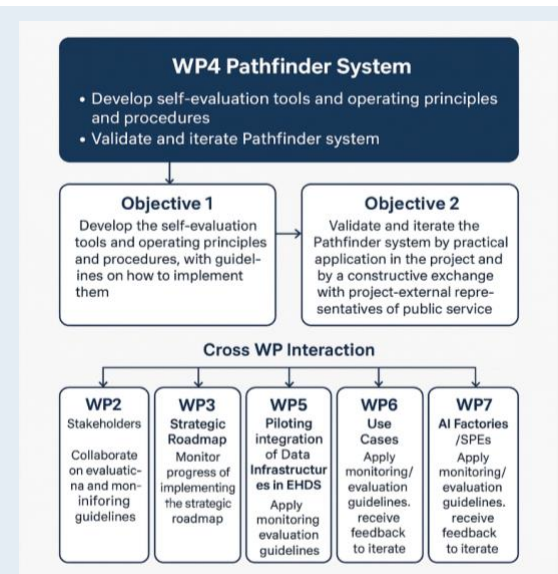
Work closely with WP3 on how to monitor the progress of implementing the strategic roadmap, within (as pilot) and beyond the project (for EC to use for future initiatives).

Support WP5 applying the Pathfinder system to Piloting integration of Data Infrastructures in EHDS and receive feedback to iterate.

Support WP6 applying the Monitoring and Evaluation Guidelines to the Case Studies and receive feedback to iterate.

Support WP7 applying the Monitoring and Evaluation Guidelines to SPEs on AI Factories and receive feedback to iterate.

Work with WP8 to ensure regulatory framework is properly reflected in Monitoring and Evaluation Guidelines and receive feedback to iterate.



Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contr. & Subcontr.
			Name	Role	
T4.1	"Pathfinder system v1": Draft Monitoring and Evaluation Guidelines	The Pathfinder Methodology builds upon the Landscape Analysis from WP3 and the Stakeholders identified in WP2 to make sure all stakeholders involved are provided with the means to identify where on the roadmap they are located and to find the tools and support to move along the roadmap. This means that we will build a comprehensive set of self-evaluation tools for locating yourself on the roadmap and a set of guiding principles to advance yourself per stakeholder. This Pathfinder Methodology will be tested, iterated, and improved by applying the first version to the use cases in WP6 and the Health Infrastructures and AI Factories in WP5 and 7. See Task 4.2 Close collaboration is foreseen with WP8 to make sure that the pathfinder tools are "road legal", i.e., regulatory constraints are considered at every recommendation the Pathfinder tools provide. <i>Addresses SO5 Produce Recommendations and Guidelines</i>	EPIDATA BSC IACS ELIXIR CERTH	BEN BEN BEN BEN BEN	Yes (subcontract the development of pathfinder tool)
T4.2	"Test Drive":	In each of WP 5, 6 and 7, we will identify one Health Infrastructure and one AI factory to validate the Pathfinder methodology. In collaboration with these participants, we will ensure that the Pathfinder	CHARITE	BEN	No

	Validate Monitoring and Evaluation Guidelines against Pilot RIs and SPEs on AI Factories	methodology will be generally applicable. Involved partners: - AI Factory: LNDs - Health data Infrastructure: ELIXIR and cancer data infrastructure (i.e. national cancer data nodes) <i>Addresses SO4 Test and Validate Real-World Use Cases, Conduct Case Studies</i>	ELIXIR EPIDATA LNDs	BEN BEN COO	
T4.3	“Pathfinder system v2”: Finalize Monitoring and Evaluation Guidelines	The original Pathfinder system will be revised after receiving three-fold feedback from practical implementations within the project. This revision will be shared with the ancillary Work Packages (WP 2,3,8) for further input to produce the final Pathfinder system, as presented in D4.2- Revise and publish Guidelines with input from WP 2,3,5,6,7,8. Enrich with examples from T4.2. <i>Addresses SO5 Produce Recommendations and Guidelines</i>	CHARITE EPIDATA BSC IACS SCIENSANO	BEN BEN BEN BEN	No

Milestones and deliverables (outputs/outcomes)

Mile-stone No	Milestone Name	WP No	Lead Beneficiary	Description	Due Date	Means of verification
MS4.1	Draft Monitoring and Evaluation Guidelines	4	EPIDATA	Develop the monitoring and evaluation guidelines, signposting tools that allow stakeholder specific perspectives on the Strategic Roadmap, and signals next possible steps to improve technical, operational, and regulatory positioning in regard to the Strategic Roadmap.	M18	Report on the monitoring and evaluation guidelines is ready under request
MS4.2	Stakeholder Event on the Strategic Roadmap & Monitoring and Evaluation Guidelines	2,4	EPIDATA	Jointly with WP2 to develop and Establish a Multistakeholder Platform and Stakeholder Forum (SO5)	M16	Report on the stakeholder forum plan with monitoring/evaluation guidelines' perspectives incorporated.
MS4.3	Evaluation Report on Use Case Studies and RIs and AI Factories	4	CHARITE	This report captures the results of the validation done in Task 4.2 in one chapter for each validation partner involved in T4.2	M24	Report on the learnings from use cases and AI factors is ready under request.
MS4.4	Final Monitoring and Evaluation Guidelines	4	CHARITE	Final integration of learnings from use cases/AI factors into the pathfinder V2	M33	Report is ready per request.

Deliverable No	Deliverable Name	WP No	Lead Beneficiary	Type	Dissem. Level	Due Date	Description
D4.1	Pathfinder System V1/demo	4	EPIDATA	(R+DEM)	SEN	M20	First version report/Demo of the Pathfinder system
D4.2	Stakeholder Event Report (including 1 st set of Guideline Revisions)	4	EPIDATA	R	PU	M18	Meeting with internal and external stakeholders: Presenting Roadmap and Evaluation Guidelines, collect Feedback
D4.3	Pathfinder Assessment	4	CHARITE	R	SEN	M27	Report on applying Pathfinder system on project activities (see T4.2)
D4.4	Finalized Guidelines	4	CHARITE	(R+DEM)	PU	M33	Implements chapters derived from Case Study Assessment and stakeholder event (MS4.2), reviewed by additional WPs (see T4.3)

Estimated budget — Resources

See detailed budget table/calculator (annex 1 to Part B).

WP5

Work Package 5: Pilot Integration of Data Infrastructures in EHDS

Duration: M1-M36 **Lead Beneficiary:** 9 - ELIXIR

Objectives

- **O5.1:** Definition, monitoring and implementation of the guidelines for the integration of Data infrastructures into EHDS (T5.1)
 - **O5.1.1:** Produce recommendations to maximise the integration and alignment on Governance and Operations (T5.2)
 - **O5.1.2:** Produce recommendations to maximise integration and alignment of Data provisioning (T5.3)
 - **O5.1.3:** Produce recommendations to maximise integration and alignment of Processing and AI Training (T5.4)
- **O5.2:** Coordinate data infrastructures involvement on biotech use cases (T5.5)

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contr. & Subcontr.
			Name	Role	
T5.1	Guidelines for integration of Data Infrastructures into EHDS	Definition and coordination of the initial guidelines for the integration of Data Infrastructures in EHDS version, monitoring, revision and adjustments Outreach and onboarding of additional data infrastructures on governance, operational and technical alignment with EHDS in close collaboration with WP2 to support and promote these activities <u>Participants role:</u>	ELIXIR LNDS SCIENSANO IISLAFE CRG BSC	BEN COO BEN BEN BEN BEN	No

		ELIXIR and LNDS will coordinate the different versions of the guidelines for the integration of Data Infrastructure into EHDS (see deliverables and milestones tables) starting by the initial guidelines to pilot the integration of Data Infrastructures in EHDS, and following with the yearly revisions and alignment with WP8 on legal guidelines SCIENSANO will contribute to the alignment with EHDS contributing inputs from TEHDAS2 discussions and open consultations IISLAFE, CRG, BSC, IACS EBRAINS, CIPH, BHDA, UT and CHARITE will contribute with the views towards HDABs, Distributed infrastructures and to ensure the outputs are fit for purpose.	IACS EBRAINS CIPH HDA UTARTU CHARITE	BEN BEN BEN BEN BEN BEN	
T5.2	Integration of Data Infrastructures into EDHS: Governance and operations	This task will implement the guidelines to strengthen the governance and operational integration of Data Infrastructures within the EHDS. The focus will be on clarifying governance roles, aligning governance models with the EHDS framework, and improving operational processes related to data findability and accessibility. Work will be carried out in close collaboration with WP8 to ensure regulatory coherence and alignment with the legal guidelines. The work will be informed by an in-depth exploration of: • Governance roles research infrastructures within the EHDS, building on work initiated under the TEHDAS2 Joint Action • Concrete governance models for the integration of (research and data infrastructures, including the roles of RIs, EHDS bodies (e.g., HDABs), and the EHDS Board • Operational aspects of data findability and accessibility, including catalogue management and coordinated handling of data access requests	SCIENSANO IISLAFE ELIXIR LNDS CRG Health-RI EBRAINS CIPH HDA UTARTU GÖG HDH	BEN BEN BEN COO BEN BEN BEN BEN BEN BEN BEN BEN	Yes, IISLAFE
T5.3	Integration of Data Infrastructures into EDHS: Data provisioning	Data provisioning through EHDS requires integration on the EHDS blueprint of digital capability workflow of Data Infrastructures, EHDS regulation and through coordinated pipeline, in order to preserve participants' privacy and maximise efficiency. This task will take care of all the aspect related to data provision and feed the results into the collective WP5 guidelines and thus to the iteration of the guidelines delivered in D51- D5.4. Each of the HDAB and DIs partners will contribute to reaching an exhaustive agreement. Those aspects include: • Data standards • Harmonization Data models used for data provision and transfer (OMOP CDM, or a different agreed one) • Data discovery (enabled at different level, with maximization of cross-reference and data linkage. Including HealthData@EU catalogue) • Data pre-processing (preparation of data before transfer to AI factories) • Quality control, using labels developed in Quantum project • Application of data minimisation principle • Data transfer to AI Factories. This task will be Informed by the piloting of: • Data infrastructures catalogue into EHDS • Data provision to users for federated analytics • Datasets quality, and utility labels for Data Infrastructures • HDABs and DIs data transfers to AI Factories for the training of computationally intensive AI models • HDABs and DIs data provision to users through SPEs	CRG Health-RI ELIXIR LNDS SCIENSANO IISLAFE BSC IACS CNR EBRAINS USC CIPH BHDA BioData.pt MUNI BAHIA UTARTU CHARITE GÖG UNILU	BEN BEN BEN COO BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN	No

		Use cases in WP6 Also, when possible, this task will tailor the infrastructures according to domain community's needs, e.g. by connecting with UNCAN.eu platform (the central platform and national cancer data nodes). Finally, the task will take care of making datasets from infrastructure into the EHDS catalogue of services. Participants role: CRG and HRI will organize structured working sessions to gather consensus from partners and produce a framework of collected recommendations for the inclusion of the aspects mentioned into the guidelines (D5.5). All the partners will contribute to the framework with their national and domain expertise. ELIXIR, LNDS, SCIENSANO, LaFE, BSC, IACS, CNR, EBRAINS, USC, CIPH, BHDA, BioData.pt, MUNI, BAHIA, UT, CHARITE, GÖG, UNILU bring critical expertise either representing HDABs or Data Infrastructures or by bringing technical expertise for the provisioning of data covering data discovery (dataset, record level, linkage), data provisioning (making data available following agreed data and semantic models via SPEs) and labeling (data quality and utility labels).			
T5.4	Integration of Data Infrastructures into EDHS: Processing and AI Training	This task will develop a set of recommendations to maximise the integration of computational sites with Data Infrastructures. Specifically, this task will define a data processing framework to allow datasets and models to be used consistently across HDABs and Data Infrastructures. This task will include common interfaces and shared processing standards to enable AI building, refinement and evaluation to run seamlessly, regardless of where data is stored or computation takes place. The work on this framework will be carried out in close collaboration with WP7 to ensure alignment with the infrastructure developments. In parallel, the task will advance the provisioning of AI Factory services within Data Infrastructures, ensuring that Secure Processing Environments (SPEs) and resource allocation mechanisms are deployed and managed efficiently. The guidelines will be informed by piloting activities that test the deployment of AI Factory Secure Processing Environments within HDABs and Data Infrastructures, the operation of the data processing framework across these environments, and the execution of AI training workflows using real data and models. The insights gained from these pilots will be used to refine and validate the recommendations, ensuring that they are technically robust, scalable, and directly applicable to health data research. Participant's role: BSC and LaFE will co-coordinate this task connecting to the activities taking place in WP6. ELIXIR, LNDS, SCIENSANO, CRG, IACS, CNR, EBRAINS, USC, CIPH, BHDA, BioData.pt, MUNI, BAHIA, UT, HDH bring critical expertise either representing HDABs or Data Infrastructures or by bringing technical expertise for the deployment of federated processing frameworks that support both analytics and learning needs.	BSC IISLAFE ELIXIR LNDS SCIENSANO CRG IACS CNR EBRAINS USC CIPH BHDA BioData.pt MUNI BAHIA UTARTU HDH	BEN BEN BEN COO BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN	No
T5.5	Piloting data infrastructure involvement in biotech use cases implementation	Coordination of data infrastructures involvement in biotech use cases to analyze the roadblocks and inefficiencies found and propose approaches to tasks 5.1 to 5.4 to produce and adjust recommendations, in close collaboration with WP6. Participant's role: ELIXIR and EBRAINS will coordinate this task with LNDS, LaFE, Health-Ri, UT, contributing to the discussion representing HDABs and Data Infrastructures involved. HDH will play a critical role also in connecting to and representing the Use Cases and their needs to improve the provision of data or processing.	ELIXIR EBRAINS LNDS LaFE HRI UTARTU HDH	BEN BEN COO BEN BEN BEN BEN	Yes, IISLAFE

Milestones and deliverables (outputs/outcomes)

Mile-	Milestone Name	WP No	Lead	Description	Due	Means of verification
-------	----------------	-------	------	-------------	-----	-----------------------

stone No			Beneficiary				Date	
MS5.1	Initial WP5 workshop to scope and define guidelines	5	ELIXIR	Preparation of the initial guidelines with presentation from EHDS, GDI, EUCAIM, EBRAINS, UNCAN, FEGA			M03	Workshop announce and news in project website
MS5.2	WP5 face to face workshop Y1	5	LNDS	Guidelines revision, monitoring and adjustments			M09	Workshop announce and news in project website
MS5.3	WP5 face to face workshop Y2	5	ELIXIR	Guidelines revision, monitoring and adjustments			M21	Workshop announce and news in project website
MS5.4	WP5 face to face workshop Y3	5	LNDS	Guidelines revision, monitoring and adjustments			M33	Workshop announce and news in project website
MS5.5	Datasets	5	CRG	Data provision framework			M18	Datasets available for provision from DIs
Delive- rable No	Deliverable Name		WP No	Lead Beneficiary	Type	Dissem. Level	Due Date	Description
D5.1	Integration of data infrastructures into EHDS: Practical Guidelines		5	ELIXIR	R	PU	M5	Initial version of the guidelines for the integration of Data Infrastructures into EHDS. This will build on the legal guidelines from WP8 over the course of the project via several iterations in the later updated versions.
D5.2	Integration of data infrastructures into EHDS: Practical Guidelines revision 1		5	LNDS	R	PU	M12	First guidelines evaluation and revision, including updated recommendations
D5.3	Integration of data infrastructures into EHDS: Practical Guidelines revision 2		5	ELIXIR	R	PU	M24	Second guidelines evaluation and revision, including updated recommendations
D5.4	Integration of data infrastructures into EHDS: Practical Guidelines revision 3		5	LNDS	R	PU	M36	Third and final guidelines evaluation and revision, including updated recommendations
D5.5	Framework for Data provision from HDBA and DIs to AI factories		5	HRI	R	PU	M18	Initial version of a framework for Data provision from HDBA and DIs to AI factories. This will include details about data standards, data models, tools for data pre-processing, and data transfer agreed among partners.
D5.6	Data Infrastructures into EDHS: Practical Framework for Governance and Operations		5	SCIENSANO	R	PU	M30	Framework presenting recommendations, guidance, and governance models for the integration of data infrastructures into the EHDS. Building on research into governance roles, operational processes, and alignment with the EHDS framework, the document outlines clarified roles for research infrastructures and EHDS bodies, proposes concrete governance models, and provides operational guidance for improving data findability, catalogue management, and coordinated data access processes. This will complement the legal guidelines from WP8 from a practical and operational

							perspective.
D5.7	Frameworks for Federated Processing across HDABs and Data Infrastructures	5	BSC	R	PU	M30	Identify and describe the approaches to facilitate the procession of data for analytics and learning across HDABs and Data Infrastructures.
D5.8	Lessons learnt from use cases for the integration of Data Infrastructures into EHDS	5	ELIXIR	R	PU	M30	Catalogue of use cases implementation issues related to the provision of data and compute from Data Infrastructures and HDABs, and recommendations to tackle them

Estimated budget — Resources

See detailed budget table/calculator (annex 1 to Part B).

Work Package 6

Work Package 6: BioTech Use Cases

Duration: M1-M36 **Lead Beneficiary:** 2 - HDH

Objectives

The primary objective of WP6 is to **serve as the project's testing and learning engine**, driving real-world validation of project outputs and ensuring compliance with EHDS standards for the biotech sector. To ensure a structured approach, the execution will divide the UCs into two distinct groups: **tooling-infrastructure UCs**, focusing on validating the essential access and computational systems, and **biotech-application UCs**, focusing on clinical and industrial value.

- **O6.1:** Manage the execution and implementation of all UCs, monitor and report on their progress and support resolution of challenges. (T6.1)
- **O6.2:** Collect lessons learnt from UCs and provide essential feedback to WPs 5, 7, and 8. (T6.2)
- **O6.3:** Select additional use cases to reinforce biotech participation and develop, test and implement additional solutions for AI in biotech. (T6.3).
- **O6.4:** Implement all UCs and attain individual objectives and maximise added value of each use case. (T6.4 to T6.11)

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contr. & Subcontr.
			Name	Role	
T6.1	UC operational management and external knowledge transfer	This task focuses on the day-to-day operational management of all UC activities. It involves tracking progress against milestones (using WP4 monitoring tools), coordinating resources, and consolidating execution results. The main goal is to prepare validated performance data and success stories for external communication . This knowledge is transferred as real-world narratives to WP2 to inform stakeholder engagement and provided as high-impact content to WP9 for project dissemination.	HDH CRG ELIXIR	BEN BEN BEN	No
T6.2	Internal refinement feedback loop	This task focuses on driving systemic improvement by managing continuous exchange with development WPs . It specifically channels operational data integration challenges, and technical requirements to WP5 (EHDS Infrastructure) and WP7 (AI Factories), ensuring their services are refined	HDH CRG UC Leads	BEN BEN BEN	No

		based on practical UC needs. Furthermore, it forwards ethical dilemmas and regulatory obstacles encountered to WP8 (Regulatory Clarity), ensuring the project provides concrete evidence to clarify compliance requirements. The output is a consolidated report of necessary lessons learnt for consortium-wide improvement.	WP Leads	BEN	
T6.3	Open call for selecting additional use cases	This task will organize an open call to select additional use cases. The rationale for this is to a) reinforce biotech interest and participation in the project and b) test additional services, tools or ideas that have come up in the initial stage of the project and c) to be able to adapt to the rapidly evolving field. The call aims to select a total of 3 additional use cases to be funded as part of the project. The open call will comply with the requirements for providing such support (e.g. the call will be open for at least 2 months, published widely, conform to EU standards, and have a clear European dimension). The task lead will draft eligibility criteria, the overall call text and the proposed detailed call procedure, including an eligibility check and expert review of eligible use case proposals. All WP leaders will review the draft text for the open call and contribute to the nomination of the reviewers evaluating the resulting use case proposals. The fund distribution is described in Section 6.	HDH LNDS WP Leads	BEN COO BEN	No
T6.4	UC1 – Generating synthetic sample for early access through EHDS metadata catalogue	Implement UC1 (see Section 2.1 for the description of the use cases)	HDH ALIA SCIENSANOC UNIC CHU Bordeaux Institut Curie	BEN BEN BEN BEN BEN BEN	Yes
T6.5	UC2 - Enabling the secondary use of data in HPC environment according to EHDS specifications	Implement UC2 (see Section 2.1 for the description of the use cases)	Findata/THL	BEN	Yes
T6.6	UC3 – Exploring cross-border Federated Learning for secondary data use	Implement UC3 (see Section 2.1 for the description of the use cases)	Findata/THL HDH	BEN BEN	Yes
T6.7	UC4 – GenAI-Powered	Implement UC4 (see Section 2.1 for the description of the use cases)	BATEA	BEN BEN	Yes

	Virtual Patients for Optimising Early-Phase Clinical Trials in Glioblastoma		BAHIA VHIO IISLAFE	BEN BEN BEN	
T6.8	UC5 – AI – driven prevention, diagnostic and management for delirium in ICU patients	Implement UC5 (see Section 2.1 for the description of the use cases)	CLIN MCN	COO BEN	No
T6.9	UC6 - Polygenic Risk Scores	Implement UC6 (see Section 2.1 for the description of the use cases)	SCIENSANO ULIEGE IACS Curewiki	BEN BEN BEN BEN	No
T6.10	UC7 - Generating high-fidelity synthetic medical images for AI development and evaluation	Implement UC7 (see Section 2.1 for the description of the use cases)	INRIA INSA LYON	COO BEN	No
T6.11	UC8 - Generating synthetic control arms to accelerate rare disease trials	Implement UC8 (see Section 2.1 for the description of the use cases)	INRIA HDH SANOFI	BEN BEN BEN	No

Milestones and deliverables (outputs/outcomes)

Mile-stone No	Milestone Name	WP No	Lead Beneficiary	Description	Due Date	Means of verification
MS6.1	Formal launch of all use cases through Use case framing and kick-off meetings	6	HDH	Formal launch and execution start of the UC (T6.3 to T6.9).	M4	Framing notes and KO meeting presentations by all use cases.

MS6.2	Launch of an open call to select additional use cases	6	HD		Publication of an open call to select additional use cases to strengthen biotech participation, relying on initial work across technical WPs.		M6	Call published	
MS6.3	Uptake and exploitation readiness		6	HDH		Interim report on lessons learnt and feedback to other WPs		M24	Preliminary version of D6.2
Delive- rable No	Deliverable Name	WP No	Lead Beneficiary	Type	Dissem. Level	Due Date	Description		
D6.1	UC framing report	6	HDH	R	PU	M09	This report documents the initial phase of Use Case execution, focusing on methodological groundwork and preparatory activities, compiling relevant information from the use case framing notes and mapping out expected contributions to other WPs.		
D6.2	Open call report and use case selection	6	HDH	R	PU	M12	Report on the results of the open call and presentation of the selected use cases to be funded, including a description of the selected projects, award dates, project durations, and final recipient legal names and countries.		
D6.3	Final UC report	6	HDH	R	PU	M36	This final report provides the validation results of all UC executed throughout the project. It includes the final assessment against pre-defined KPIs, detailed lessons learnt concerning technical interoperability (WP5/7) and regulatory hurdles (WP8).		

Estimated budget — Resources

See detailed budget table/calculator (annex 1 to Part B).

WP7**Work Package 7: Integration of AI Factories with EHDS (computation and services)**

Duration:	M1-M36	Lead Beneficiary:	8 - BSC
------------------	--------	--------------------------	---------

Objectives

- **07.1:** Map existing capacities, challenges and proposed solutions for enabling secure processing in the context of the EHDS (T7.1)
- **07.2:** Definition of the technical requirements for AI models construction, validation and deployment. (T7.2)
- **07.3:** Enable the use of SPEs and AI Regulatory Sandboxes to support compliant, secure AI across the network (T7.3)
- **07.4:** Deliver a roadmap for a federated network of AI Factories to support biotechnology use cases. (T7.4)

Activities and division of work (WP description)

Task	Task Name	Description	Participants	In-kind Contr.
------	-----------	-------------	--------------	----------------

No			Name	Role	& Subcontr.
T7.1	Critical evaluation of secure technologies for processing Health Data	This task evaluates the technological and operational requirements for secure processing of Health Data in the EHDS. Different models of processing, e.g. centralized, federated, and hybrid, would be examined and discussed to ensure data can be processed across Europe. To this end, this task will build on existing efforts (such as TEHDAS2 and EOSC-ENTRUST) to identify and catalogue existing SPEs across Europe by analyzing and comparing their technical and functional capacities and security conditions with respect to EHDS regulatory requirements, focusing on the aspects relevant for AI. This effort should contribute to identifying potential gaps and misalignments and propose solutions to address them, explicitly including AI Factories and Antennas. The task should provide a set of recommendations and guides on how to securely process data following the progressive implementation of the EHDS regulation.	LNDS IACS SCIENSANO BSC CIPH MU	COO BEN BEN BEN BEN	No
T7.2	Technical and methodological toolkit for building AI models in the EHDS	This task develops a common and practical toolkit that guides how AI models are built, trained, validated, deployed, and maintained within EHDS-compliant infrastructures. To ensure its applicability, it builds parallel efforts across European Research Infrastructures, the European Open Science Cloud, and other community-driven initiatives and projects. Joint work with case studies would be prioritized to ensure the applicability of the developments carried out in the context of this task. The toolkit will include: • Definition of a minimum set of core elements across computational infrastructures, such as common interfaces, standards, and security and computational requirements. • Support for the onboarding of AI factories and Antennas by providing clear practices and documentation requirements that ensure their consistent integration into secure computational infrastructures. Importantly, such support will follow a progressive model to cover different technological scenarios and needs. • Methods for data quality and bias assessment, ensuring accuracy and reliability, and including provenance and bias mitigation checks. • AI explainability methods, offering guidelines and tools for trustworthy AI suitable for health and research contexts. • AI Model Assessment framework, covering performance benchmarking and evaluation, and ethical and regulatory compliance.	CERTH HDH BSC CESGA BAHIA MU	BEN BEN BEN BEN BEN BEN	No
T7.3	Use and reuse of Health Data through secure environments in the context of AI factories and EHDS	This task operationalises health data use and reuse within secure processing environments outlined in task 7.1. The task will include the connection with European Research Infrastructures as complementary providers of research data. The focus would be on the use of AI Factories and Antennas as the main computational reference framework. However, it will not exclude alternative sites for securely processing health-related data. To ensure applicability and facilitate gaining real world experience, this task includes: • Establishing maturity levels of infrastructures based on specific case-study requirements (data type, model complexity, security needs). • Distinguishing between generic SPEs, specialized SPEs and Regulatory AI sandboxes, with criteria for selecting appropriate environments.	BSC LNDS HDH IACS ELIXIR CIPH CHARITE GOEG MoH-ES	BEN COO BEN BEN BEN BEN BEN BEN	No

		- Defining a minimal technical setup required for any AI application, covering computational needs, security controls, and regulatory aspects to ensure alignment with EHDS. - Proposing good practices for onboarding case studies, including documentation and risk evaluation assessments.			
T7.4	Technical solutions for bringing forward use-cases and case-studies	This task validates the technical solutions developed in T7.1-T7.3 by applying them to WP6 use-cases and partner clinical case studies, including secure data transfer, analysis federation, and secure processing environments. The task coordinates secure connections to AI factories and orchestrates deployment of federated analysis capabilities, testing these solutions in operational settings. This enables collaborative AI model development on distributed sensitive health data while maintaining data sovereignty within EHDS. Lessons learned will refine the toolkit for broader integration.	MU CESGA BSC ELIXIR	BEN BEN BEN BEN	No

Milestones and deliverables (outputs/outcomes)

Milestone No	Milestone Name	WP No	Lead Beneficiary	Description			Due Date	Means of verification
MS1	Stakeholder validation workshop	WP7	CERTH	Workshop with AI Factory, Antenna, and EHDS stakeholders to validate SPE mapping and initial toolkit concept			M16	Workshop announced and news available on the project website.
Deliverable No	Deliverable Name		WP No	Lead Beneficiary	Type	Dissem. Level	Due Date	Description
D7.1	SPE mapping and evaluation report		WP7	LNDS	R	PU	M18	Comprehensive inventory and assessment of SPEs, identifying gaps, capacities and alignment with EHDS specifications
D7.2	AI development toolkit and data quality guidelines		WP7	CERTH	R	PU	M24	Consolidated toolkit for building EHDS-compliant AI models, covering onboarding, interoperability, data quality, bias, model assessment, and explainability
D7.3	Framework for health data use and reuse		WP7	BSC	R	PU	M30	Framework for secure health data use and reuse, defining maturity levels, minimal setup, interoperability, and good practices for case-study deployment
D7.4	Lessons learned and final recommendations		WP7	MU	R	PU	M36	Final report with validated technical solutions, good practices, and recommendations for EHDS integration

Estimated budget — Resources

See detailed budget table/calculator (annex 1 to Part B).

Work Package 8

Work Package 8: Regulatory Clarity and Ethics

Duration:	M1-M36	Lead Beneficiary:	17 – TMF
------------------	--------	--------------------------	----------

Objectives

- **O8.1:** Identify legal/regulatory issues slowing down the development and marketing process of biotech products
- **O8.2:** Explore the roles for data infrastructures provided by the EHDS Regulation in detail from the legal requirements perspective and provide guidance for the infrastructures themselves, the data providers, HDABs and national legislators as well as AI factories, regarding how they need to prepare and to address the complex regulatory landscape for bringing biotech products to the market; come up with concrete guidance and recommendations for the legislator regarding the up-coming Biotech Act
- **O8.3:** Explore sandboxes as means to speed up the development process
- **O8.4:** Demonstrate how the EHDS infrastructure (HealthData@EU) helps to comply with regulatory requirements and to provide guidance to HDABs what services are expected in this respect
- **O8.5:** Discuss with patients and industry from the ethical perspective minimum IP requirements regarding AI models and trade secrets regarding health data to foster investment in development of biotech products balanced with public interest
- **O8.6:** Support the other WPs, especially the use cases in their compliance with ethical standards and approvals

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contr. & Subcontr.
			Name	Role	
T8.1	Regulatory problem inventory	Industry regulatory representatives from use cases plus relevant professional associations put together the main hurdles in their daily practice regarding data access as well as regulatory compliance, so that the tasks 8.2, 8.3 and 8.4 can develop a more concrete workplan focused on the actual issues. The partners will conduct different online workshops and run surveys with different stakeholders to come up with the inventory.	DE TMF LNDS IACS ALIA CLIN BAT	BEN BEN COO BEN BEN BEN BEN	No
T8.2	The legal integration of existing and future data infrastructures in the EHDS	This task will determine what the appropriate roles the EHDS regulation provides for existing and upcoming data sharing infrastructures are and how the infrastructures themselves, the data providers (data holders), HDABs and national legislators and AI factories need to prepare in legal terms. Thus, it will describe the data access procedure from the application to the access decision and the data delivery for different roles different infrastructures might choose. It will build on the work already done within TEHDAS2 and in UNCAN-Connect. In addition, it will build on the landscape analysis done in WP3. Subtask 8.2.1 will discuss the implication of the EHDS regulation with the infrastructures themselves, categorise them along their legal status (international organisation, private or public entity under national law) and analyse their statutes and data sharing policies to provide guidance how to adapt. Subtask 8.2.2 will build on 8.2.1 and explore under which conditions the data providers (data holders) are able to share data through infrastructures taking the EHDS regulation into account and how useful this is for data users, especially the industry partners. Contractual clauses for data sharing agreements with respect to	TMF LNDS IACS CRG HDA CERTH HDH FINDATA GÖG CLIN DE	BEN COO BEN BEN BEN BEN BEN BEN BEN BEN BEN BEN	No

		the EHDS regulation will be provided. Subtask 8.2.3 will explore with HDABs what national legislators needs to provide for in terms of HDAB establishment to complete the integration of infrastructures. Subtask 8.2.4 will develop with existing AI factories how they can provide SPE Services from the legal perspective building on the requirements set up in TEHDAS2. The data user needs will be integrated. All subtasks will integrate the user perspective provided by the industry partners. The result will be a guideline about how to prepare for connecting to EHDS along the lines of possible choices and its legal consequences in terms of effort and cost. The guideline will include how to phrase certain clauses in the statutes of the infrastructure to be fit for the chosen role. A checklist for the other stakeholders will be included, assessed by the industry with a user perspective. The results will constantly inform WP5 and WP7. A workshop with all stakeholders will be held in Brussels to discuss the draft guideline in collaboration with WP2.			
T8.3	Clarifying the regulatory framework for bringing products to the market	This task will examine the European legal, regulatory, and policy framework concerning the AI-driven biotechnology product development lifecycle. Special emphasis will be placed on clarifying the alignment between the EHDS Regulation and other components of the overall legal/regulatory framework, including the AI Act, IVDR/MDR, the Health Technology Assessment Regulation, and the key forthcoming legislative pieces (e.g., the Biotech Act), among other components. The task will follow a stepwise, phased approach to analysing the applicable framework, allowing for the scope of this analysis to adapt in response to the evolution of the legislative changes. The task will consist of three sub-tasks: <i>T8.3.1. Preliminary legal and regulatory landscape mapping.</i> This sub-task will provide a high-level overview of the current and forthcoming EU laws and regulations pertaining to various aspects of AI-enabled biotech innovation lifecycle. While not a comprehensive analysis, this preliminary overview serves the practical purpose of enabling regulatory requirement mapping for the Use Cases and Case Studies during the first year of the project. Additionally, this preliminary overview will lay the groundwork for a more comprehensive analysis to be carried out in the next sub-task. <i>T8.3.2. Interplay of the EHDS Regulation with the laws, regulations, and EU-level policy frameworks pertaining to AI-driven biotech innovation.</i> This sub-task will provide regulatory clarity around the interplay between EHDSR concepts (such as Data Holders, HDABs, SPEs, data and metadata requirements), and the actors, roles, and processes defined by the laws/regulations pertaining to AI-driven biotechnology innovation. Potential synergies, overlaps, and interdependencies among the relevant frameworks will be highlighted, pinpointing opportunities for improved efficiencies and enhanced cooperation by different actors across the relevant frameworks. <i>T8.3.3. Regulatory compliance path throughout the biotechnology innovation lifecycle.</i> Building primarily on the outputs of the previous sub-task, T8.3.3. will outline detailed user journeys along the biotech innovation lifecycle. These illustrative user journeys will capture the roles and duties of the key actors in the ecosystem, including industry players, infrastructure providers, and regulators. When designing the user journeys, this sub-task will also draw on the insights and learnings from the implementation of the Use Cases reported in WP6.	LNDS IACS TMF CLIN BAT CRG BAH ASEBIO HAD CERTH HDH CRG	COO (CoL) BEN BEN BEN BEN BEN BEN BEN BEN BEN	No
T8.4	Stakeholder Involved ethics framework for trusted data and	A multidisciplinary panel with WP2 support (n= +-15) with patients at the centre (n= >33%), featuring legal, policy, ethics representatives will codesign (through deliberative workshops and design thinking methods) a practical ethical framework to facilitate good practices in the biotechnology innovation lifecycle within the EHDS ecosystem, complementary to the regulatory framework. It will focus on topics like AI use and transparency,	Sciensano ELLOK POLA DE	BEN BEN BEN BEN	No

	AI practices	information provision, commercial interests and public good, guiding principles for synthetic data generation and use, proportionality in IP claims and privacy protection. Depending on needs identified throughout the project this list will be focused. The main purpose of the task is to foster trust between patients and industry in a shared ethical framework to leverage the EHDS for biotechnology innovation. For example, by developing fair standards for the development of products under IP based on a public good.	GÖG LNDS TMF	BEN COO BEN	
T8.5	Ethical and Legal support and oversight for WP1 and WP6	This task will provide support to other WPs towards meeting the partners' ethical, legal, regulatory, and institutional compliance obligations in the context of the project. While the institutional compliance is ultimately the responsibility of each partner, this task will ensure the efforts are coordinated and aligned across the WPs and that respective partners are informed about their duties and timelines, especially in the use cases. This task will also act as the main contact point for the legal departments, data protection officers, and other compliance personnel of the relevant consortium partners who require guidance and contextual clarifications when pursuing their compliance work. This task will mainly render support to the consortium members in the following WPs: WP1 in setting up the DMP and conducting DPIAs (where necessary), and WP6 in navigating through different compliance obligations across the multiple legal frameworks.	LNDS TMF	COO BEN	No

Milestones and deliverables (outputs/outcomes)

Mile-stone No	Milestone Name	WP No	Lead Beneficiary	Description	Due Date	Means of verification
MS8.1	Problem inventory group organised	8	DE	At the start of the project, a working group dedicated to compiling the regulatory problem inventory will be convened. This will formally kick-start the process of mapping the regulatory problem inventory	M2	Meeting minutes or meeting summary notes drafted
MS8.2	Panel established	8	SCIENSANO	All members contacted and confirmed participation, workplan and topic priority setting established	M6	Confirmation emails and first design thinking online workshop organized
MS8.3	Overview of the applicable legal & regulatory framework	8	LNDS	This will be an internal report intended to: i) inform the mapping of the compliance requirements for the defined use cases and case studies (to be carried out as part of MS8.4); and ii) lay the groundwork for an extensive legal analysis under D.8.2	M6	Internal report submitted via email to the project coordinator
MS8.4	Compliance requirements and ethics clearance needs assessment for use cases	8	LNDS	This internal report will summarise the legal, contractual, and institutional requirements (including research ethics committee and/or institutional review board approvals) for the WP6 use cases. Additionally, it will describe the preliminary steps taken towards obtaining these requirements or approvals, alongside any hurdles identified.	M9	Internal report submitted via email to the coordinator.

MS8.5	Guideline on how to navigate the complex regulatory framework	8	LNDS	This guideline will be developed based on the findings of the in-depth legal analysis to be carried out in the context of D8.2. The guideline will be used as input for deliverable D8.5 describing user journeys throughout the AI- and data-driven biotechnology innovation lifecycle.			M32	The document submitted via email to the coordinator and D8.5 lead.
Deliverable No	Deliverable Name	WP No	Lead Beneficiary	Type	Dissem. Level	Due Date	Description	
D8.1	Report on regulatory problems from the industry perspective	8	DE	R	PU	M08	This report summarizes the practical regulatory issues biotech industry faces in developing their products and bringing them to the market.	
D8.2	Legal analysis of the Interplay between EHDS Regulation and other regulatory frameworks governing AI-driven biotech innovation lifecycle	8	LNDS	R	PU	M30	This deliverable will examine the interplay between EHDS Regulation concepts (e.g., the SPE and its associated processes, as well as the roles and responsibilities of Health Data Holders and HDABs) with related concepts under other EU laws, regulations and policy frameworks (e.g., AI Regulatory Sandboxes under the AI Act, the forthcoming “trusted processing environments” to be established by Data Labs in the context of the recently announced European Data Union Strategy; concrete activities where these Data Labs could support EHDS actors such as Health Data Holders and HDABs).	
D8.3	Legal guideline how to integrate infrastructures and AI factories into the EHDS	8	TMF	R	PU	M36	This guideline will be the result of the work done in Task 8.2 and provide legal guidance from the perspective of infrastructures, HDABs/national legislators, data holders, AI factories and biotech industry how data sharing infrastructures can be integrated into the EHDS.	
D8.4	Recommendations for an ethical application of the EHDS framework for biotechnology innovation	8	SCIENSANO	R	PU	M36	The panel will continuously produce recommendations, which will be bundled in a final report.	
D8.5	Archetypal user journeys in the biotech industry illustrating regulatory pathways for bringing products to the market	8	IACS	R	PU	M36	Illustrative user journeys within the AI-driven biotechnology innovation lifecycle will be developed, building on the outcomes of D8.2 (legal analysis) and MS8.5, while also incorporating learnings from the WP6 Use Cases. This deliverable will describe pathways for bringing AI- and data-driven products to the market, clarifying the roles and responsibilities of the key actors (such as industry and various regulatory authorities).	

Estimated budget — Resources

See detailed budget table/calculator (annex 1 to Part B).

Work Package 9**Work Package 9: Dissemination, Communication and Exploitation**

Duration:	M1-M36	Lead Beneficiary:	27 - DE
------------------	--------	--------------------------	---------

Objectives

- **09.1:** Increase visibility and understanding of the multistakeholder ecosystem around AI and biotechnology. Strengthen awareness and engagement of biotech industry actors, data holders, authorities, AI developers and patient organisations through targeted communication explaining the project's purpose, benefits, and opportunities for participation.
- **09.2:** Create the project branding and disseminate progress and results on the identification and prioritisation of digital/AI applications. Ensure transparency and visibility of the methodologies, criteria, and outcomes used to identify high-impact AI/biotech applications, including communication of prototypes and validations produced with AI Factories.
- **09.3:** Raise awareness about challenges and solutions for AI development in biotechnology. Communicate findings related to regulatory, ethical, technical and scientific barriers, as well as strategies to overcome them with a strong link to the role of the EHDS.
- **09.4:** Ensure uptake of project recommendations in policy, strategic and scientific debates. Promote the project's recommendations to policymakers, industry leaders, research infrastructures and EU networks to support coordinated EHDS development.
- **09.5:** Promote long-term adoption, replicability and sustainability of project results. Ensure long-lasting visibility of outputs and create mechanisms to support continuous knowledge exchange beyond the project's lifetime.

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contr. & Subcontr.
			Name	Role	
T9.1	Dissemination and communication plan and strategy monitoring (M1-M36)	This task establishes a targeted strategy to guide all dissemination and communication actions throughout the project. A dissemination and communication plan will be prepared by Month 3 to ensure structured, coherent and effective outreach across all partners. The plan will define the project's communication objectives, key audiences, core messages, channels, responsibilities and measurable indicators, enabling a consistent approach across the entire consortium. It will support partners in planning and coordinating their activities and will ensure alignment with scientific, technical, regulatory and stakeholder-engagement outputs produced across the project. This task ensures accuracy, coherence and adherence to European Union visibility requirements , with close coordination between the communication team, the stakeholder engagement work, and the legal and ethical work.	DE AseBio	BEN BEN	No
T9.2	Dissemination and communication	This task implements the communication and dissemination measures defined under the strategy. Activities will include establishing the project's visual identity , maintaining a public website , developing regular news and updates, preparing thematic content reflecting project findings, and ensuring continuous visibility through established communication channels. Tailored measures for	AseBio DE All beneficiaries	BEN BEN BEN	No

	activities (M1-M36)	dissemination and communication are foreseen, including: (i) logo, branding package and website creation (led by DE); (ii) website content management (led by ASEBIO); (iii) social media creation and active presence (ASEBIO); (iv) bi-annual newsletter (ASEBIO); ad-hoc press releases and publications (ASEBIO); (v) external conferences and events (DE); as well as, (vi) policy briefs and infographics (DE). A report on the progress of our dissemination and communication activities as well as on their results will be consolidated in yearly reports at M12, M24 and M36 respectively.			
T9.3	Innovation management, exploitation and sustainability (M1-M36)	This task ensures that the project's scientific, technical and organisational results are systematically captured, assessed, protected where appropriate, and prepared for long-term use. An exploitation and sustainability plan will be established in the project to set out how background and new knowledge will be identified, documented and managed. The plan will include: (i) the project's overall approach to innovation and knowledge management, including the handling of background and foreground results; (ii) the catalogue of project assets and their potential routes for exploitation during and after the project; (iii) the target groups who may benefit from the uptake of each asset and the conditions required for successful adoption; (iv) the relevant intellectual-property considerations, the applicable provisions for access or use, and any joint exploitation interests among partners; and (v) the mechanisms and procedures through which the results may be exploited, including pathways through existing European infrastructures, digital research environments and recognised platforms for results sharing. The task will examine regulatory, operational and market factors influencing sustainability, and will work closely with scientific, technical and legal work strands to ensure that exploitable results are recognised early and supported with realistic strategies for long-term continuation. This process will include assessing opportunities for integration into health-data infrastructures, secure processing environments and relevant public or private initiatives, ensuring that the project's outputs remain usable and impactful beyond its 36-month duration.	DE AseBio	BEN BEN	No
T9.4	Synergies and knowledge sharing with relevant initiatives, platforms, and stakeholders (M1-M36)	This task establishes focused synergies with European initiatives , infrastructures, and other initiatives to promote knowledge exchange and ensure that the project is fully aligned with ongoing efforts in responsible data use and biotechnology innovation. Synergies will be prioritized with initiatives already connected to consortium partners and directly relevant to health-data governance, secondary use and artificial-intelligence-enabled biotechnology (e.g HealthData@EU Pilot, TEHDAS2, HealthyCloud, the Health Data Access Bodies Community of Practice, Genomics Data Infrastructure and the 1+ Million Genomes actions, Cancer Image Europe (EUCAIM), ELIXIR, EBRAINS, UNCAN-Connect, CANDLE, QUANTUM and SHAIPEd, as well as emerging national and regional AI Factories). Communication channels with these initiatives will be established in coordination with partners engaged in them, and concrete synergies will be defined after direct exchanges with their coordinators. These may include mutual participation in events, joint webinars, coordinated dissemination, or shared thematic sessions. The outcomes of this task will be reported in the project's dissemination and communication reports.	DE AseBio All beneficiaries	BEN BEN BEN	No

Milestones and deliverables (outputs/outcomes)

Mile-stone No	Milestone Name	WP No	Lead Beneficiary	Description	Due Date	Means of verification
---------------	----------------	-------	------------------	-------------	----------	-----------------------

MS9.1	Project branding available	9	DE	Branding manual to be applicable by all partners.			M04	Branding document.
MS9.2	Project website	9	AseBio	The project's main website is live.			M04	Project website content.
Deliverable No	Deliverable Name	WP No	Lead Beneficiary	Type	Dissem. Level	Due Date	Description	
D9.1	Dissemination communication and stakeholder engagement plan	9	DE	R	PU	M03	The dissemination and communication plan sets a framework that ensures a consistent and coherent communication and dissemination activities throughout the project lifetime.	
D9.2	Dissemination and communication report v1	9	AseBio	R	PU	M12	First report to brief on the dissemination and communication actions, events and publications.	
D9.3	Dissemination and communication report v2	9	AseBio	R	PU	M24	Second report to brief on the dissemination and communication actions, events and publications.	
D9.4	Dissemination and communication report v3	9	AseBio	R	PU	M36	Final report to brief on the dissemination and communication actions, events and publications.	
D9.5	Exploitation and sustainability plan v1	9	DE	R	PU	M14	A report outlining the growth strategy and sustainability plan to capture new sources of value.	
D9.6	Exploitation and sustainability plan v2	9	DE	R	PU	M36	An updated report outlining the growth strategy and sustainability plan to capture new sources of value.	

Estimated budget — Resources

See detailed budget table/calculator (annex 1 to Part B).

TIMELINE	YEAR 1				YEAR 2				YEAR 3			
	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4
T1.1 – Project Management												
T1.2 – Administrative and financial management												
T1.3 – Data Management												
T2.1 – Stakeholder Mapping and Mobilisation												
T2.2 – Platform Establishment (Governance & Digital Hub)												
T2.3 – Community Building & Fora Organisation												
T2.4 – Strategic Communication and Impact												
T3.1 – Landscape Analysis of Health Data Use for AI in the Biotech Sector												

TIMELINE	YEAR 1				YEAR 2				YEAR 3			
	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4
T3.3 – Strategic Roadmap for Scaling AI in the Biotech Sector												
T4.1 – “Pathfinder system v1”: Draft Monitoring and Evaluation Guidelines												
T4.2 – “Test Drive”: Validate Monitoring and Evaluation Guidelines against Case Studies and Pilot RIs and SPEs on AI Factories												
T4.3 – “Pathfinder system v2”: Finalize Monitoring and Evaluation Guidelines												
T5.1 - Guidelines for integration of Data Infrastructures into EHDS												
T5.2 - Integration of Data Infrastructures into EDHS: Governance and operations												
T5.3 - Integration of Data Infrastructures into EDHS: Data provisioning												
T5.4 - Integration of Data Infrastructures into EDHS: Processing and AI Training												
T5.5 - Piloting data infrastructure involvement in biotech use cases implementation												
T6.1 - UC operational management and external knowledge transfer												
T6.2 – Internal refinement feedback loop												
T6.3 - Open call for selecting additional use cases												
T6.4 - UC1 – Generating synthetic sample for early access through EHDS metadata catalogue												
T6.5 - UC2 - Enabling the secondary use of data in HPC environment according to EHDS specifications												
T6.6 - UC3 – Exploring cross-border Federated Learning for secondary data use												
T6.7 - UC4 – GenAI-Powered Virtual Patients for Optimising Early-Phase Clinical Trials in Glioblastoma												
T6.8 - UC5 – AI – driven prevention, diagnostic and management for delirium in ICU patients												
T6.9 - UC6 - Polygenic Risk Scores												
T6.10 - UC7 - Generating high-fidelity synthetic medical images for AI development and evaluation												
T6.11 - UC8 - Generating synthetic control arms to accelerate rare disease trials												
T7.1 - Critical evaluation of secure technologies for processing Health Data												
T7.2 - Technical and methodological toolkit for building AI models in the EHDS												
T7.3 - Use and reuse of Health Data through secure environments in the context of AI factories and EHDS												
T7.4 - Technical solutions for bringing forward use-cases and case-studies												
T8.1 - Regulatory problem inventory (M1-M12)												
T8.2 - The legal integration of existing and future data infrastructures in the EHDS (M7-M36)												
T8.3 - Clarifying the regulatory framework for bringing products to the market (M1-M36)												

TIMELINE	YEAR 1				YEAR 2				YEAR 3			
	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4
T8.4 - Stakeholder Involved ethics framework for trusted data and AI practices (M1-M36)												
T8.5 - Ethical and Legal support and oversight for WP1 and WP6 (M1-M36)												
T9.1 – Dissemination and communication plan and strategy monitoring (M1-M36)												
T9.2 – Dissemination and communication activities (M1-M36)												
T9.3 – Innovation management, exploitation and sustainability (M12-M36)												
T9.4 – Synergies and knowledge sharing with relevant initiatives, platforms, and stakeholders (M1-M36)												

#\$WRK-PLA-WP\$#

Subcontracting (n/a for prefixed Lump Sum Grants)**Subcontracting**

Give details on subcontracted project tasks (if any) and explain the reasons why (as opposed to direct implementation by the Beneficiaries/Affiliated Entities).

Subcontracting — Subcontracting means the implementation of ‘action tasks’, i.e. specific tasks which are part of the EU grant and are described in Annex 1 of the Grant Agreement.

Note: Subcontracting concerns the outsourcing of a part of the project to a party outside the consortium. It is not simply about purchasing goods or services. We normally expect that the participants have sufficient operational capacity to implement the project activities themselves. Subcontracting should therefore be exceptional.

Include only subcontracts that comply with the rules (i.e. best value for money and no conflict of interest; no subcontracting of coordinator tasks).

Work Package No	Subcontract No (continuous numbering linked to WP)	Subcontract Name (subcontracted action tasks)	Description (including task number and BEN/AE to which it is linked)	Estimated Costs (EUR)	Justification (why is subcontracting necessary?)	Best-Value-for-Money (how do you intend to ensure it?)
6	S6.1	Subcontractor A	UC2 (Enabling the secondary use of data in HPC environment according to EHDS specifications) technical expertise and development work	169,000	THL/Findata's ICT strategy is based on organizing the necessary secondary use infrastructure and data centre services by subcontracting development and maintenance work. We use the service providers available on the market and highly sophisticated cloud service-based solutions.	Through a competitive, market-based selection process in accordance with the beneficiary's internal procurement rules and the EU4Health MGA, The assessment shall consider proven competence on secondary use services and infrastructure, knowledge of pertinent implementation and stakeholder/legal environment, price, technical quality, compliance with EHDS specifications, security and data protection guarantees, and proven experience in HPC-based environments for secondary use of health

						data.S
6	S6.2	Subcontractor A	UC3 (Exploring cross-border Federated Learning for secondary data use) technical expertise and development work	155,000	THL/Findata's ICT strategy is based on organizing the necessary secondary use infrastructure and data centre services by subcontracting development and maintenance work. We use the service providers available on the market and highly sophisticated cloud service-based solutions.	Ensuring both cost-effectiveness and technical excellence by using existing contracts or if needed, by applying a transparent selection process. The assessment shall consider proven competence on secondary use services and infrastructure, pricing, scalability, compliance with EHDS and GDPR requirements, interoperability, and demonstrated experience in federated learning and cross-border data processing infrastructures.
6	S6.3	Subcontractor A	Sequencing services	30,000	Batea will integrate imaging-based patient-related data with tumour characteristics using single cell RNAseq and/or other methodologies to optimize model performance and predictive value.	Transparent selection process that evaluates price, technical quality, sequencing performance and data analysis and release. Demonstrated experience in sequencing.
6	S6.4	Subcontractor A	UC1 - Regulatory management of data (DPO)	2,000	UC1 - Alia Santé involves a Data Protection Officer (DPO) to ensure that data access, sharing, and usage within the project fully comply with GDPR and EHDS requirements. The DPO oversees secure access controls, monitors data usage purposes, and ensures that both real and synthetic data are processed in a lawful, ethical, and privacy-preserving manner, enabling trusted and compliant innovation.	Best value for money will be ensured through the reuse of existing, mature AI components and infrastructure, limiting development costs while maximizing impact. Alia Santé relies on a proven synthetic data platform and experienced multidisciplinary teams, reducing technical risk and time-to-delivery. Open standards, interoperability with EHDS, and scalable deployment models further ensure cost efficiency and long-term sustainability beyond the project.
6	S6.5	Subcontractor A	Task 6.6 - UC3 (Exploring cross-border Federated Learning for secondary data use) technical expertise and	75,000	Task 6.6 includes technical work potentially requiring support from a subcontractor. This work depending on the exact set up to be put in place this amount is an	Transparent selection process among qualified providers, strictly evaluating both cost-effectiveness and technical excellence. The assessment will prioritize proven experience in FL protocol

			development work		estimation.	implementation, guaranteed EHDS/GDPR compliance, and demonstrated scalability and interoperability within distributed SPE infrastructure.
4	S4.1	Subcontractor A	Development of Pathfinder Tool	145,000	As one of the key deliverables of the SCAILED project, the Pathfinder system should be a tool that is widely used by stakeholders, in and beyond the project. To ensure the guidelines of recommendation and monitoring system be a user-friendly tool, tailored according to stakeholders' needs, Epidata will seek expertise in tool development to collaborate with, as subcontractor to develop the pathfinder system.	By applying a transparent selection process among qualified providers, evaluating both cost-effectiveness and technical excellence. The assessment will consider pricing, scalability, compliance with EHDS and GDPR requirements, interoperability, and demonstrated experience in tool development on international market.

Other issues:

If subcontracting for the project goes beyond 30% of the total eligible costs, give specific reasons.

N/A

5. OTHER

5.1 Ethics

Ethics

If the Call document contains a section on ethics, describe ethics issues that may arise during the project implementation and the measures you intend to take to solve/avoid them.

As the SCAILED consortium involves multiple partners from different EU countries, and incorporates biotech use cases, the consortium could face ethical challenges related to the governance and secondary use of health/biomedical data. Variations in national practices, such as requirements and conditions for secondary use of health data in research projects, may lead to inconsistencies and pose challenges to the project's implementation. For example, procedural and institutional aspects of research ethics could give rise to challenges in coordinating approvals, oversight mechanisms, and compliance across multiple jurisdictions. Different partners might be subject to diverging ethical review processes, timelines, or institutional expectations, potentially slowing progress or creating gaps in oversight.

The consortium will address these challenges by mapping all applicable institutional (e.g., ethical review) requirements, aligning procedures across partners, and supporting the institutional compliance work of the relevant partners every step of the way. First, the consortium will implement a harmonised data governance framework (WP1), laying the foundation for a streamlined implementation of the planned activities. Second, the relevant WPs, namely WP1 and WP6, will be supported by the dedicated Ethics & Data Protection Manager alongside the ongoing Task T8.5. Through these resources, the consortium will ensure the partner institutes are adequately supported in their ethical, legal, and compliance efforts as they pursue the implementation of the project.

Considering that the use cases selected for this project heavily utilise AI approaches, the consortium might encounter ethical issues associated with the governance, training, and deployment of AI systems. These issues will be explored and addressed in Task T8.4, which revolves around trusted data and AI practices. By leveraging diverse methods ranging from conceptual analyses of the relevant ethical frameworks to stakeholder engagement activities, this task will examine the core principles of trustworthy AI through the lens of biotechnology innovation lifecycle. The concepts of explainability, transparency, algorithmic bias and human-in-the-loop requirements will be revisited from an ethical perspective. The insights generated from these activities will be complemented by the regulatory analyses to be carried out in other tasks of WP8, most pertinently Task T8.3.

Finally, the consortium will establish an independent ethics advisory board composed of three leading ethics and legal experts in the fields of health and biomedical data governance. The ethics advisory board will provide continuous high-level feedback to the consortium on various ethical and legal matters, helping steer the project towards ethically sound and legally compliant implementation.

#§ETH-ICS-EIS# #@SEC-URI-SU@#

5.2 Security

Security

If the Call document contains a section on security, describe security issues that may arise during the project implementation and the measures you intend to take to solve/avoid them.

Indicate if there is need for EU classification of information (Decision [2015/444](#)) or any other specific security measures.

N/A

#§SEC-URI-SU# #@DEC-LAR-DL@#

6. DECLARATIONS

Higher funding rate

Do you fulfil the conditions set out in the Call document for a higher funding rate?

If YES, explain and provide details.

YES

Condition met by the Action

“Actions with bodies from at least 14 EU countries and where at least four are from EU countries whose GNI per inhabitant is less than 90% of the EU average.”

The SCAILED partners included organisations from **17 EU countries. 8 of the EU countries fulfil the 90% GNI criteria.**

Double funding	
Information concerning other EU grants for this project ⚠️ Please note that there is a strict prohibition of double funding from the EU budget (except under EU Synergies actions).	NO
We confirm that to our best knowledge neither the project as a whole nor any parts of it have benefitted from any other EU grant (including EU funding managed by authorities in EU Member States or other funding bodies, e.g. EU Regional Funds, EU Agricultural Funds, etc). If NO, explain and provide details.	YES
We confirm that to our best knowledge neither the project as a whole nor any parts of it are (nor will be) submitted for any other EU grant (including EU funding managed by authorities in EU Member States or other funding bodies, e.g. EU Regional Funds, EU Agricultural Funds, etc). If NO, explain and provide details.	YES

Financial support to third parties
<i>If in your project the maximum amount per third party will be more than the threshold amount set in the Call document, justify and explain why the higher amount is necessary in order to fulfil your project's objectives.</i>

The project plans to launch an open call to select additional Use Cases, funded as financial support to third parties. This mechanism is crucial to reinforce biotech interest and participation in the project and to test new services or tools that emerge from the rapidly evolving field of AI and health data. The rationale is to ensure the project remains relevant, flexible, and responsive to the latest market and technological needs, thereby maximizing the validation scope of the EHDS and AI Factory infrastructures. The call aims to select a total of three additional UCs. The open call procedure will comply strictly with all EU requirements regarding transparency, publication, and expert evaluation to ensure a clear European dimension.

The management of the open call will be carried out in Task 6.3 with support from WP1. The funding for the selected third parties will be managed and distributed by parter 1-LNDS and 2-HDH, each organisation will oversee distributing 50% of the funds.

Sovereignty Seal (if applicable)	
If provided in the Call document, all eligible proposals that exceed the evaluation thresholds will be awarded a Sovereignty Seal. The Sovereignty Seal is a quality label which aims at facilitating access to additional public and/or private funding.	
Do you agree that certain information about your proposal (i.e. acronym; title; abstract of the project; name, contact e-mail and address of the coordinator; requested EU contribution; and grant amount awarded, if any) will be published on the STEP online Portal (with open public access) to increase the visibility of your project?	YES
Do you agree that this information is shared with national managing authorities of EU funding that recognise and support Sovereignty Seal projects to facilitate funding for your project under other EU programmes and funds?	YES

#\$DEC-LAR-DL\$#

List of Annexes:

- Annex 1 Detailed budget table/Calculator
- Annex 2 CVs
- Annex 3 Annual activity reports
- Annex 4 List of previous projects